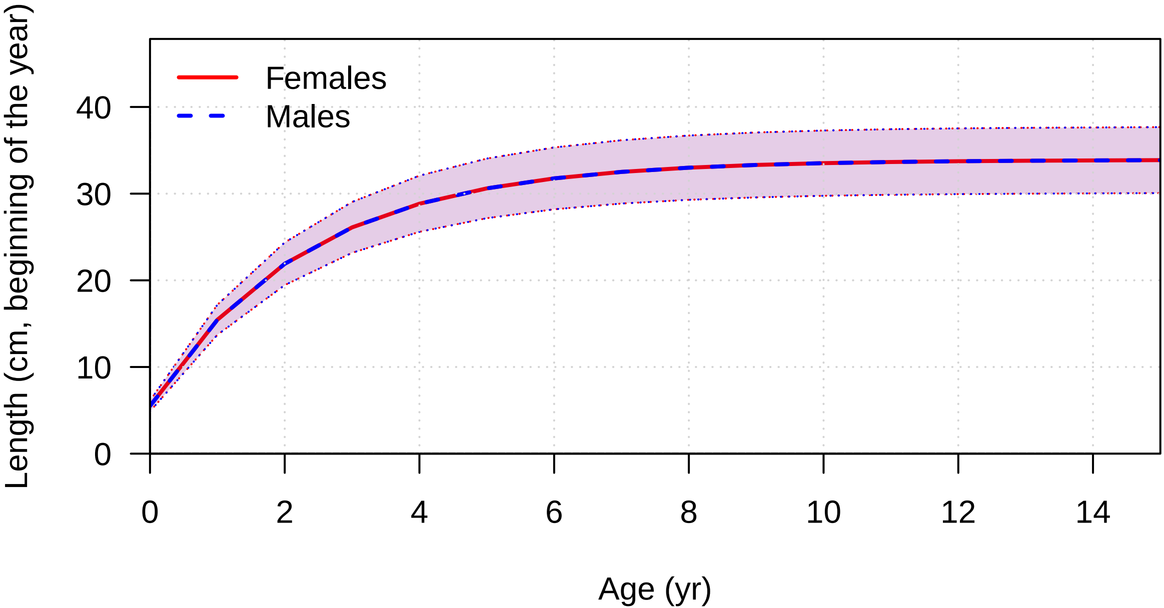
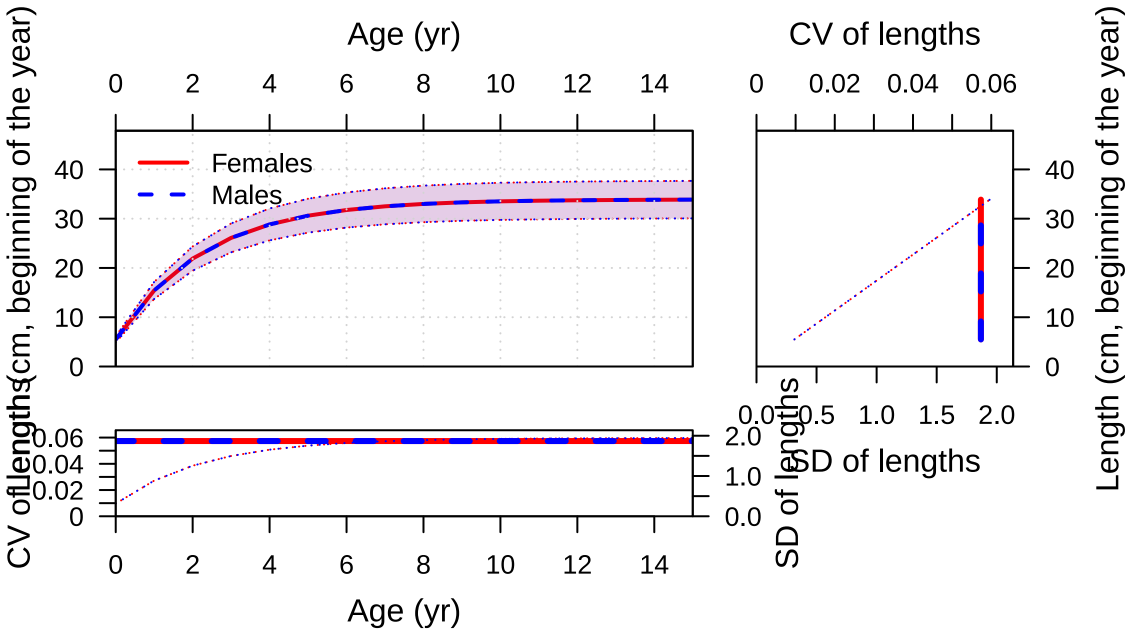
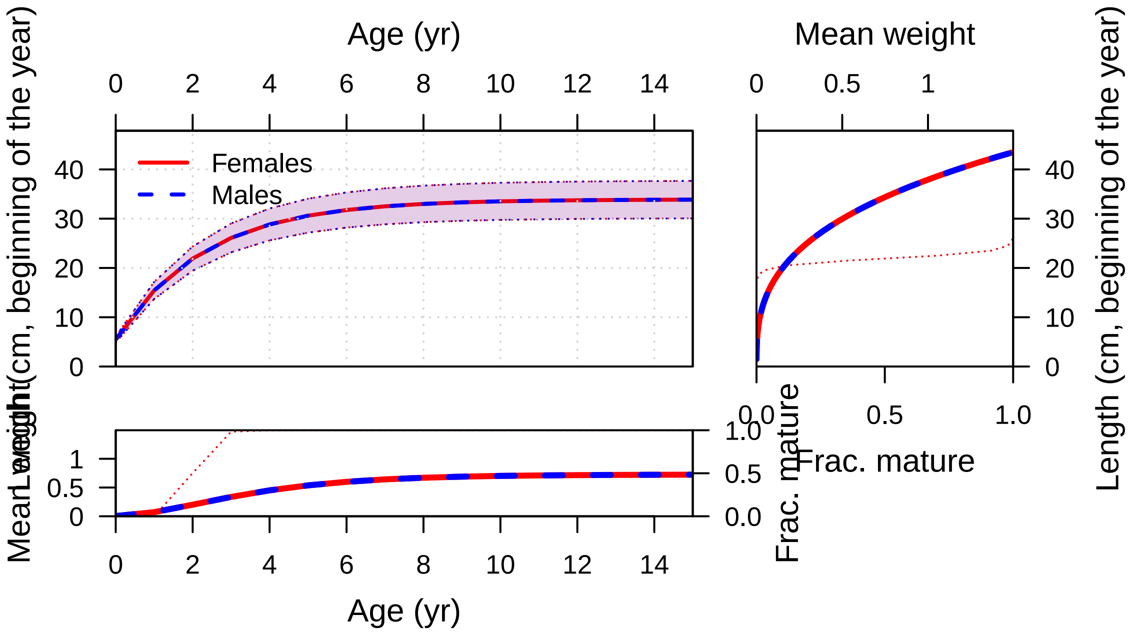
Plots created using the 'r4ss' package in R
Stock Synthesis version: 3.30.19.0
StartTime: Thu Dec 11 15:22:48 2025
Data_File: data.ss
Control_File: control.ss

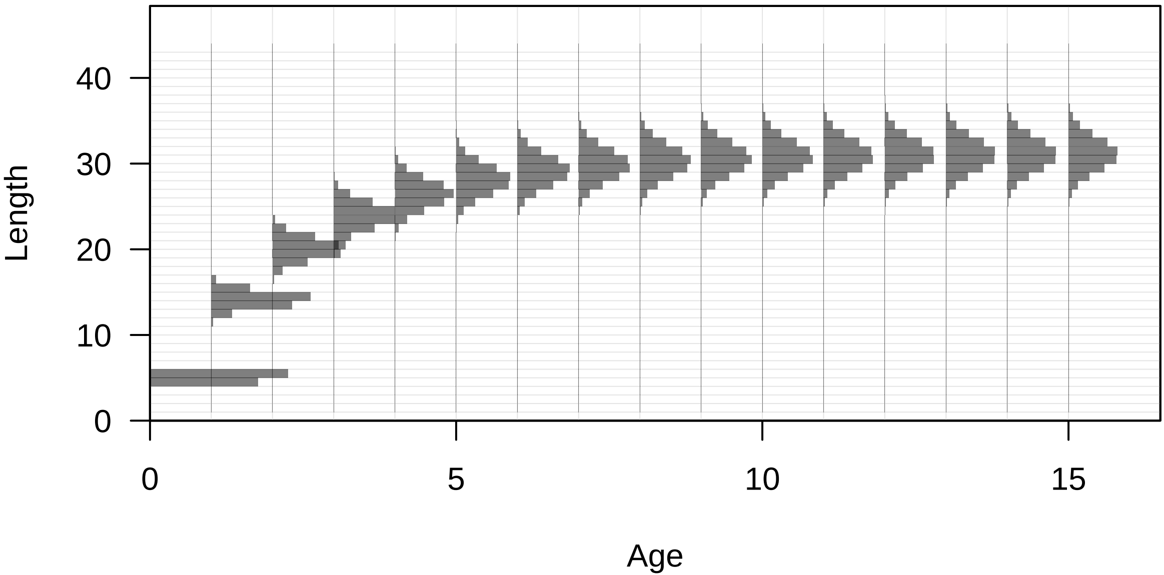

Females
Males
Length (cm, beginning of the year)
Age (yr)
0
10
20
30
40
0
2
4
6
8
10
12
14



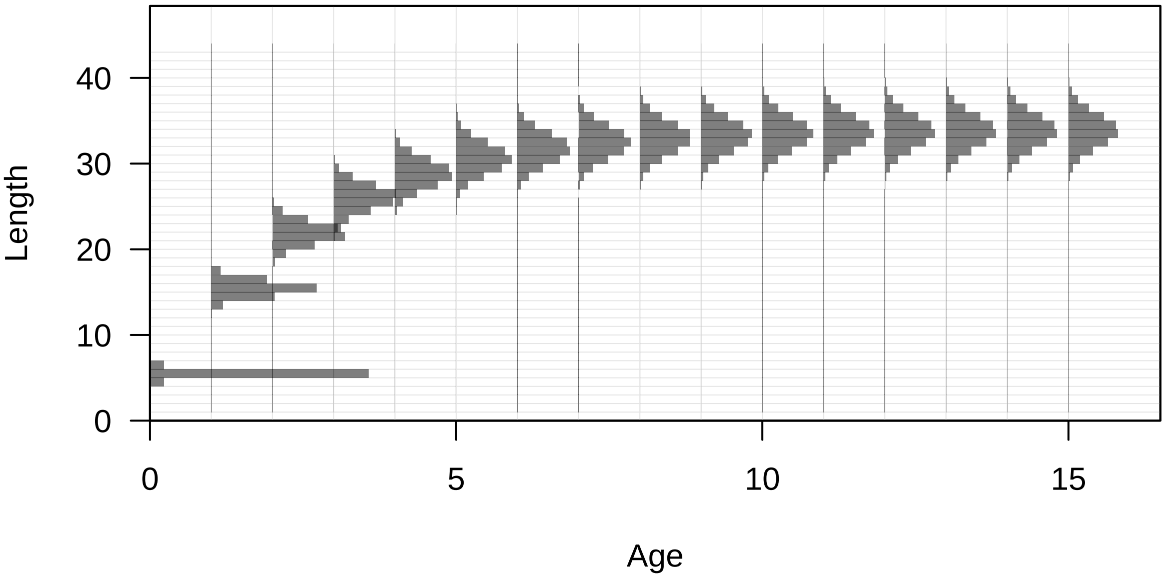
Age (yr)
CV of lengths
Females
Males
Length (cm, beginning of the year)
CV of lengths
SD of lengths
SD of lengths
Age (yr)



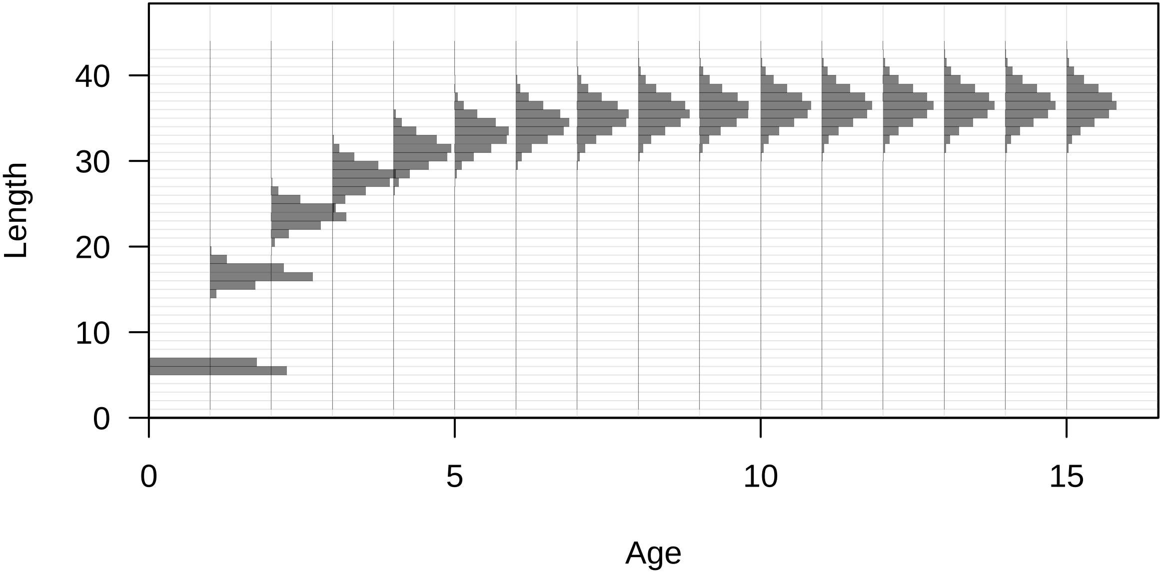
Age (yr)
Mean weight
Females
Males
Length (cm, beginning of the year)
Mean weight
Frac. mature
Age (yr)
0
2
4
6
8
10
12
14
0
0.5
1
0
10
20
30
40
0.0
0.5
1.0



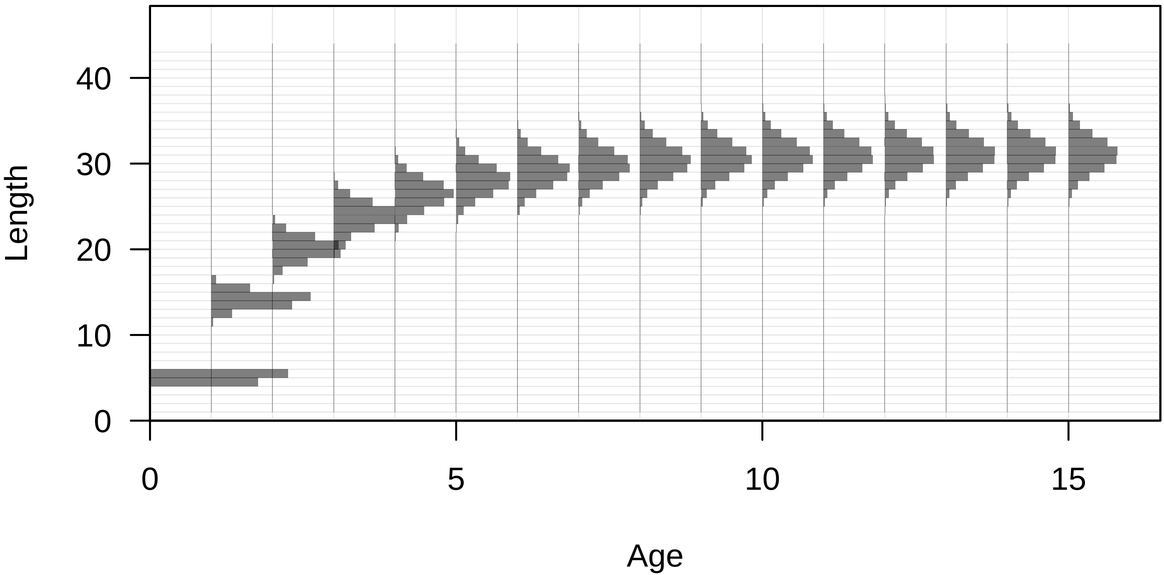
Length
Age
0
10
20
30
40
0
5
10
15



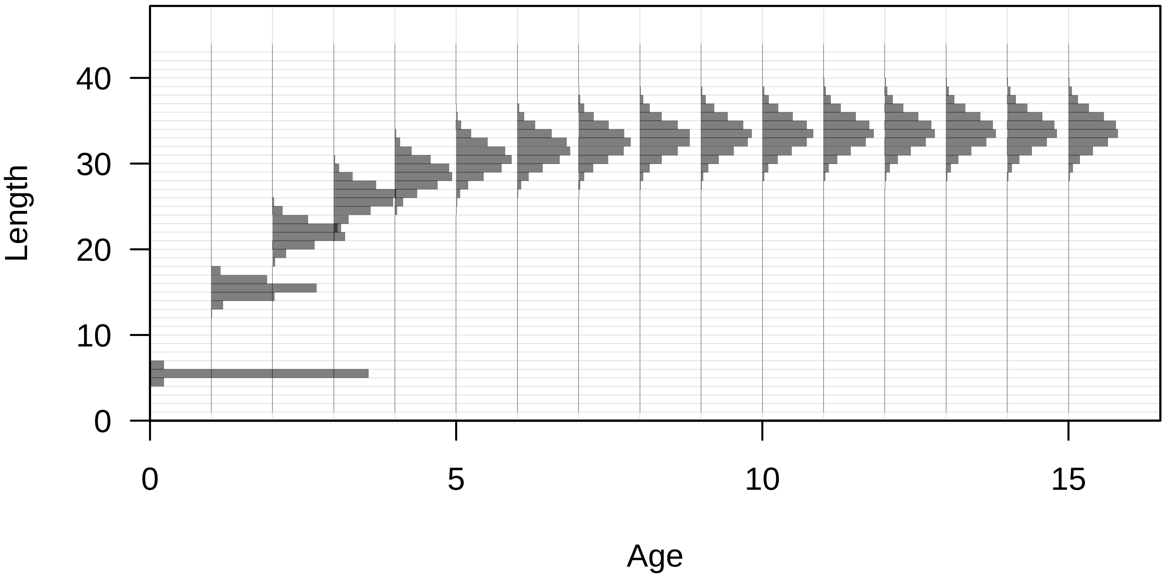
Length
Age
0
10
20
30
40
0
5
10
15



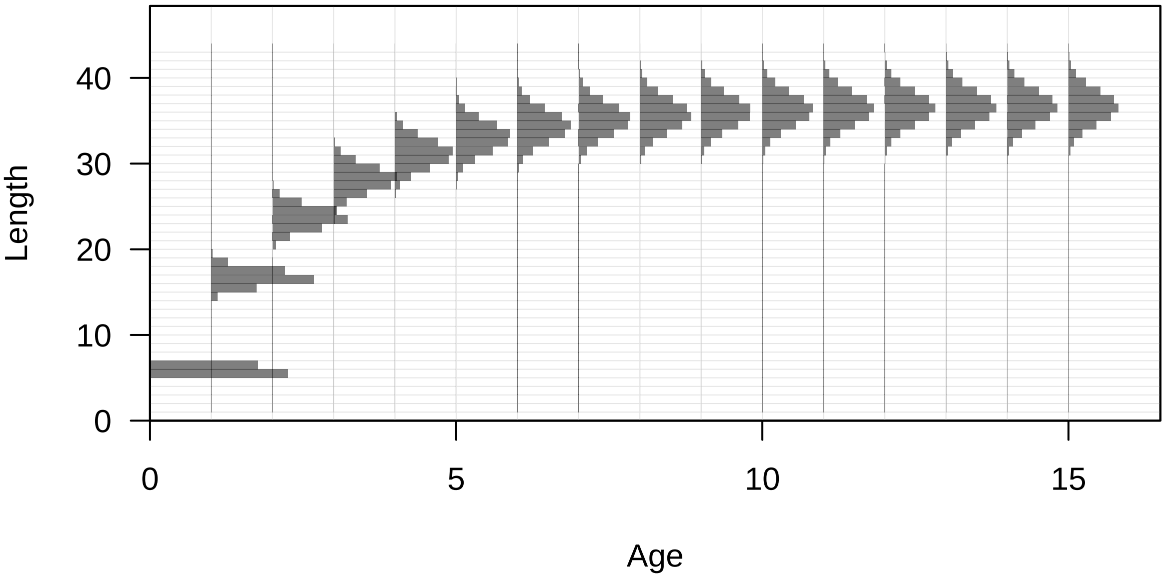
Length
40
30
20
10
0
0
5
10
15
Age

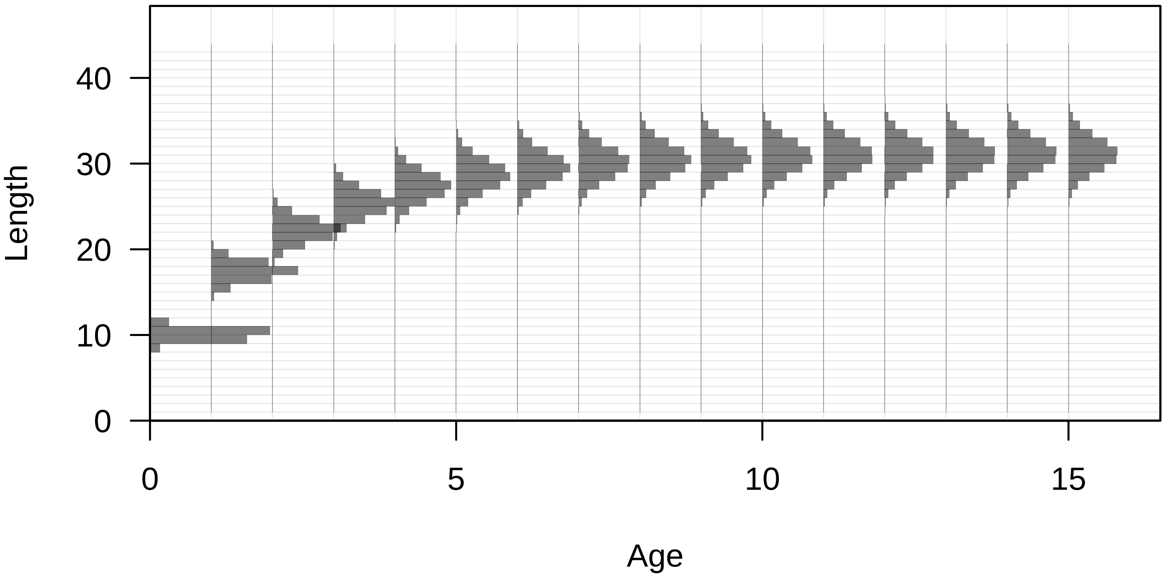


Length
0
10
20
30
40
Age
0
5
10
15

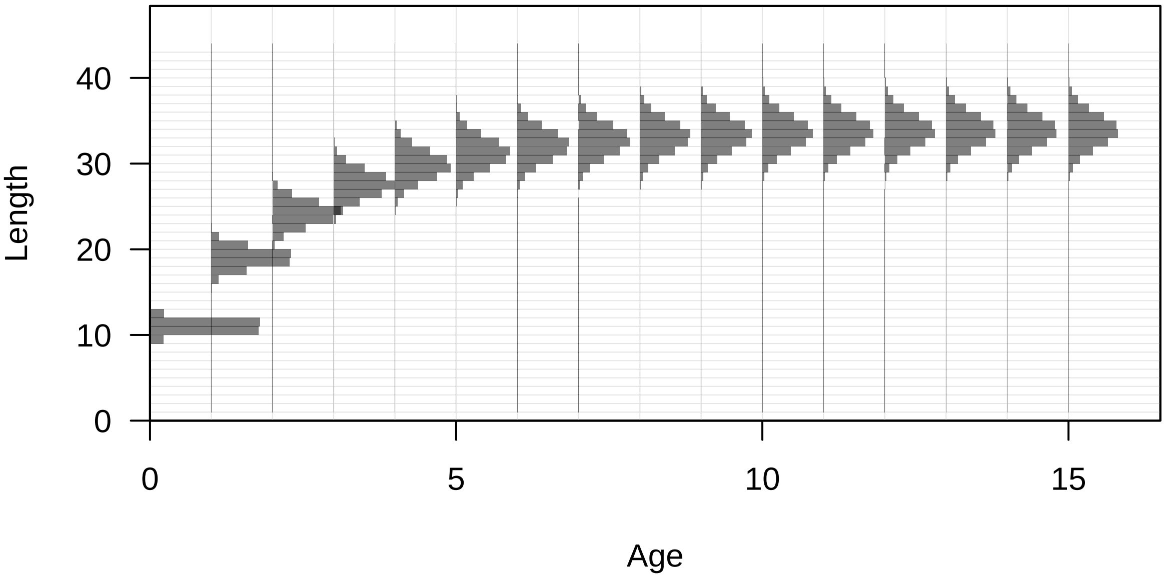


Length
Age
0
10
20
30
40
0
5
10
15

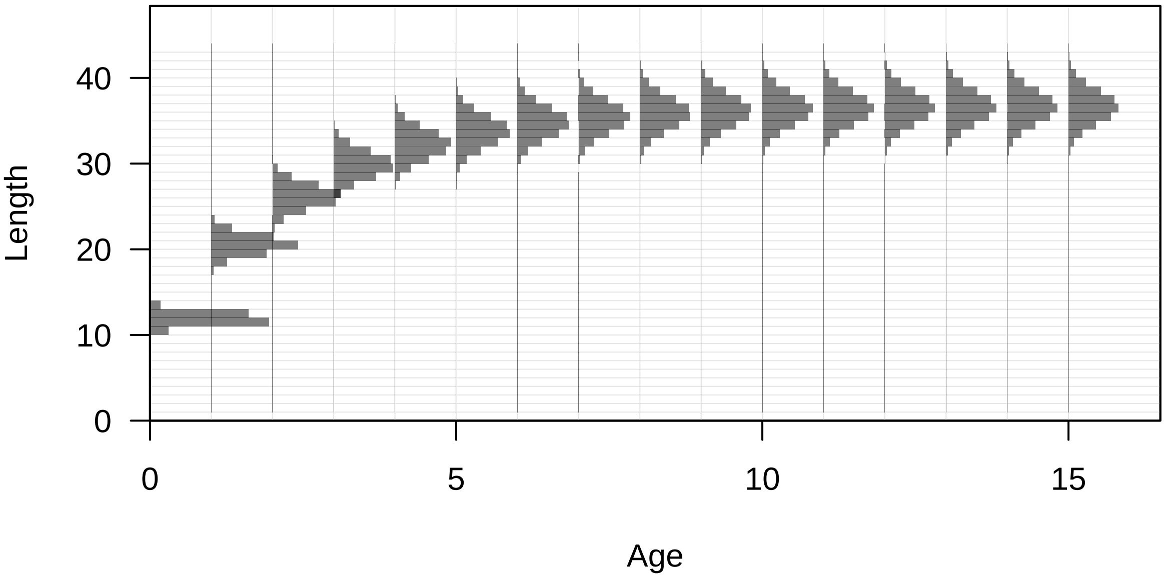

Length
Age
0
10
20
30
40
0
5
10
15



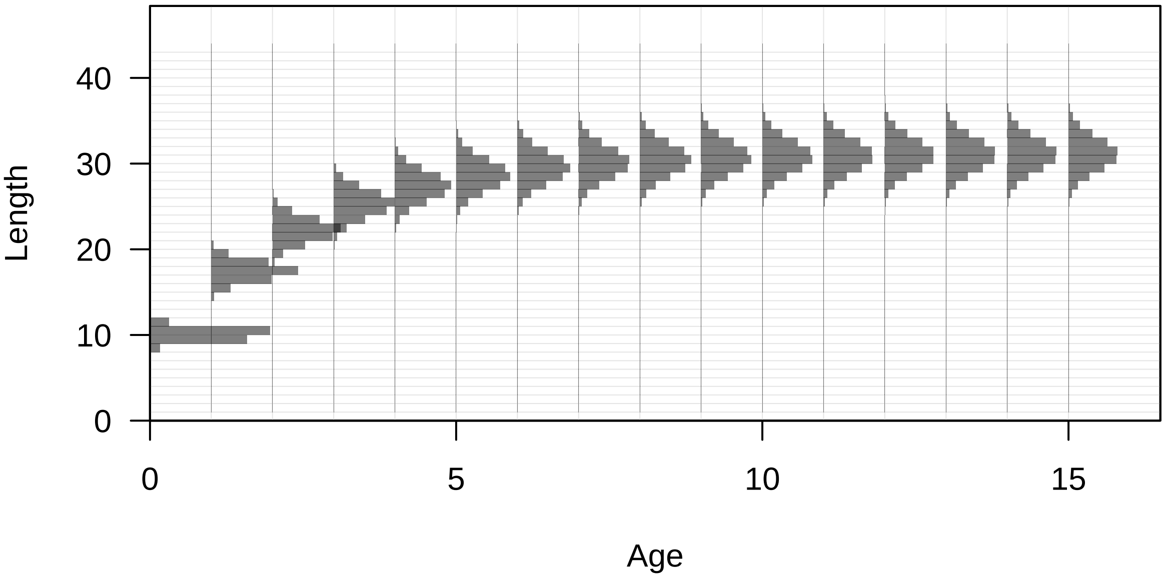
Length
40
30
20
10
0
0
5
10
15
Age



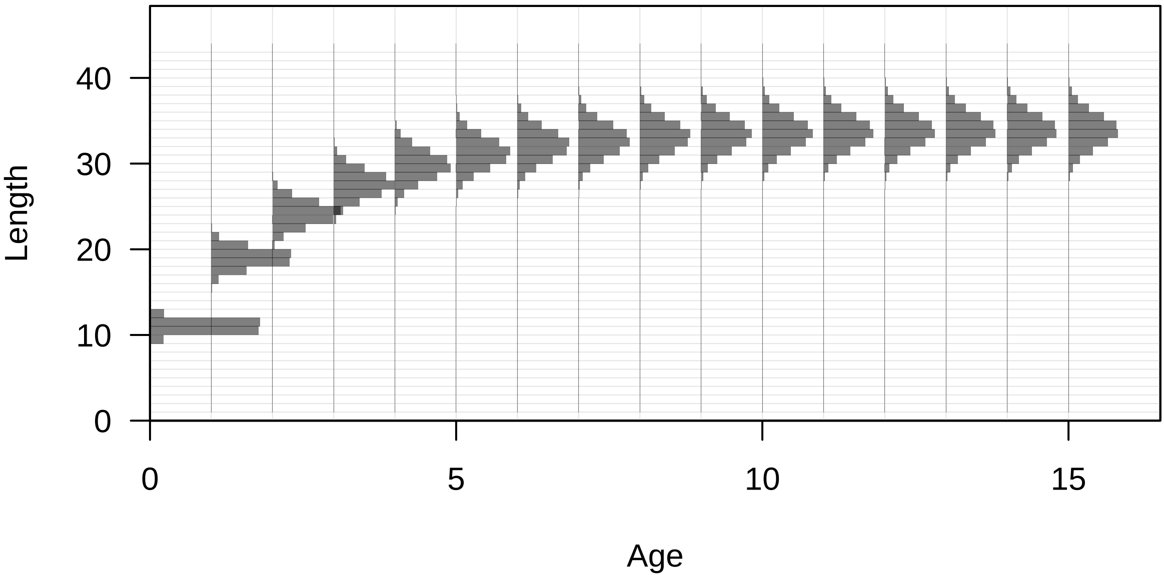
Length
Age
0
10
20
30
40
0
5
10
15



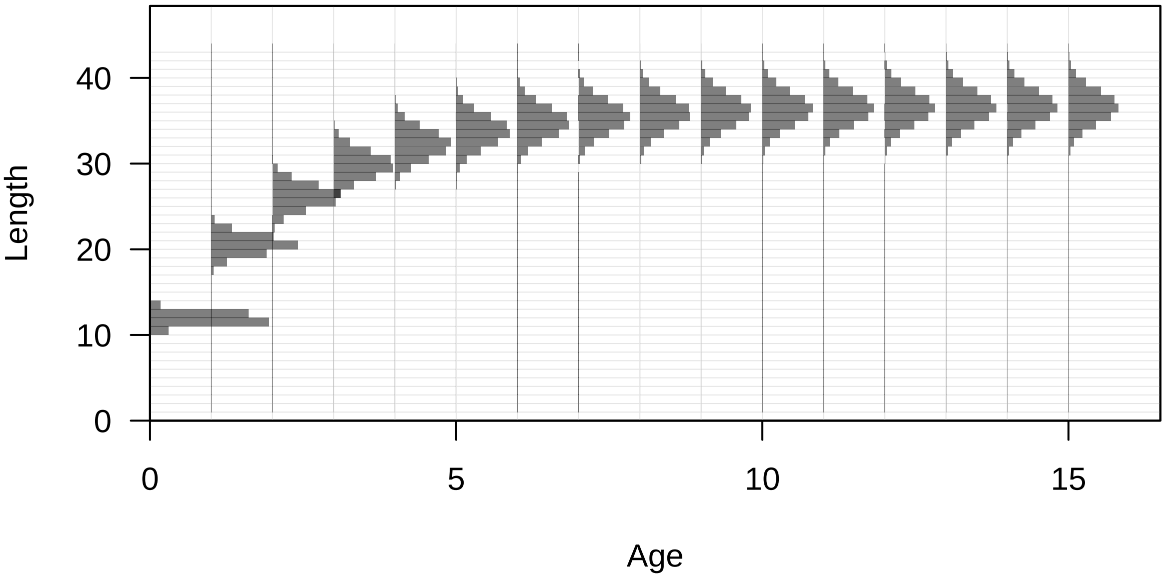
Length
40
30
20
10
0
0
5
10
15
Age

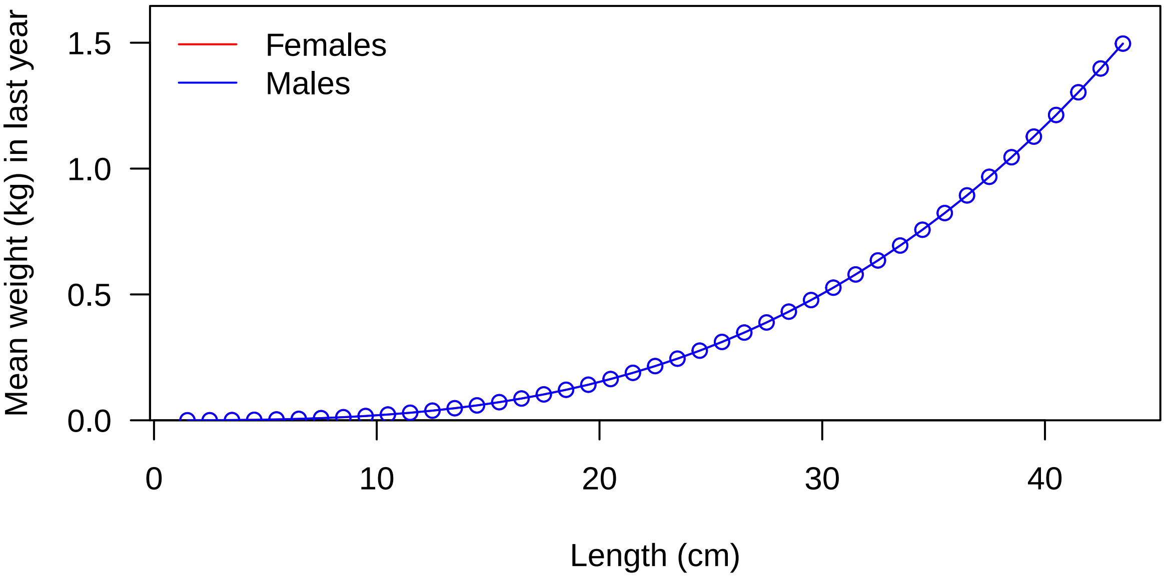


Length
0
10
20
30
40
Age
0
5
10
15

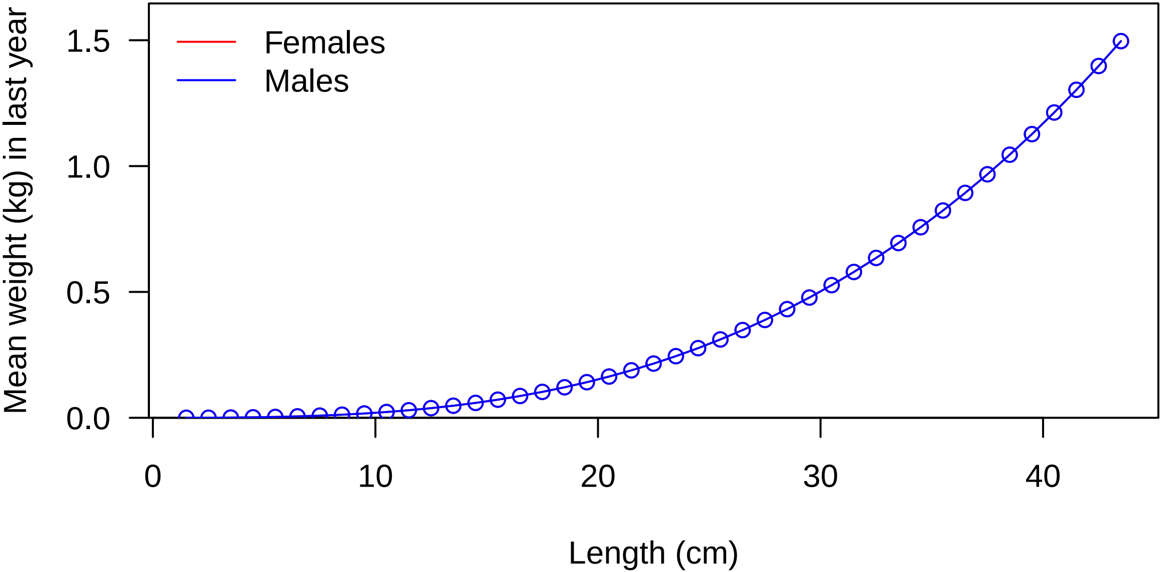


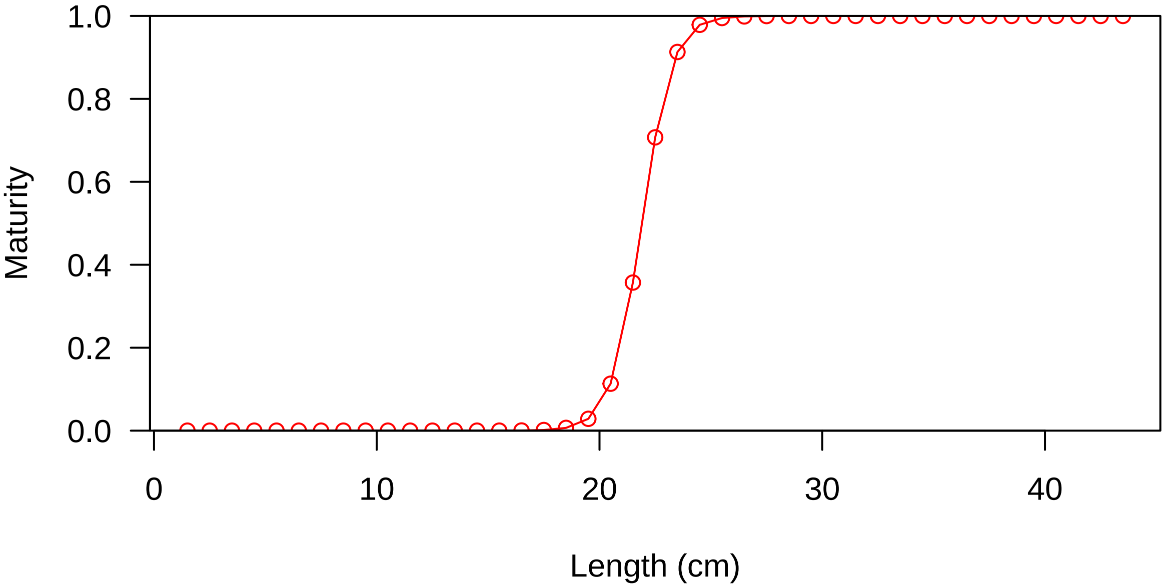
Length
Age
0
10
20
30
40
0
5
10
15


Length
40
30
20
10
0
0
5
10
15
Age

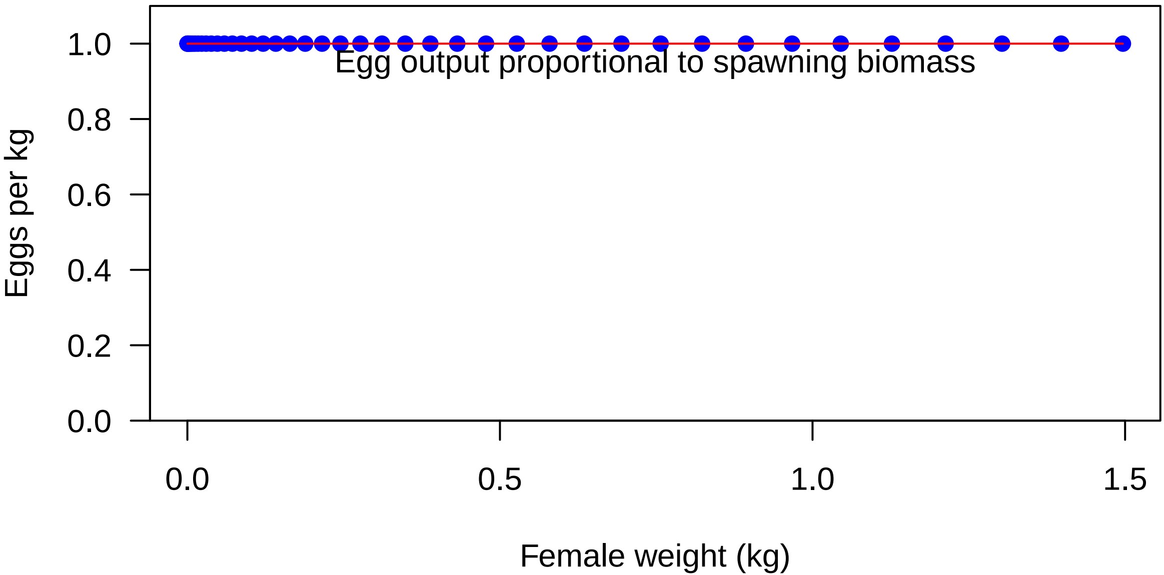


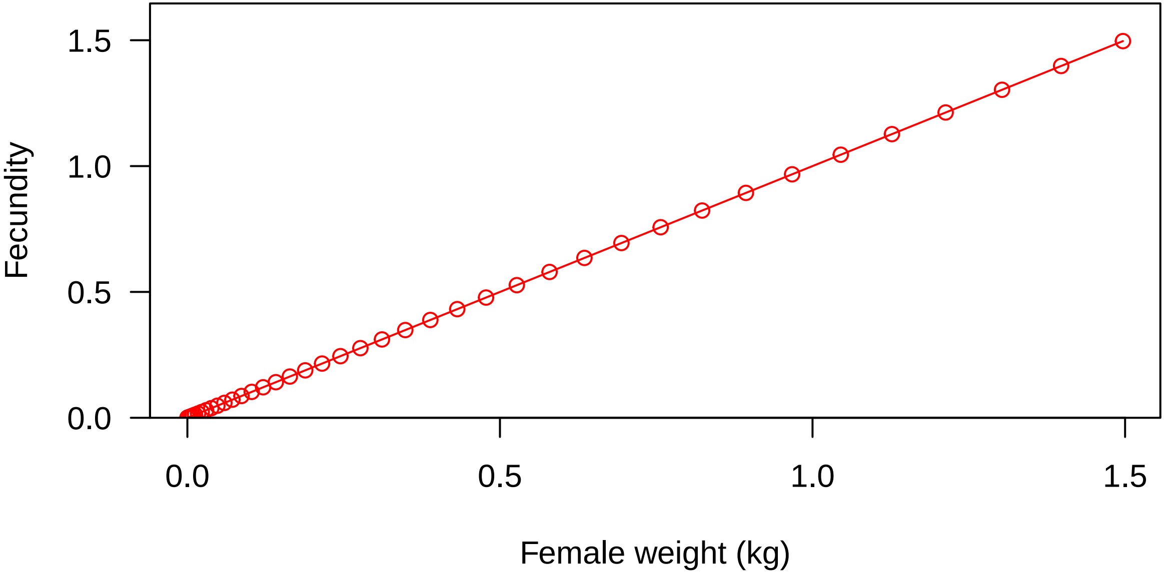
Females
Males
Mean weight (kg) in last year
Length (cm)
0.0
0.5
1.0
1.5
0
10
20
30
40


Mean weight (kg) in last year
Length (cm)
Females
Males
0.0
0.5
1.0
1.5
0
10
20
30
40

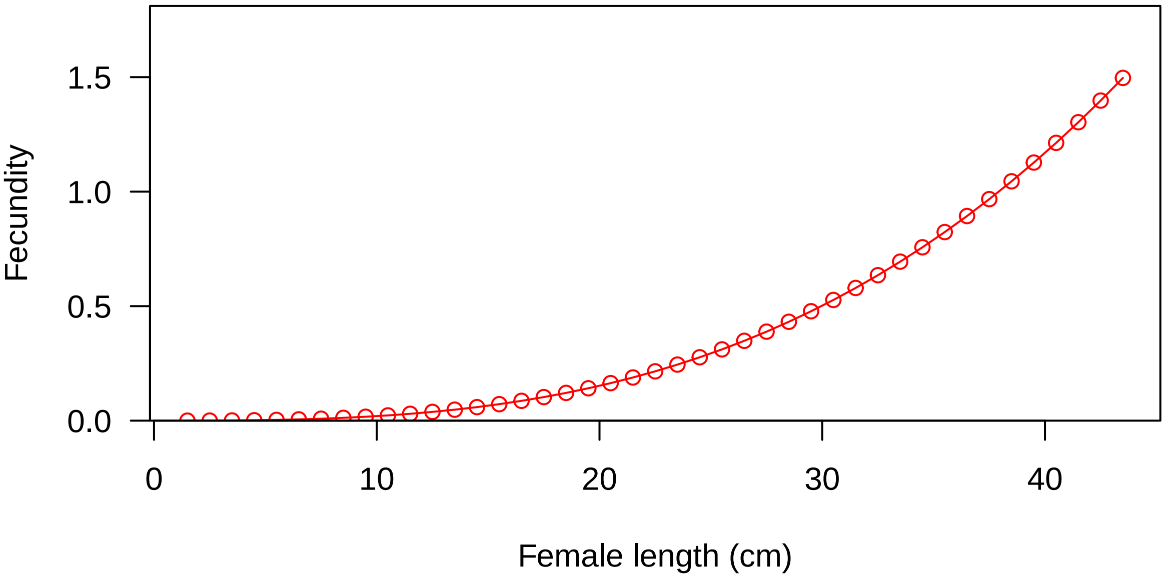


Maturity
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
Length (cm)

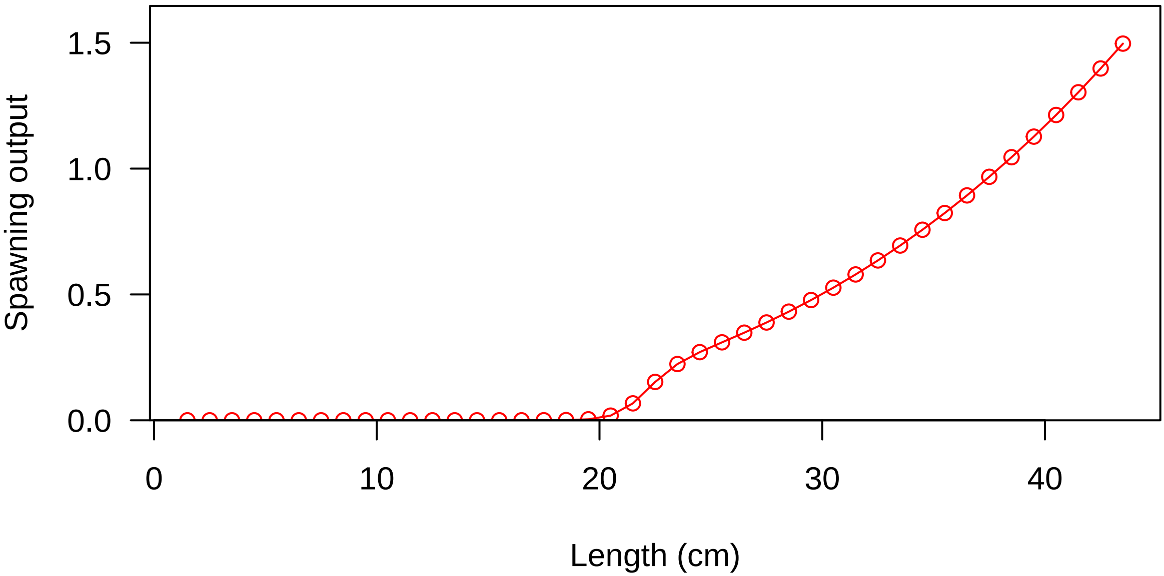

Egg output proportional to spawning biomass
Eggs per kg
Female weight (kg)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5



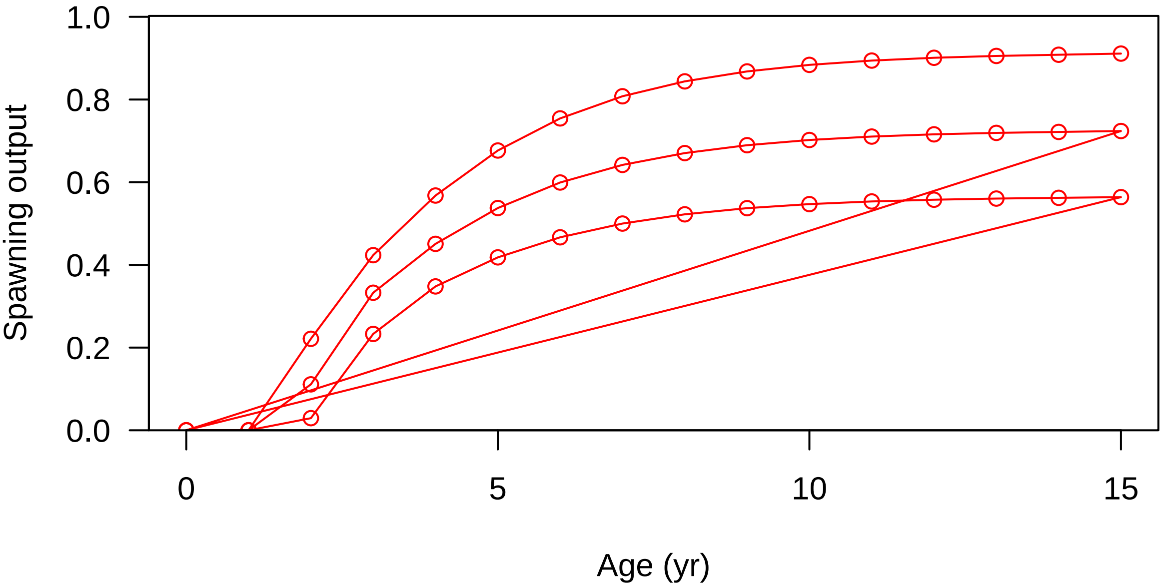
Fecundity
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
Female weight (kg)



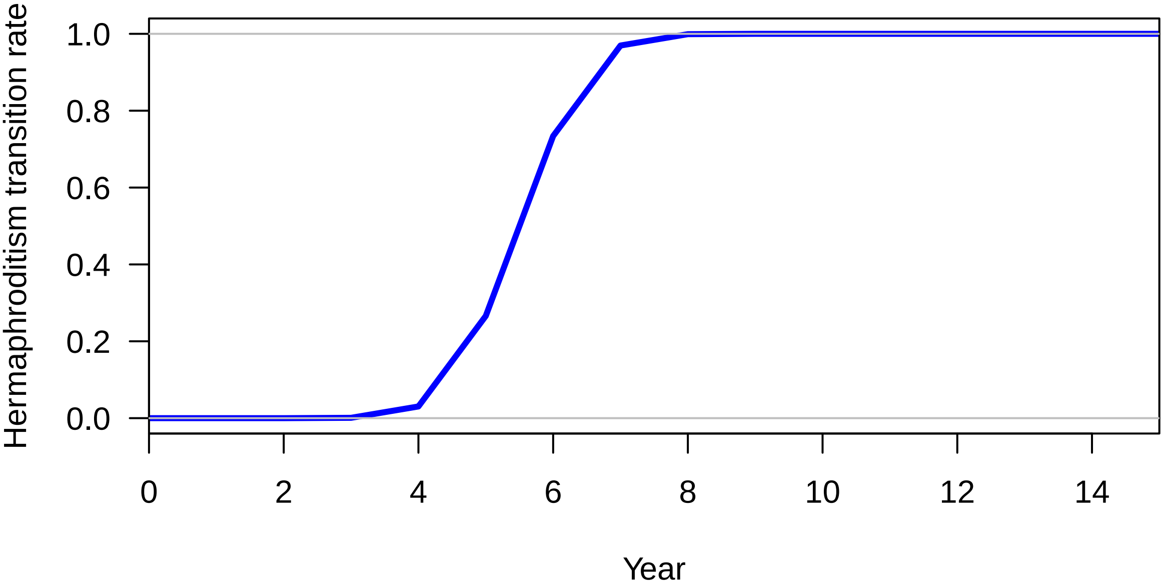
Fecundity
1.5
1.0
0.5
0.0
0
10
20
30
40
Female length (cm)

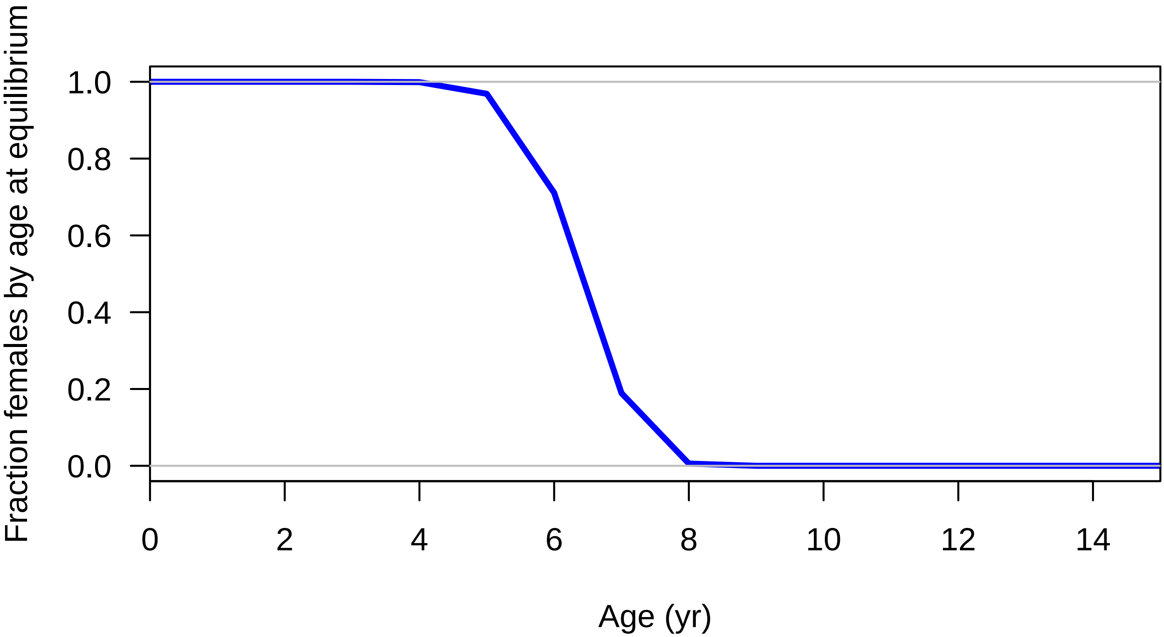


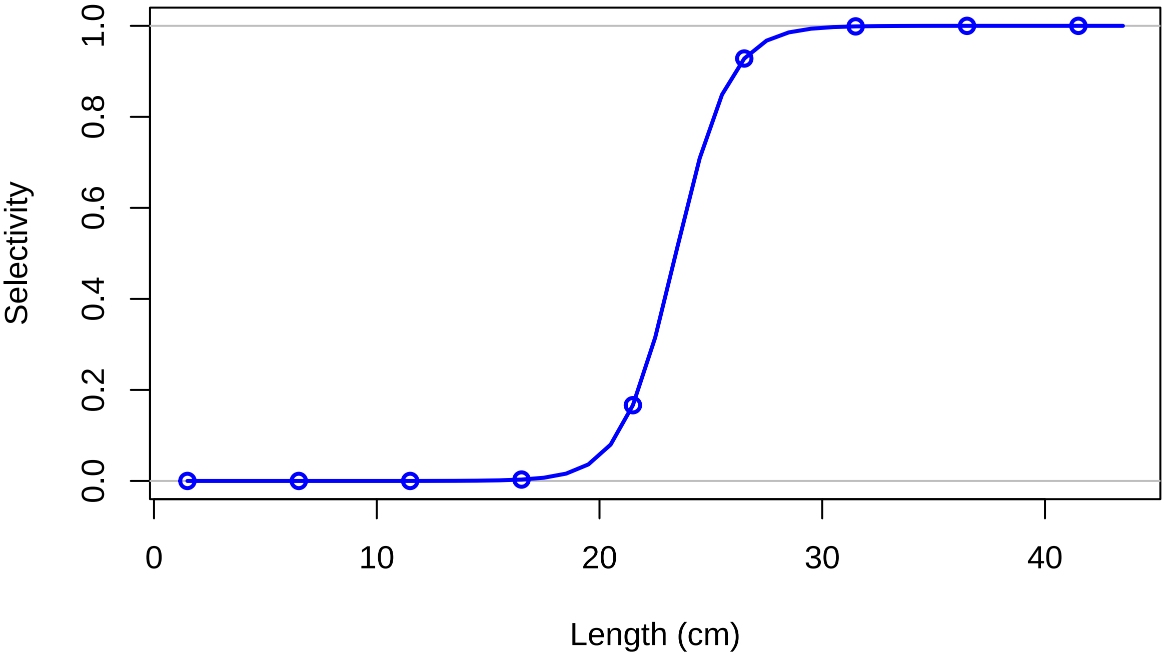
Spawning output
1.5
1.0
0.5
0.0
0
10
20
30
40
Length (cm)



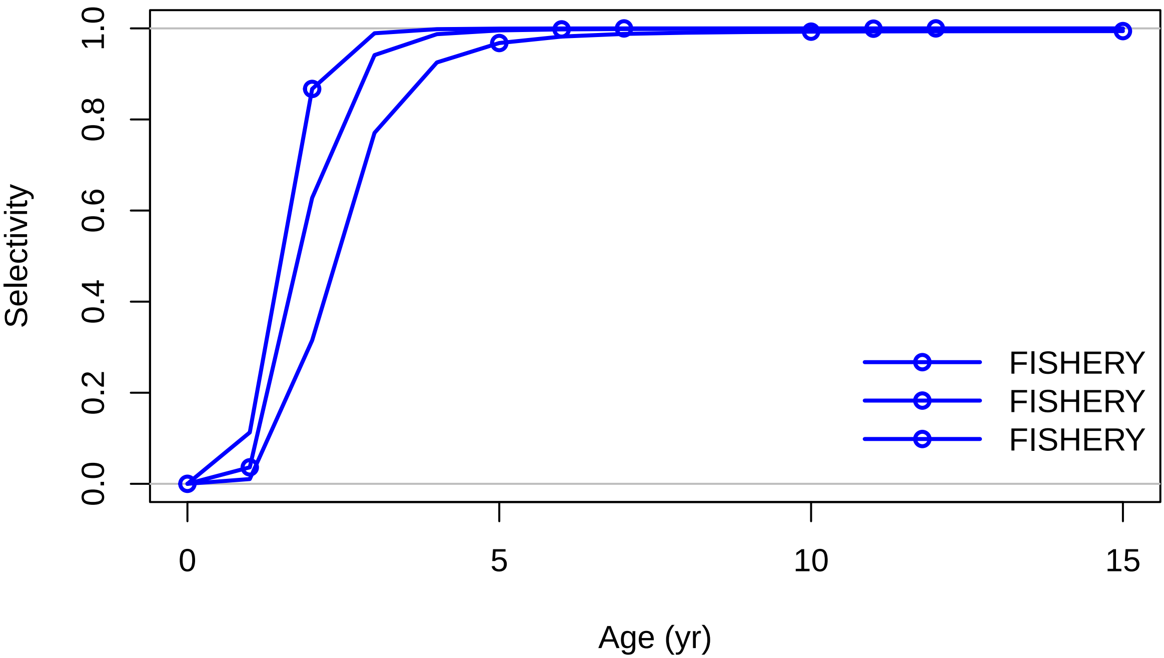
Spawning output
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
Age (yr)


Hermaphroditism transition rate
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
14
Year

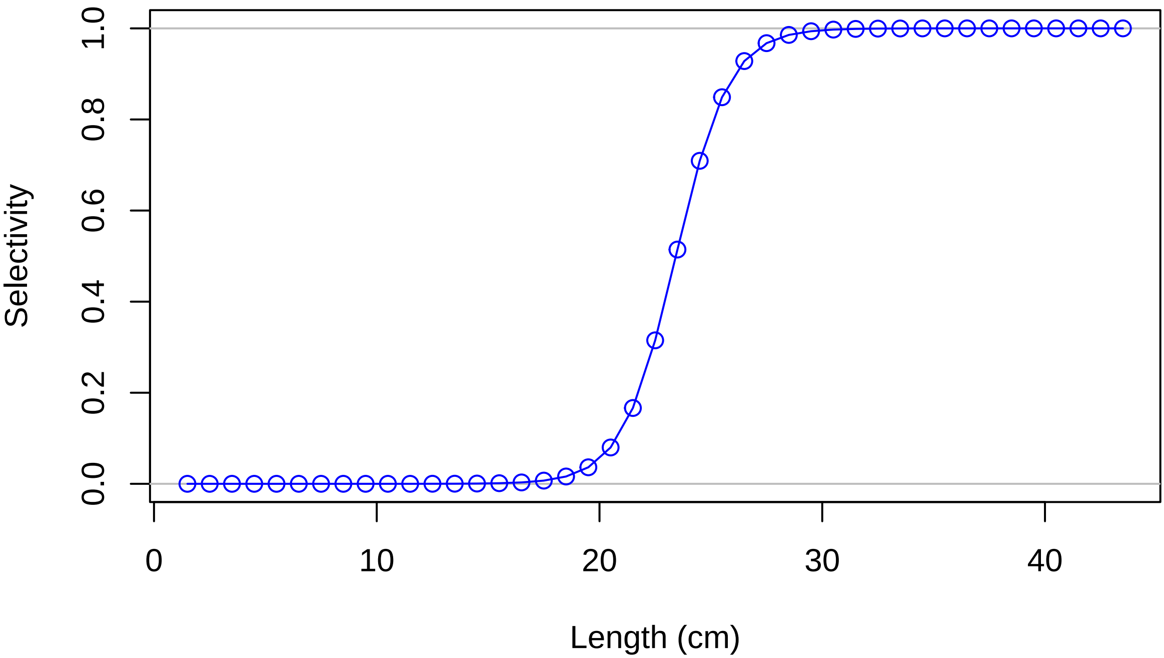

Fraction females by age at equilibrium
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
14
Age (yr)



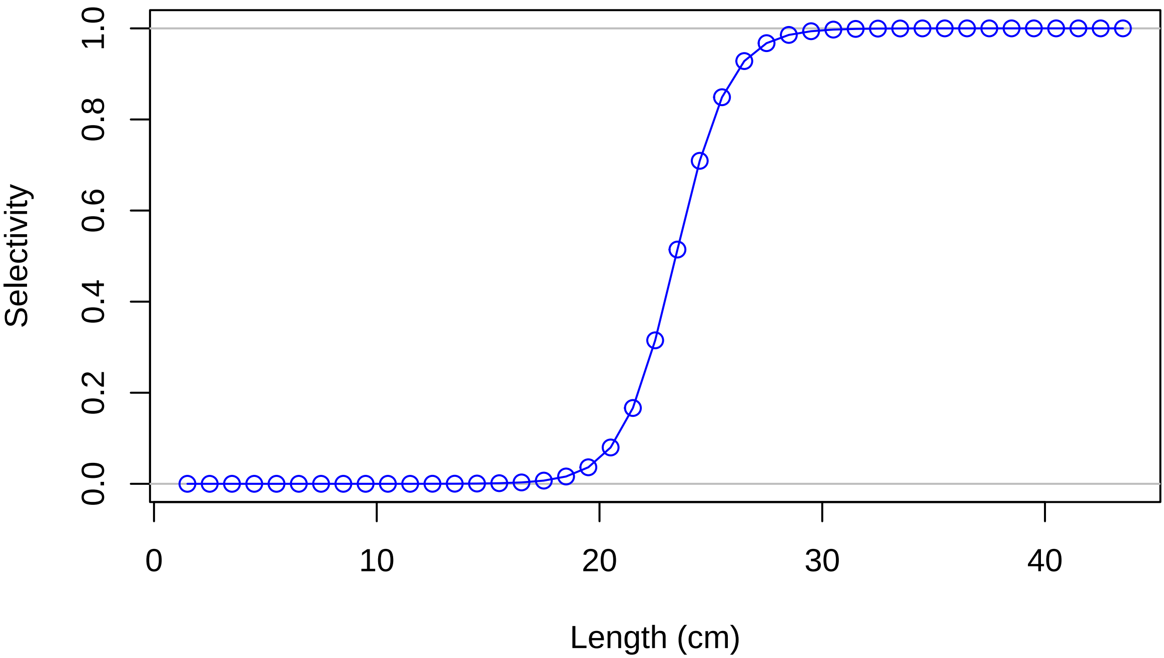
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
Length (cm)

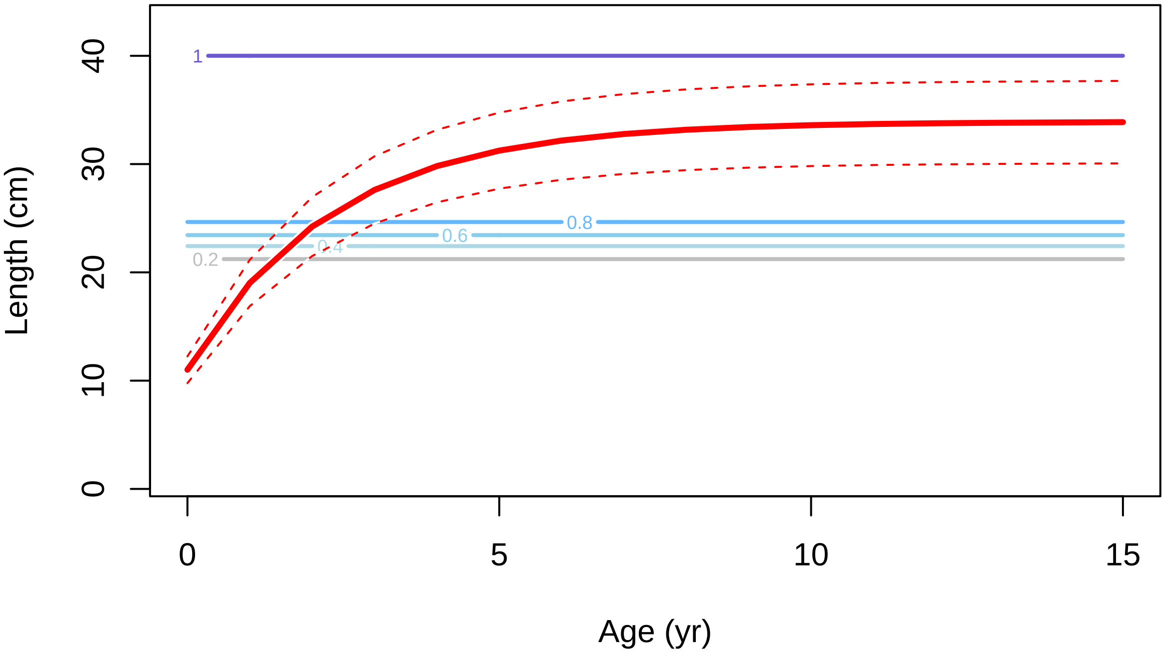


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
Age (yr)
FISHERY
FISHERY
FISHERY

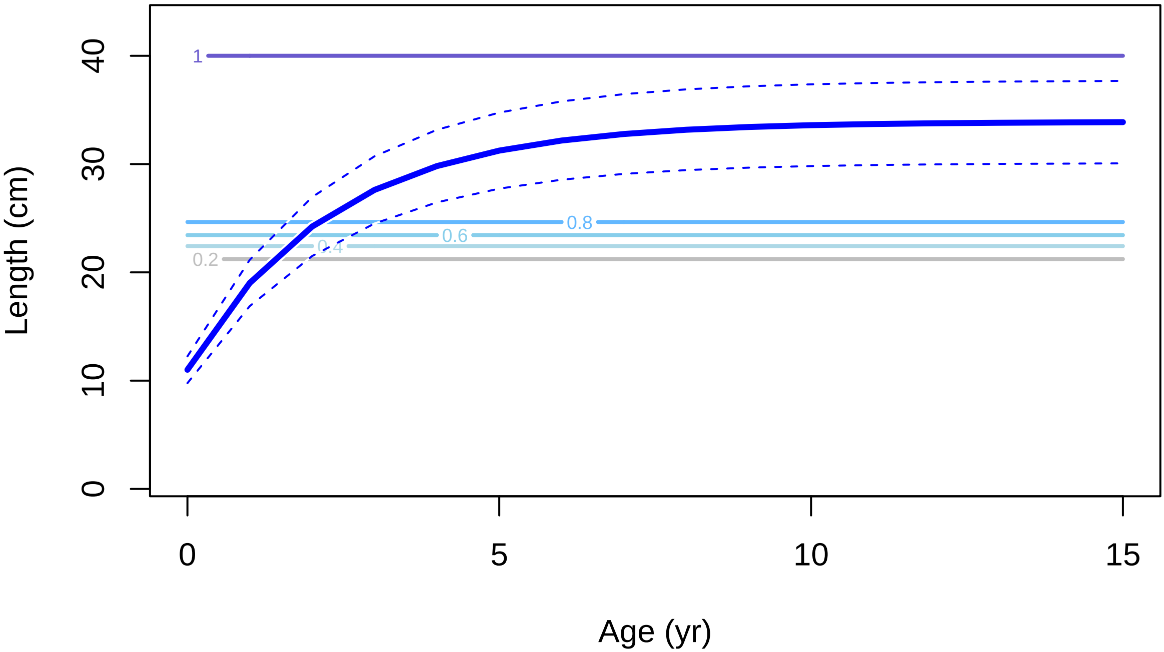


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
Length (cm)

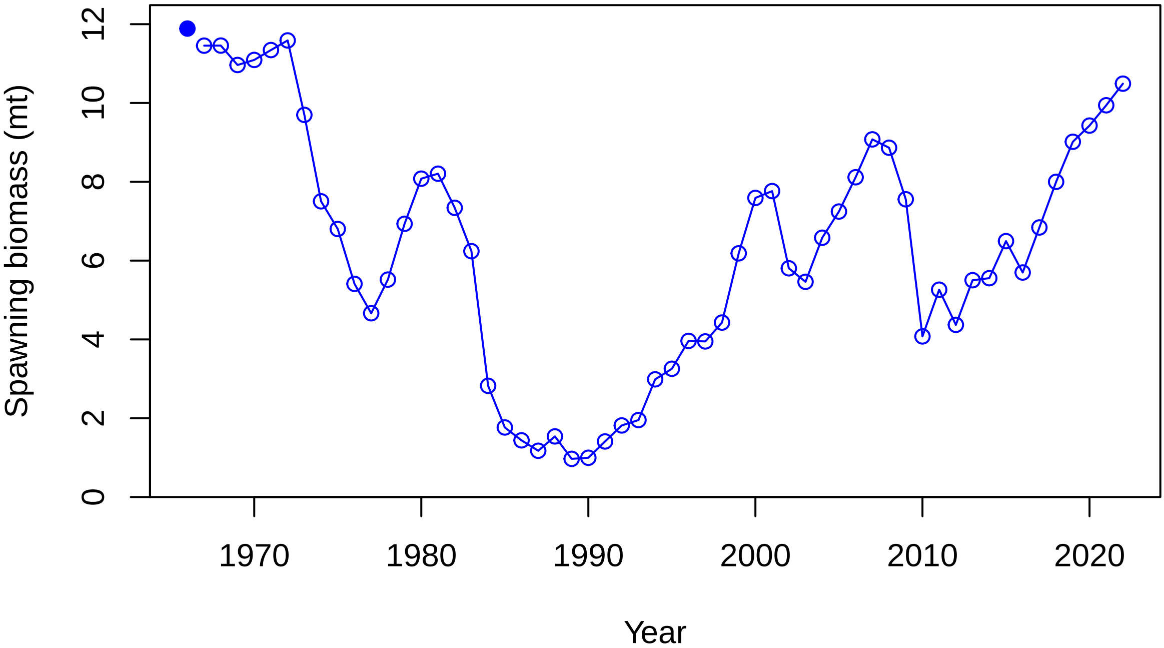

Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
Length (cm)



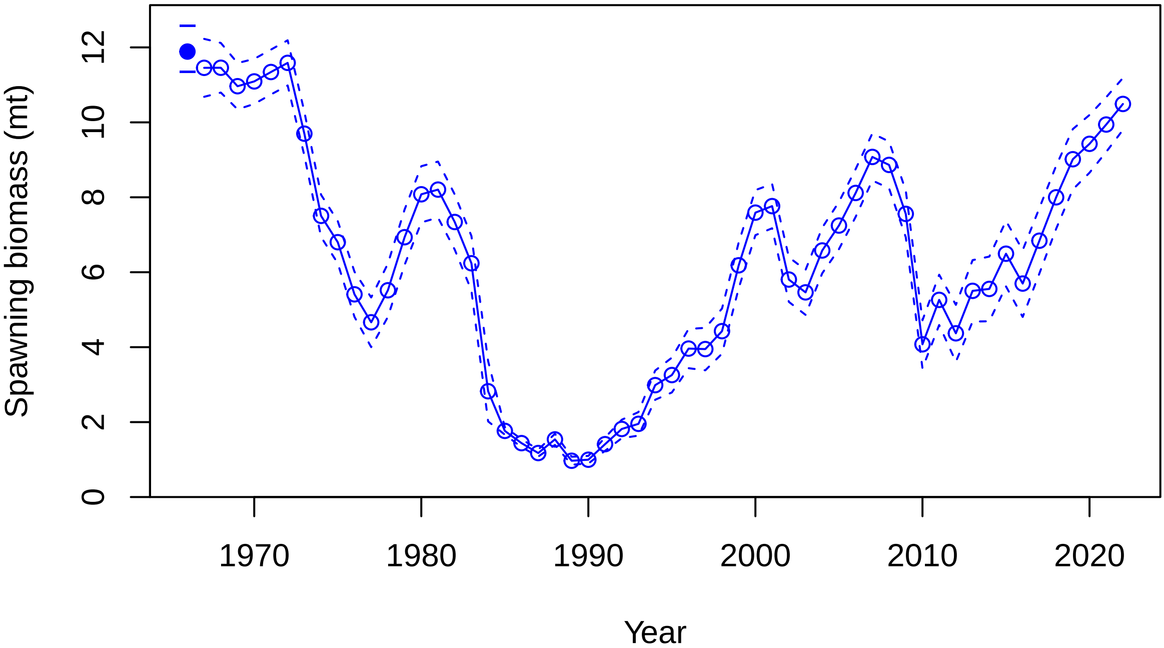
Length (cm)
0
10
20
30
40
1
0.8
0.6
0.4
0.2
0
5
10
15
Age (yr)



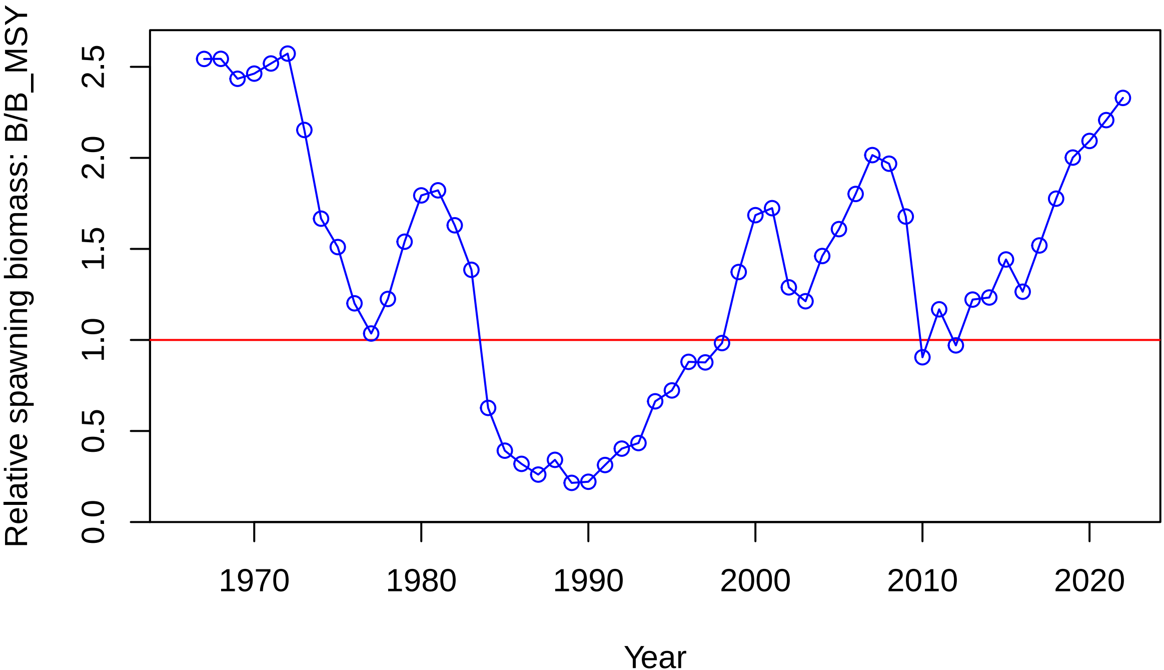
Length (cm)
0
10
20
30
40
1
0.8
0.6
0.4
0.2
0
5
10
15
Age (yr)



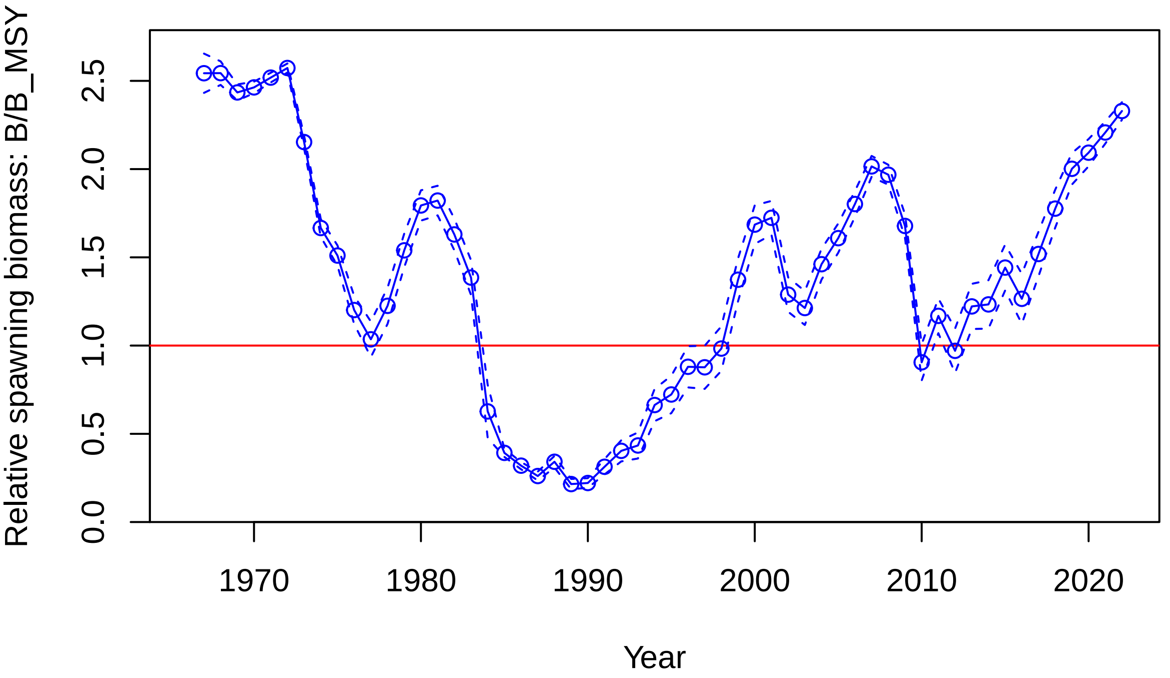
Spawning biomass (mt)
0
2
4
6
8
10
12
1970
1980
1990
2000
2010
2020
Year

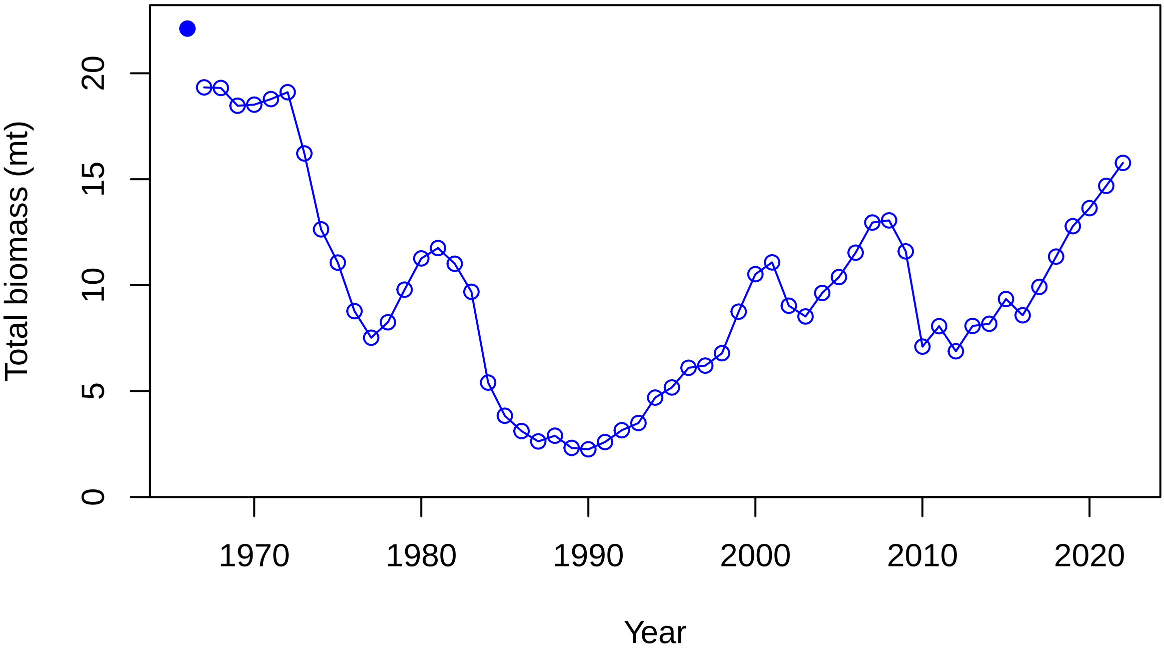


Spawning biomass (mt)
0
2
4
6
8
10
12
1970
1980
1990
2000
2010
2020
Year

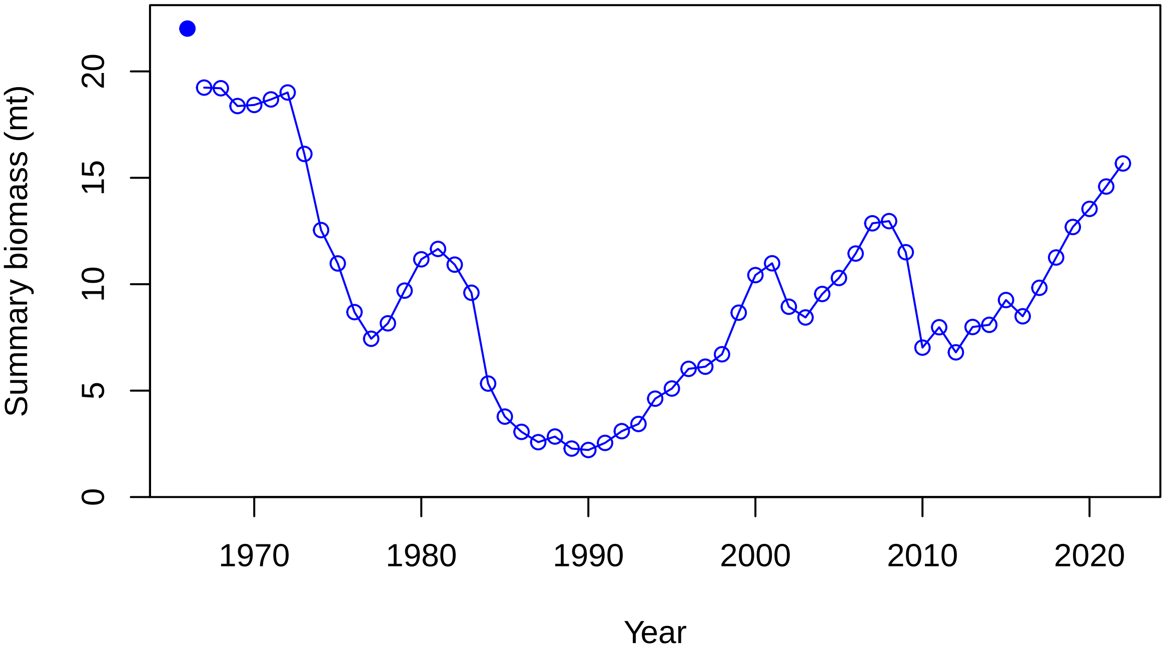


Relative spawning biomass: B/B_MSY
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year

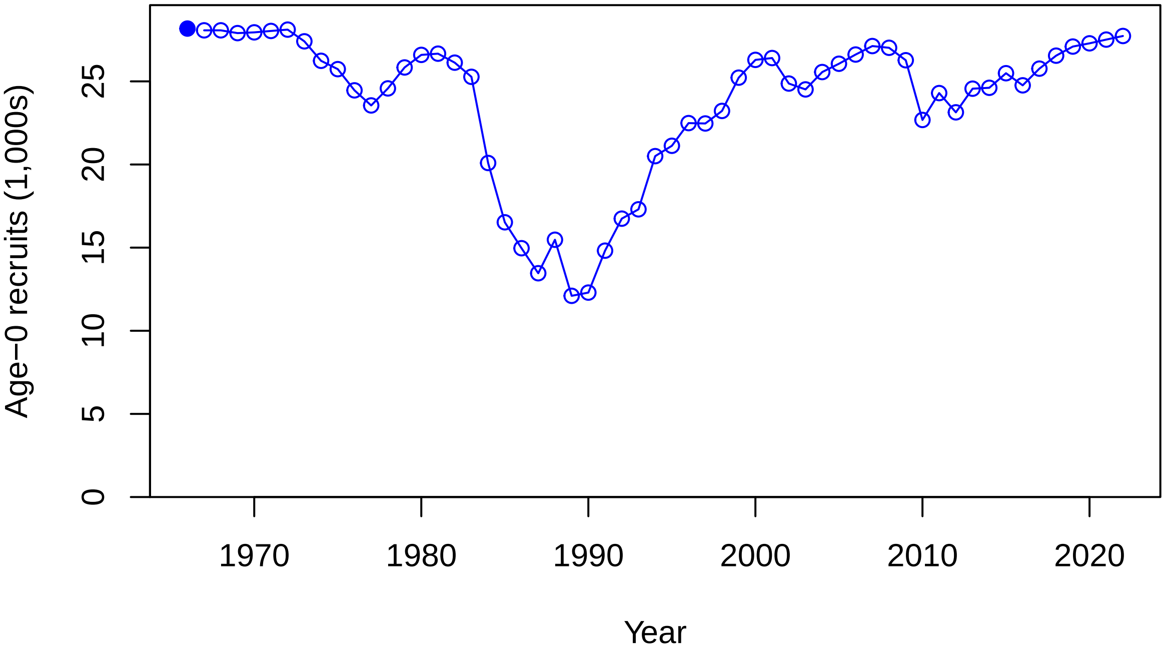

Relative spawning biomass: B/B_MSY
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year

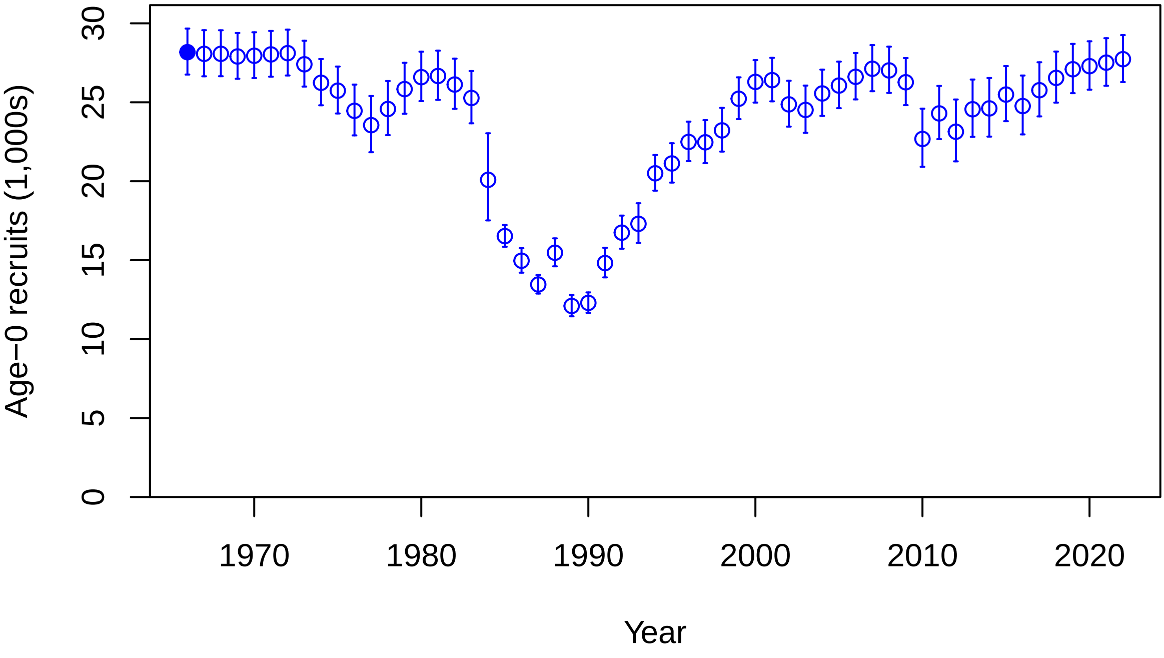


Total biomass (mt)
0
5
10
15
20
1970
1980
1990
2000
2010
2020
Year

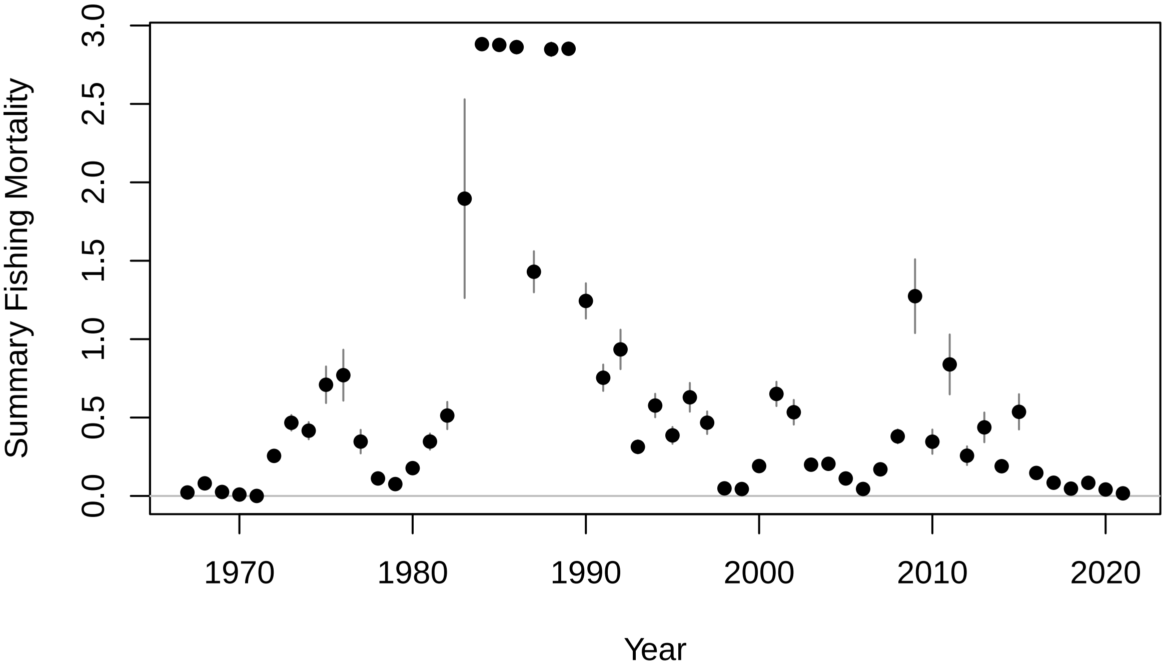


Summary biomass (mt)
0
5
10
15
20
1970
1980
1990
2000
2010
2020
Year

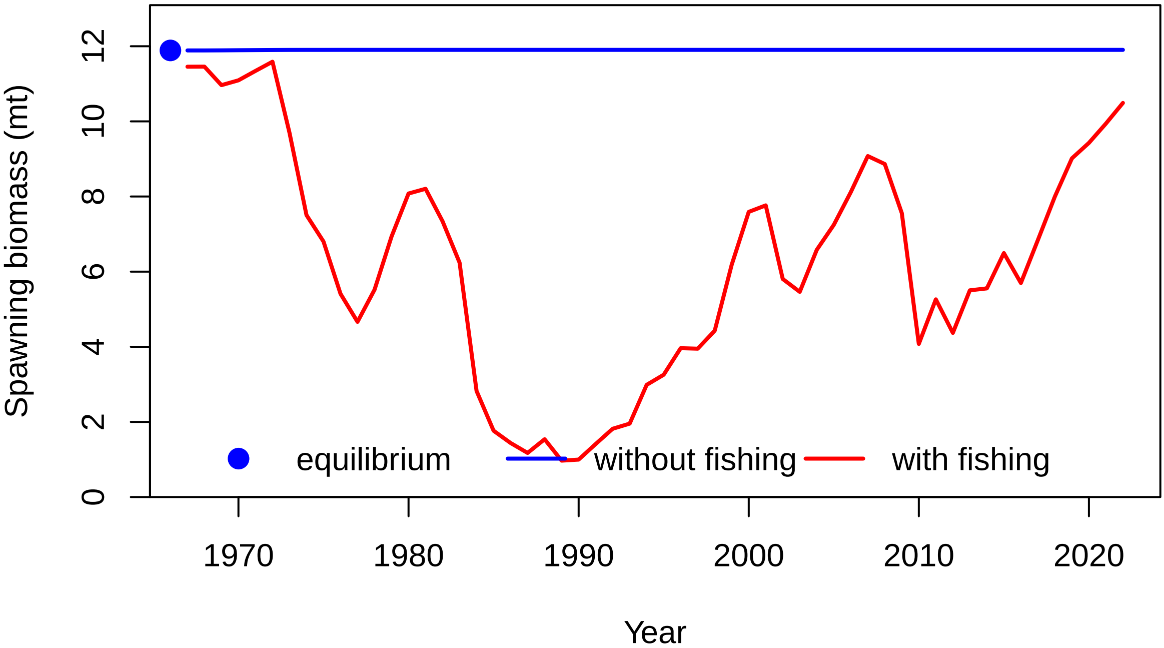

Age−0 recruits (1,000s)
0
5
10
15
20
25
1970
1980
1990
2000
2010
2020
Year



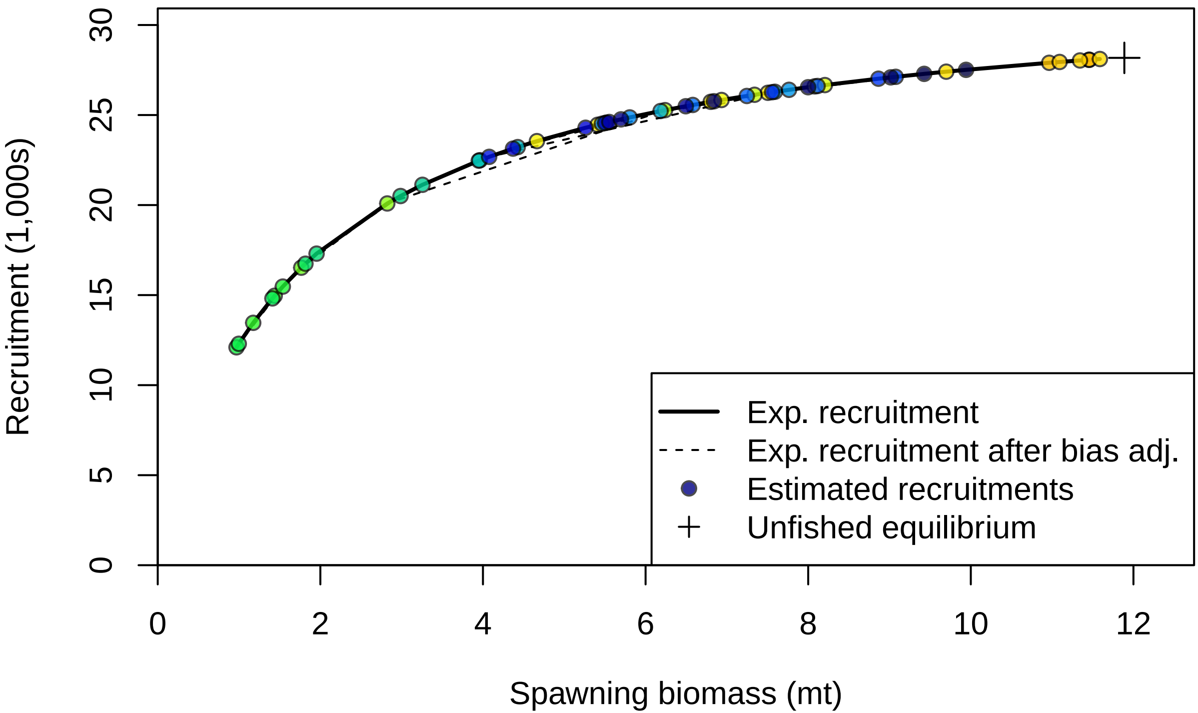
Age−0 recruits (1,000s)
0
5
10
15
20
25
30
1970
1980
1990
2000
2010
2020
Year

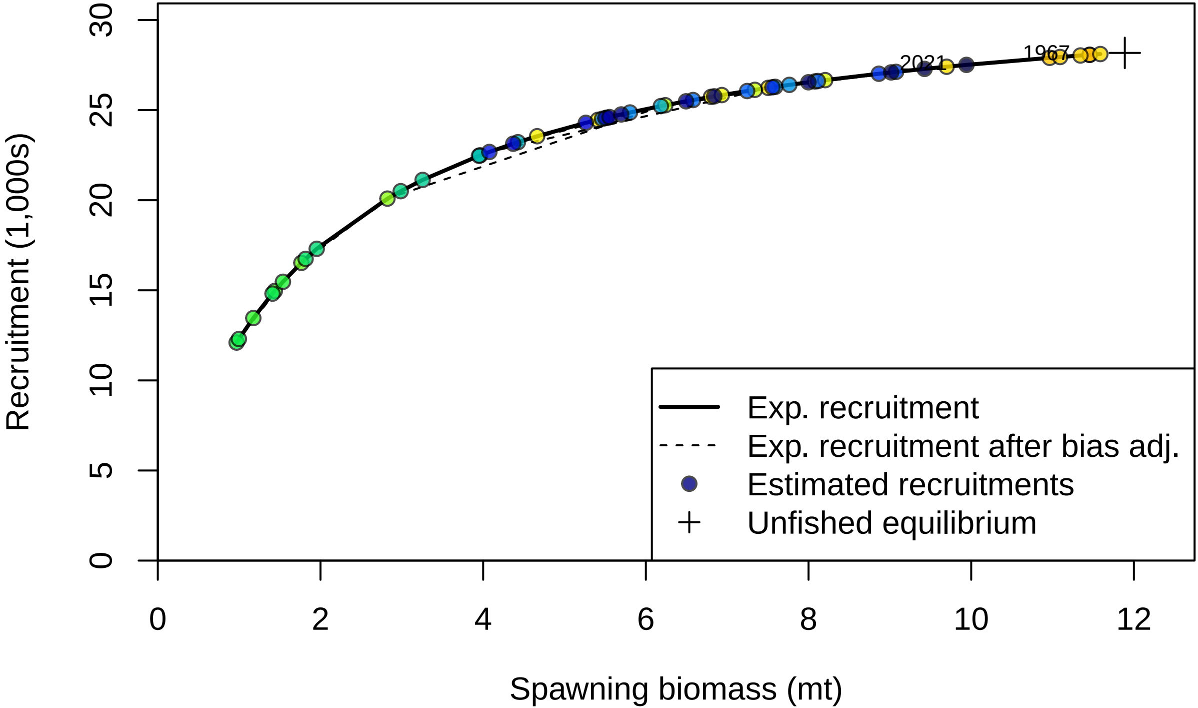


Summary Fishing Mortality
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year

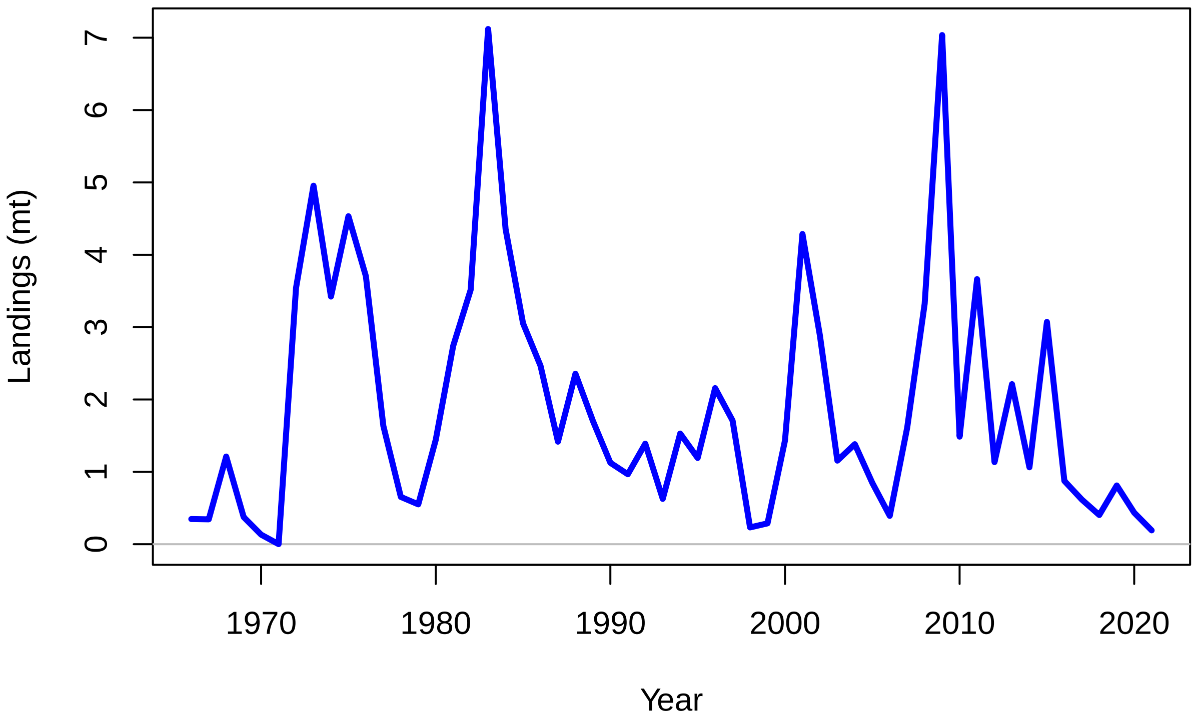


Spawning biomass (mt)
0
2
4
6
8
10
12
1970
1980
1990
2000
2010
2020
Year
equilibrium
without fishing
with fishing

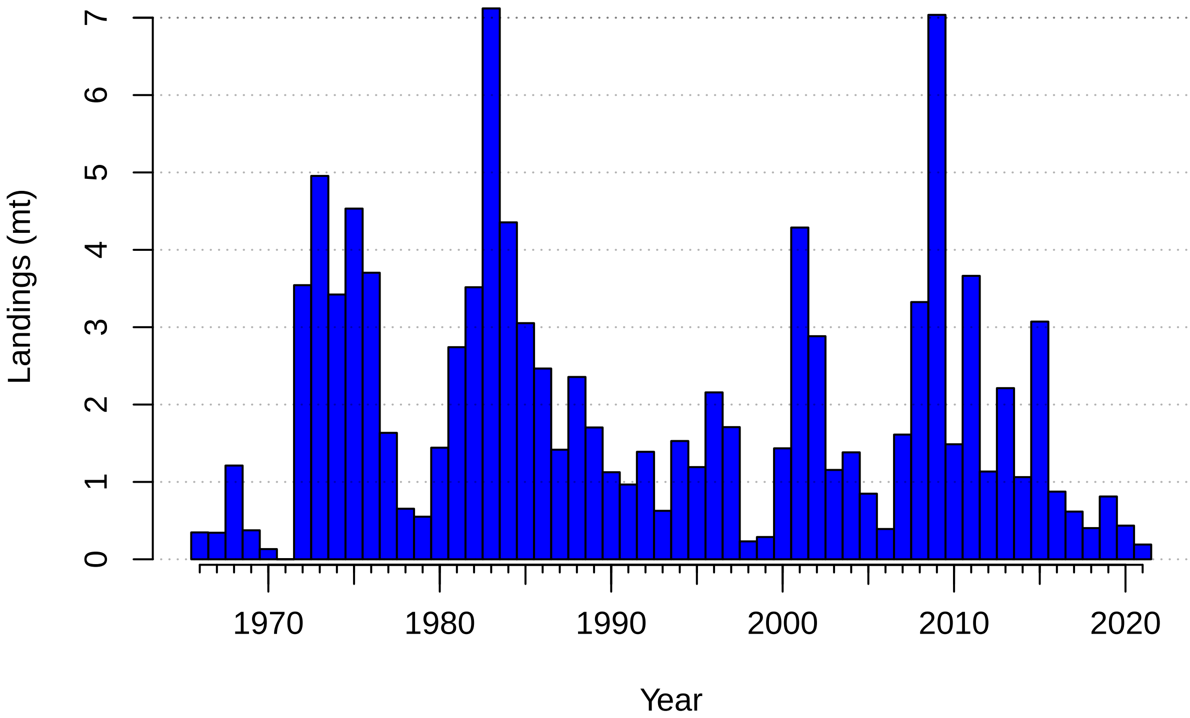

Recruitment (1,000s)
0
5
10
15
20
25
30
Spawning biomass (mt)
0
2
4
6
8
10
12
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



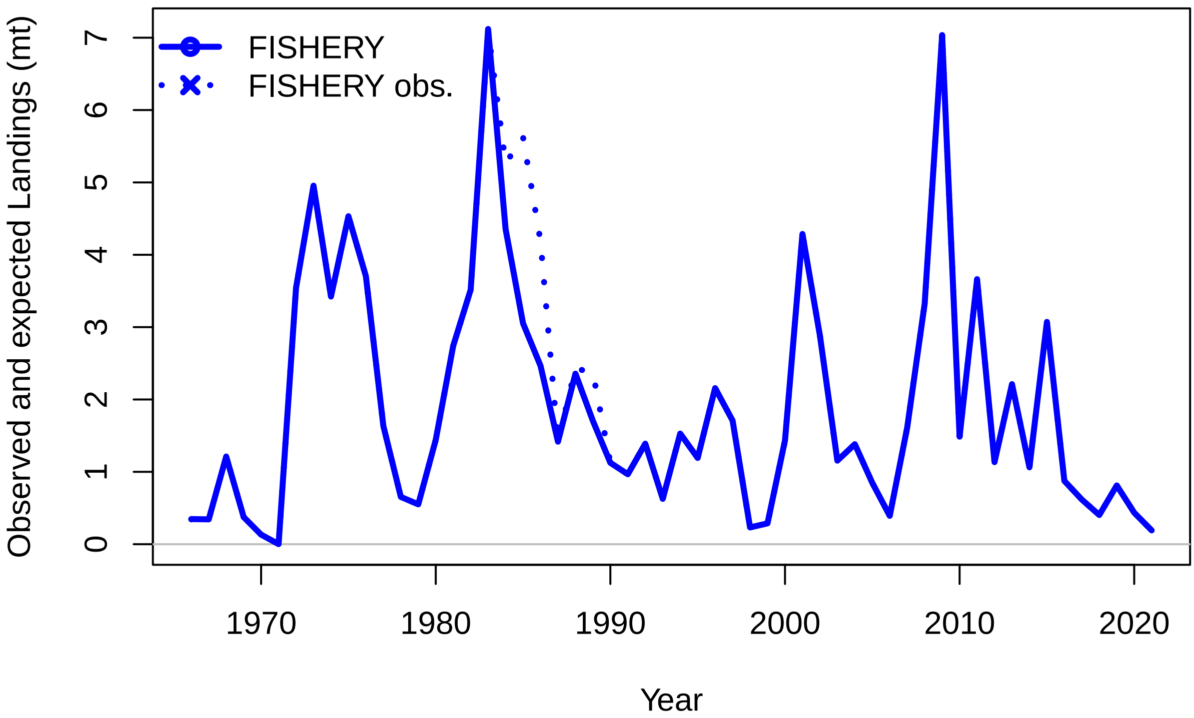
Recruitment (1,000s)
0
5
10
15
20
25
30
Spawning biomass (mt)
0
2
4
6
8
10
12
2021
1967
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



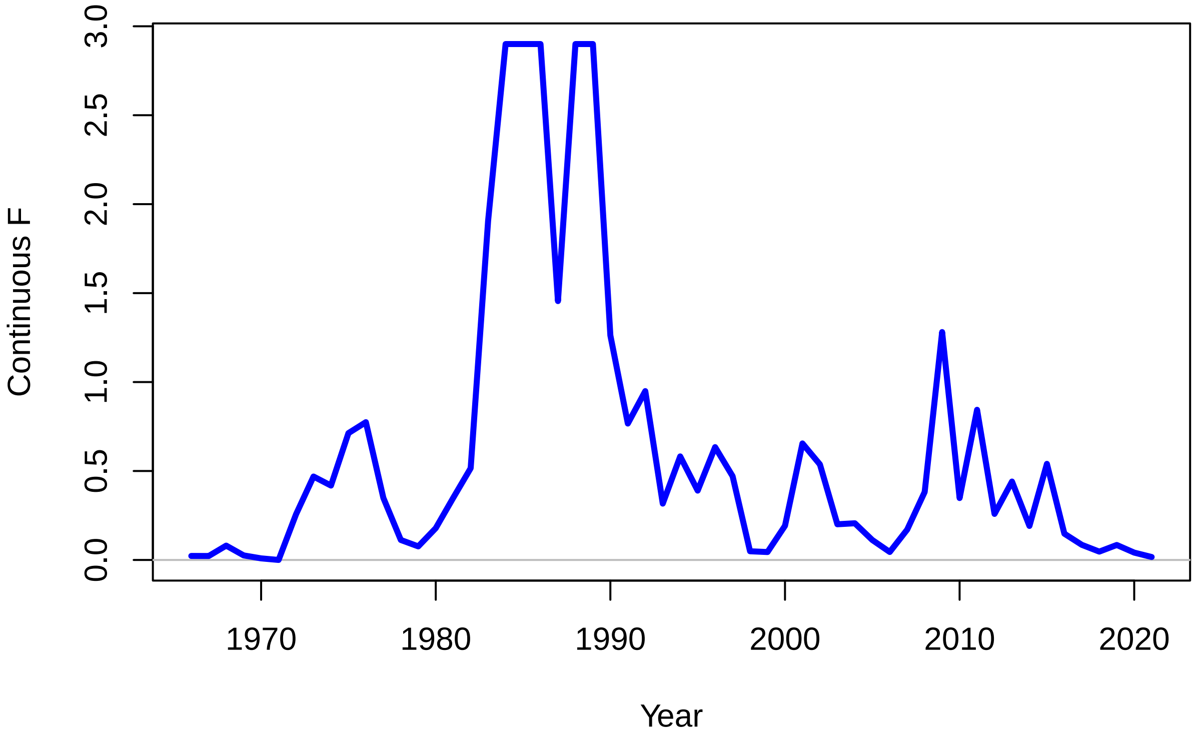
Landings (mt)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year



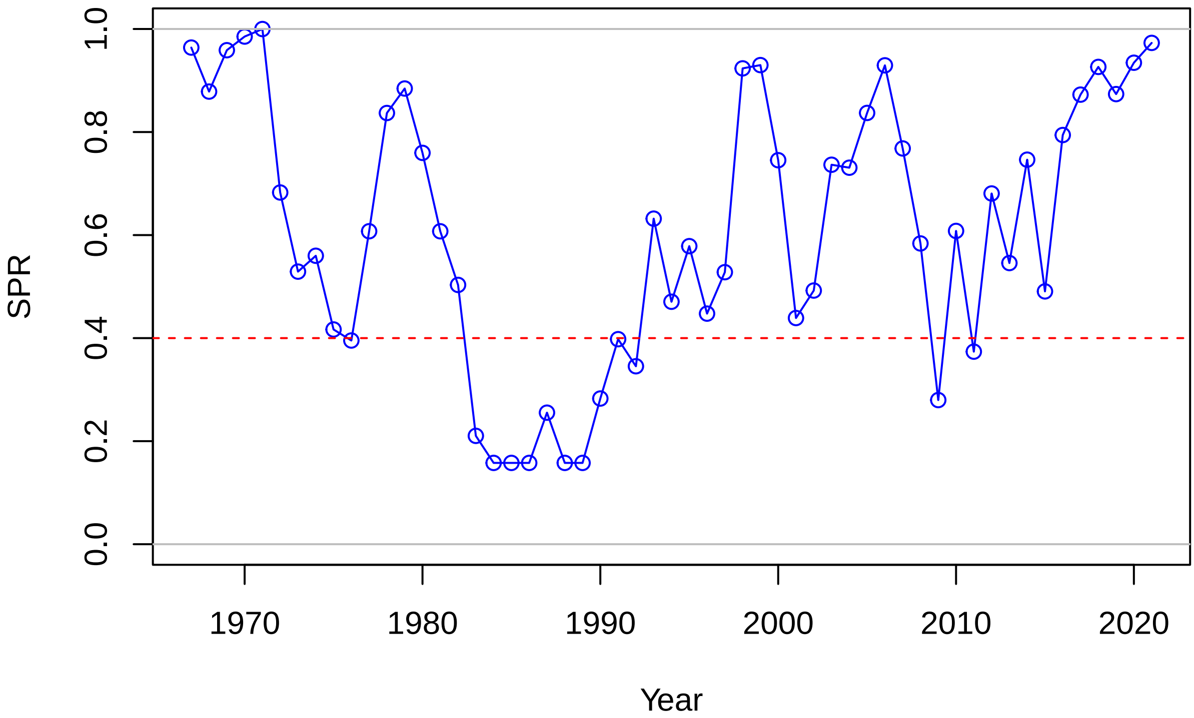
Landings (mt)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year

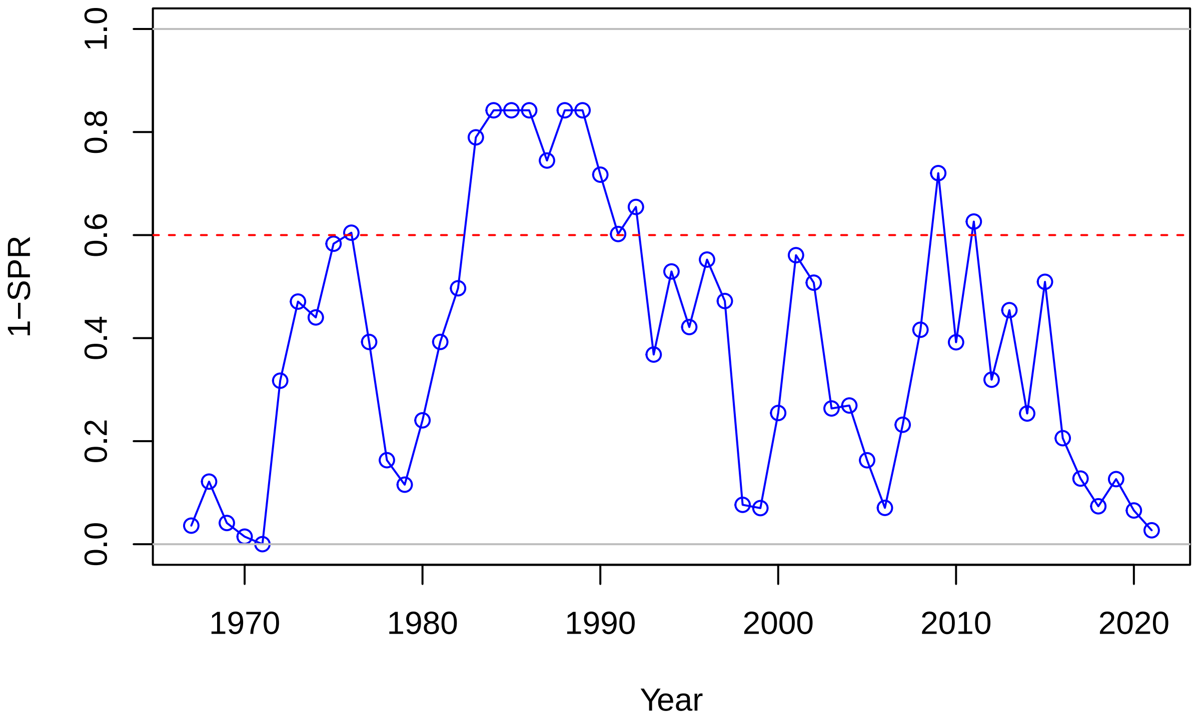


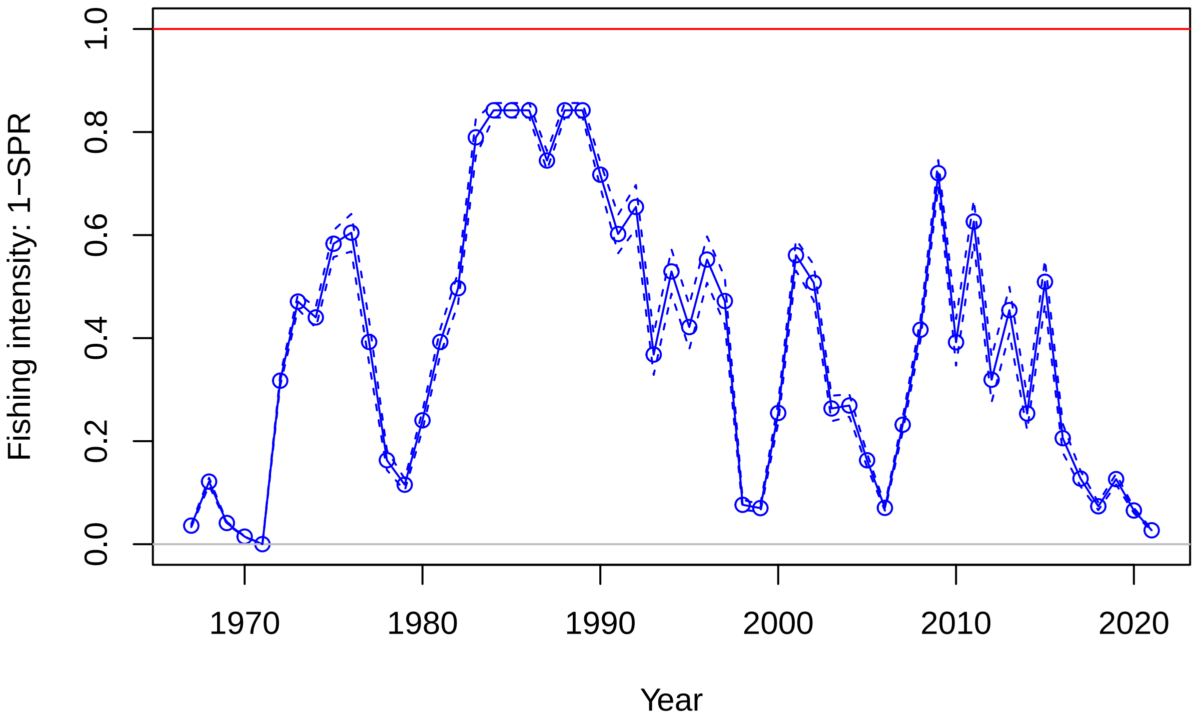
FISHERY
FISHERY obs.
Observed and expected Landings (mt)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year

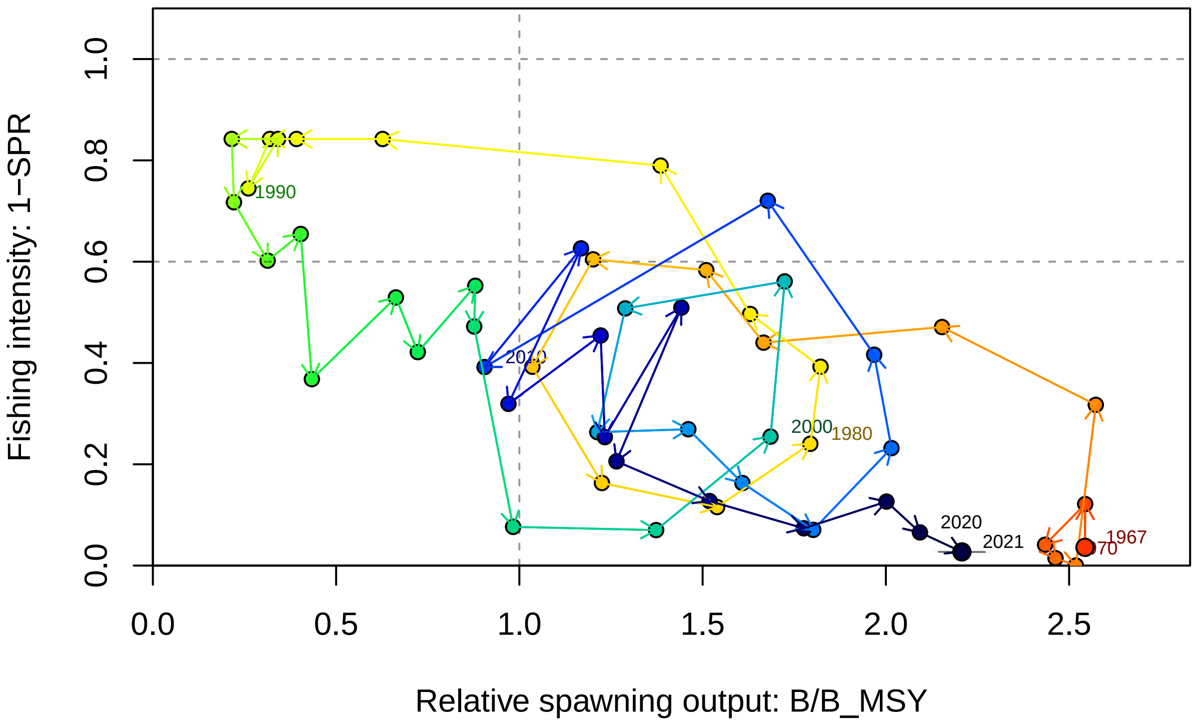


Continuous F
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year

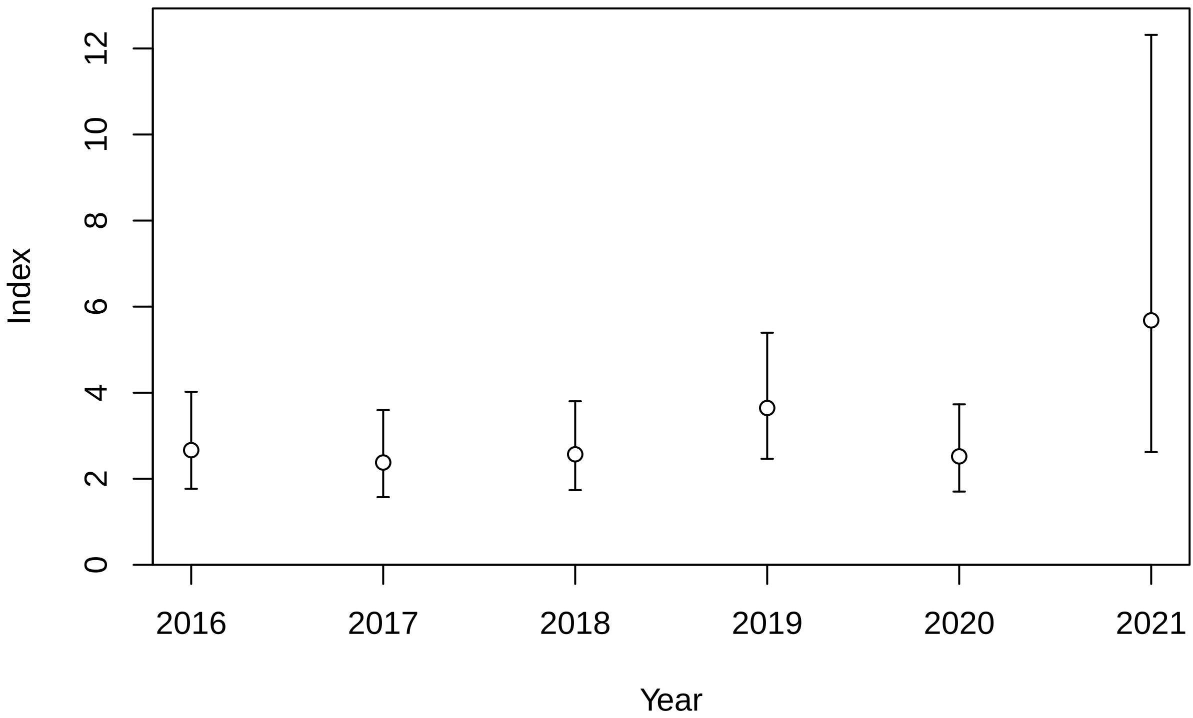

SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year


1.0
0.8
0.6
0.4
0.2
0.0
1−SPR
1970
1980
1990
2000
2010
2020
Year

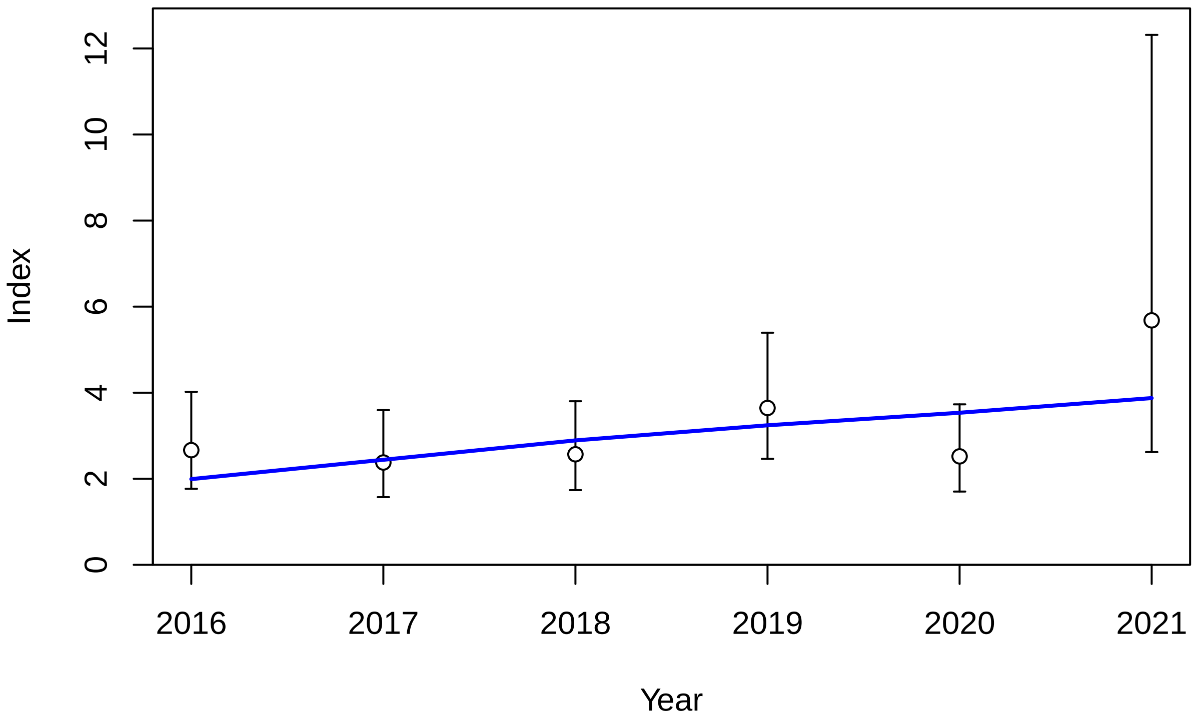

Fishing intensity: 1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

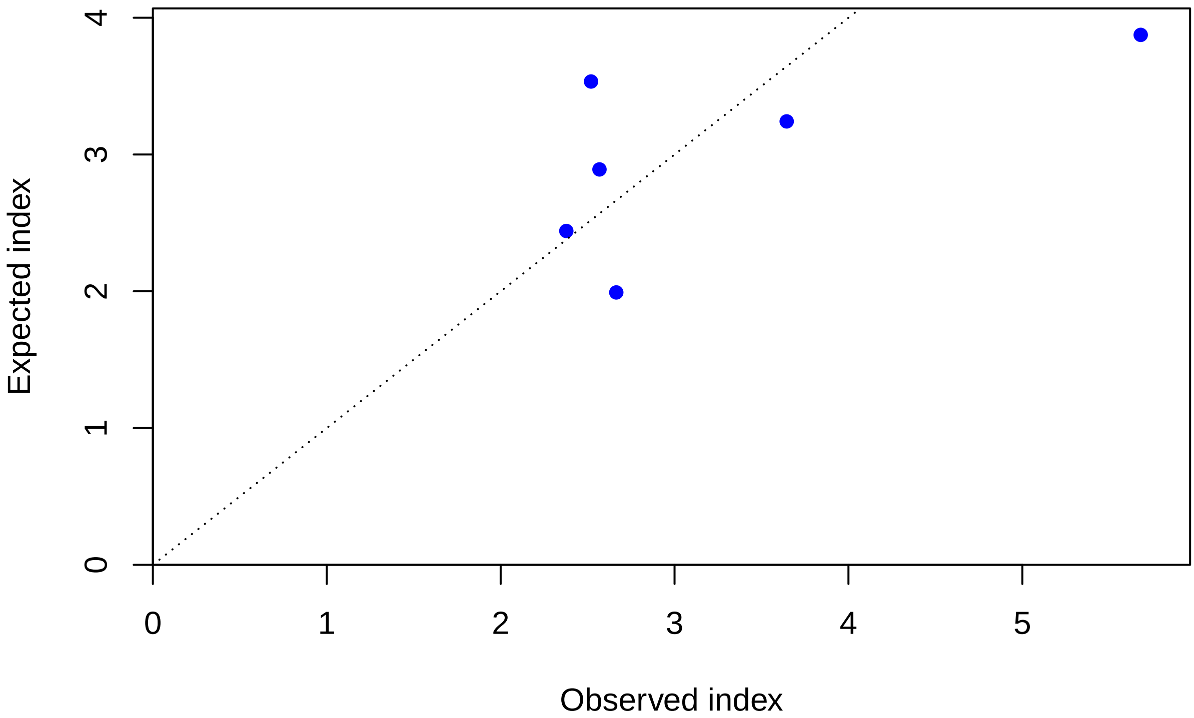


Fishing intensity: 1−SPR
Relative spawning output: B/B_MSY
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
1990
2010
2000
1980
2020
2021
1967

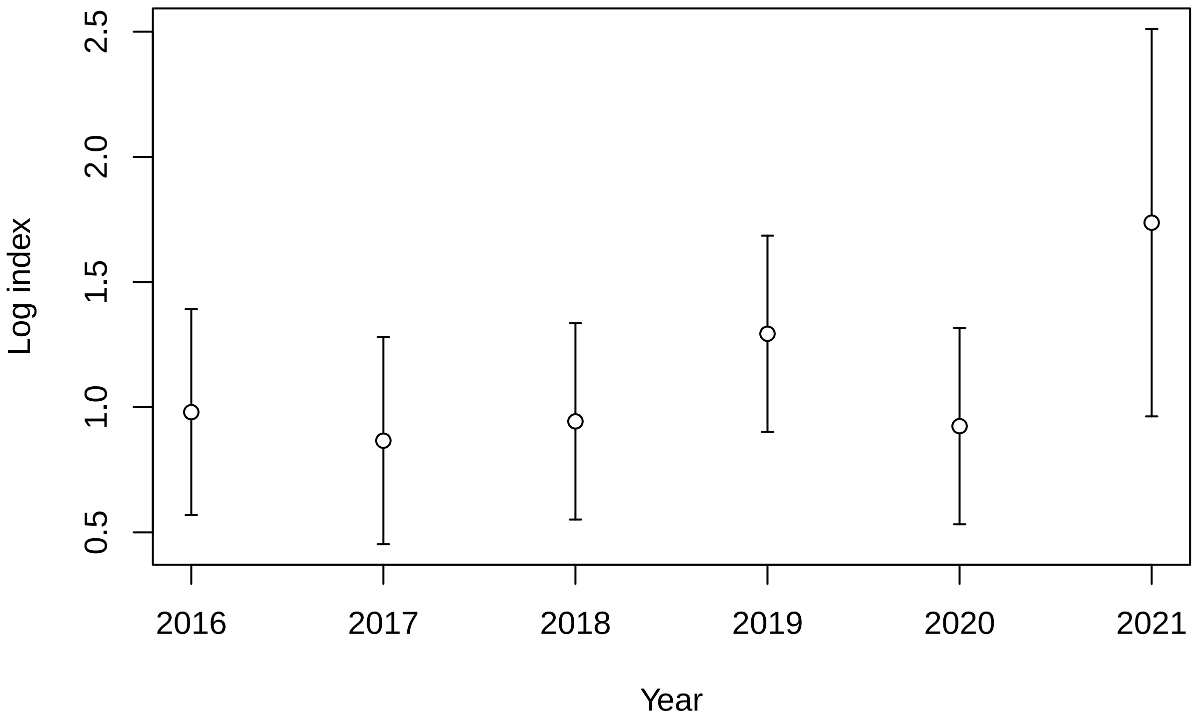


Index
0
2
4
6
8
10
12
2016
2017
2018
2019
2020
2021
Year

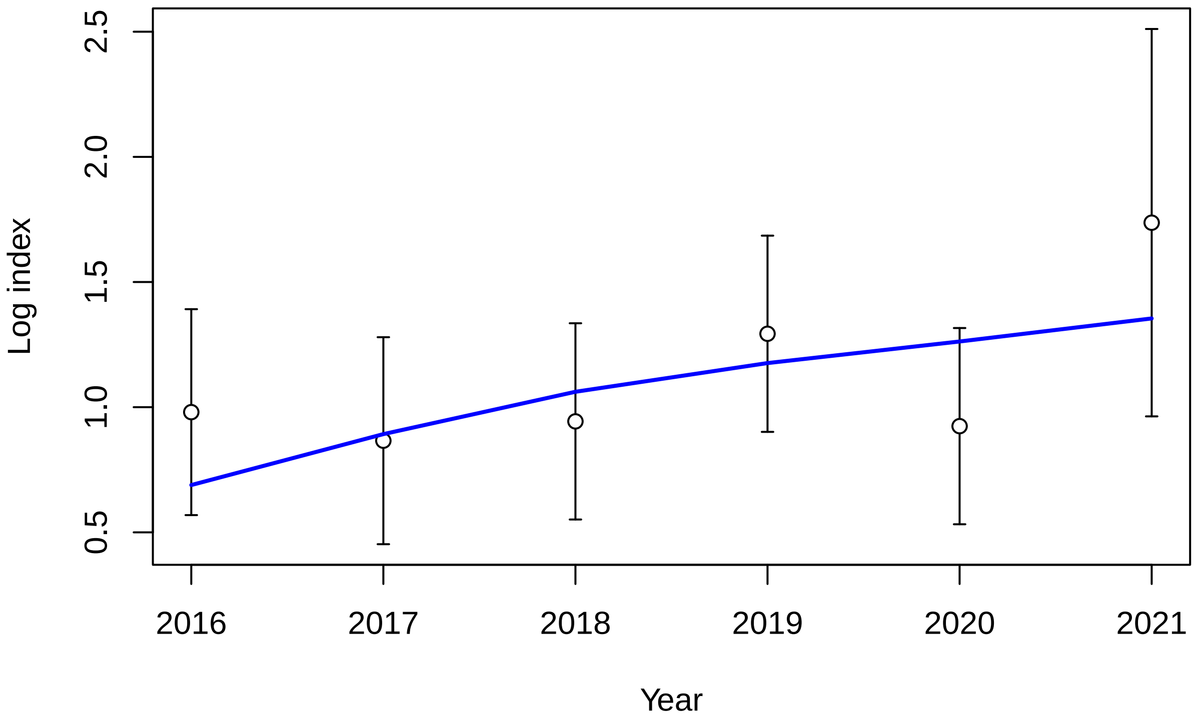

Index
0
2
4
6
8
10
12
2016
2017
2018
2019
2020
2021
Year



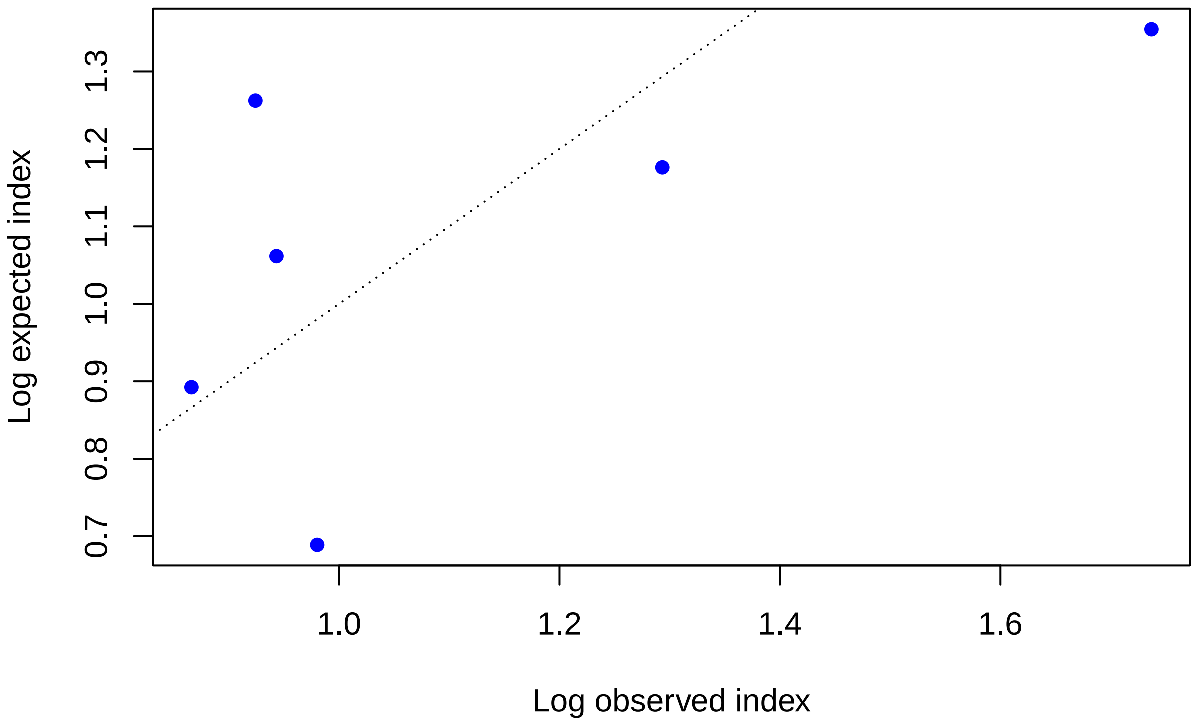
Expected index
Observed index
0
1
2
3
4
5



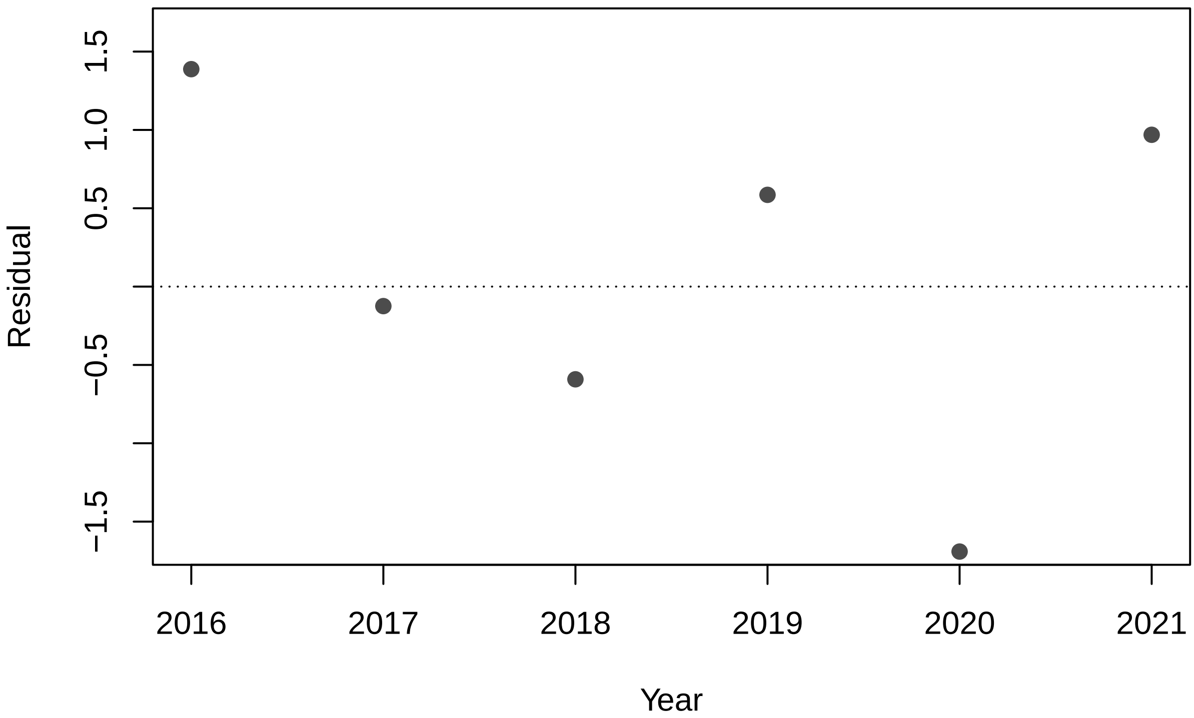
Log index
0.5
1.0
1.5
2.0
2.5
2016
2017
2018
2019
2020
2021
Year

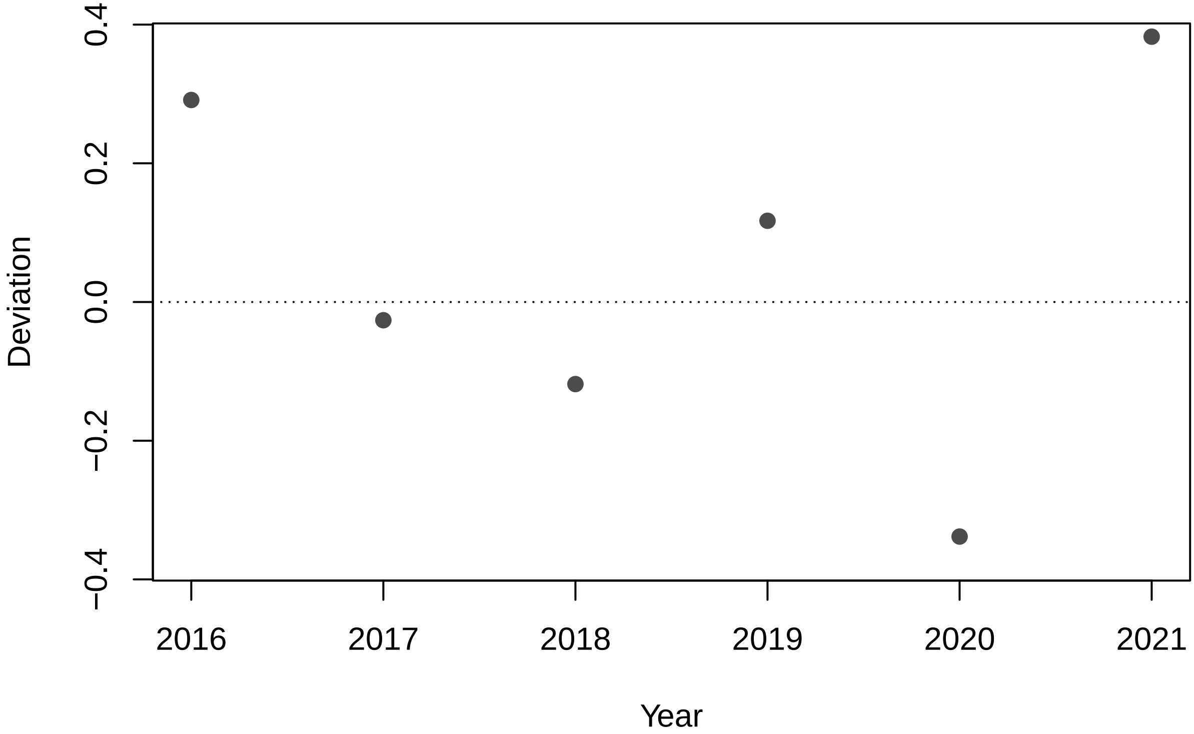


Log index
0.5
1.0
1.5
2.0
2.5
2016
2017
2018
2019
2020
2021
Year

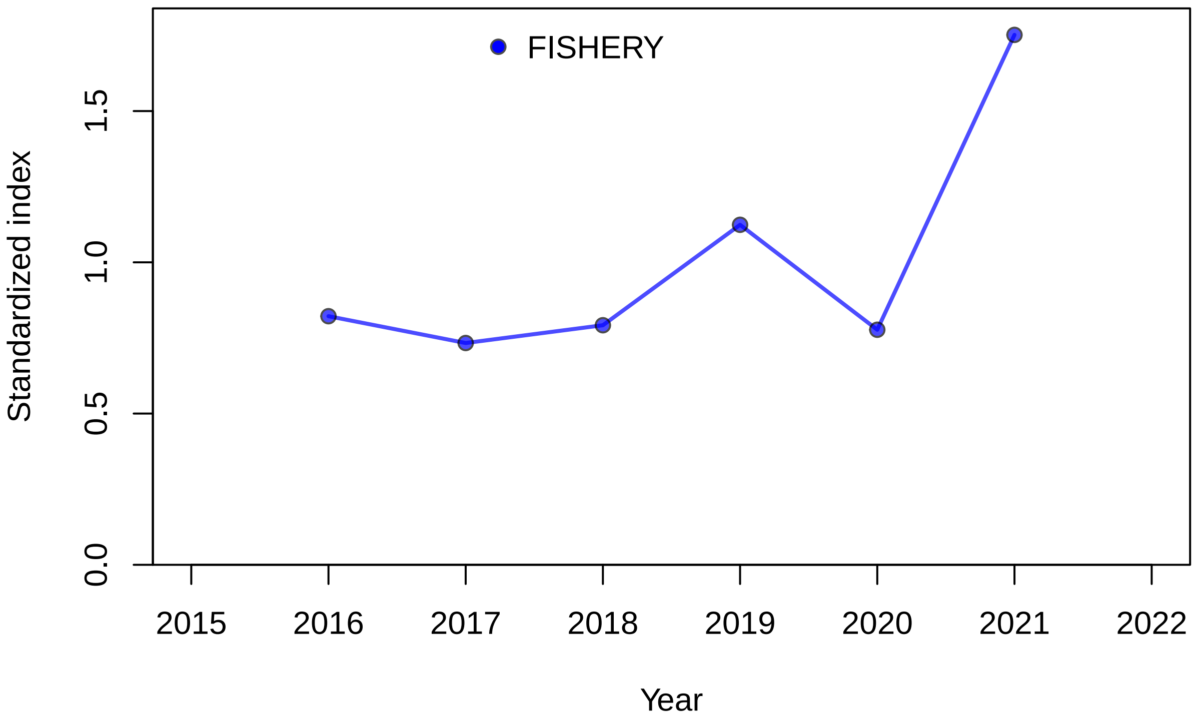


Log expected index
0.7
0.8
0.9
1.0
1.1
1.2
1.3
1.0
1.2
1.4
1.6
Log observed index

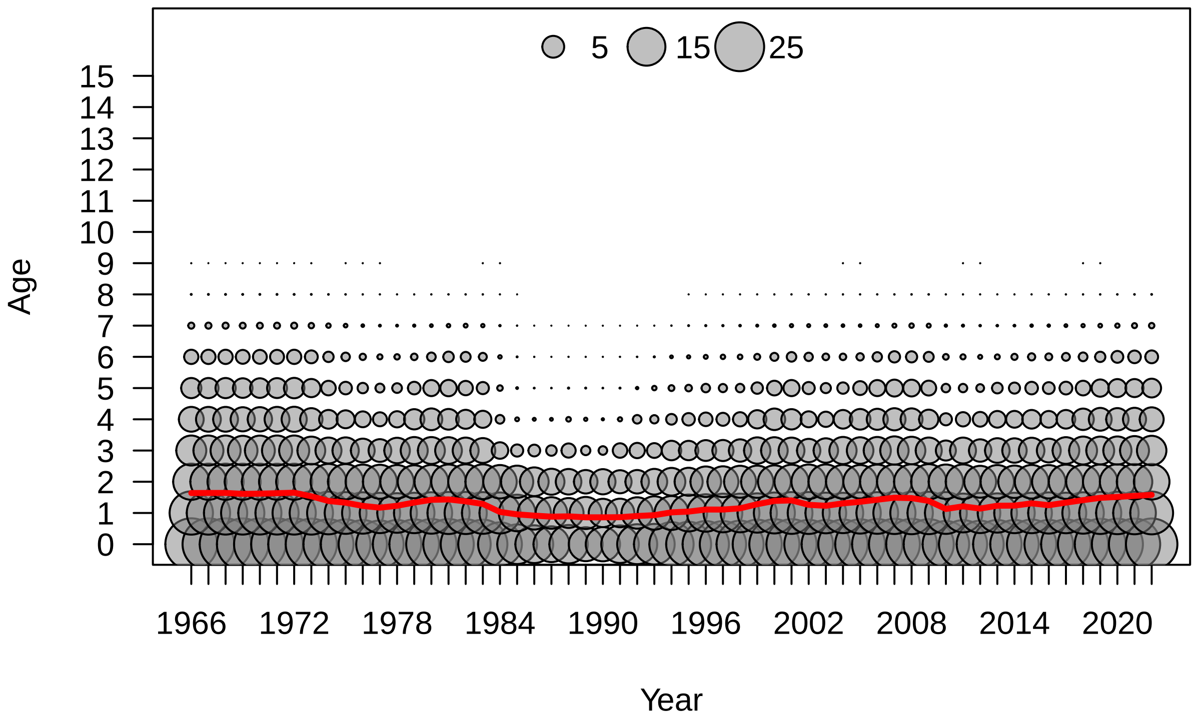

Residual
1.5
1.0
0.5
−0.5
−1.5
2016
2017
2018
2019
2020
2021
Year



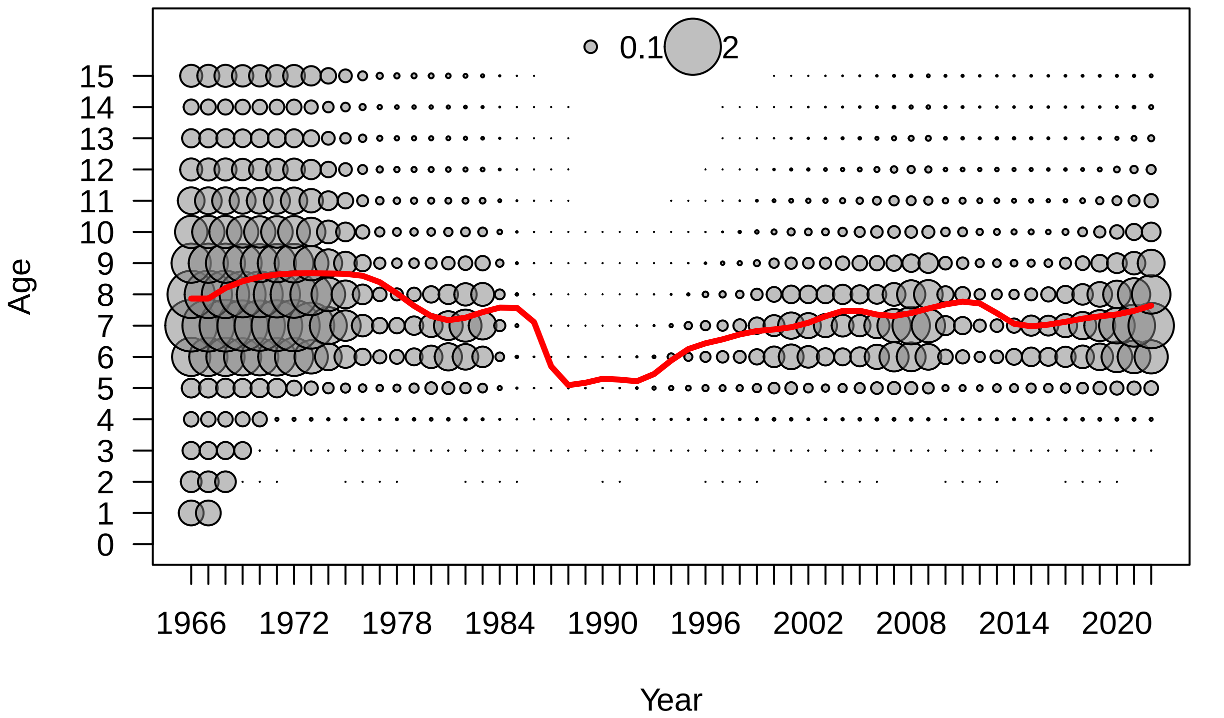
Deviation
0.4
0.2
0.0
−0.2
−0.4
2016
2017
2018
2019
2020
2021
Year



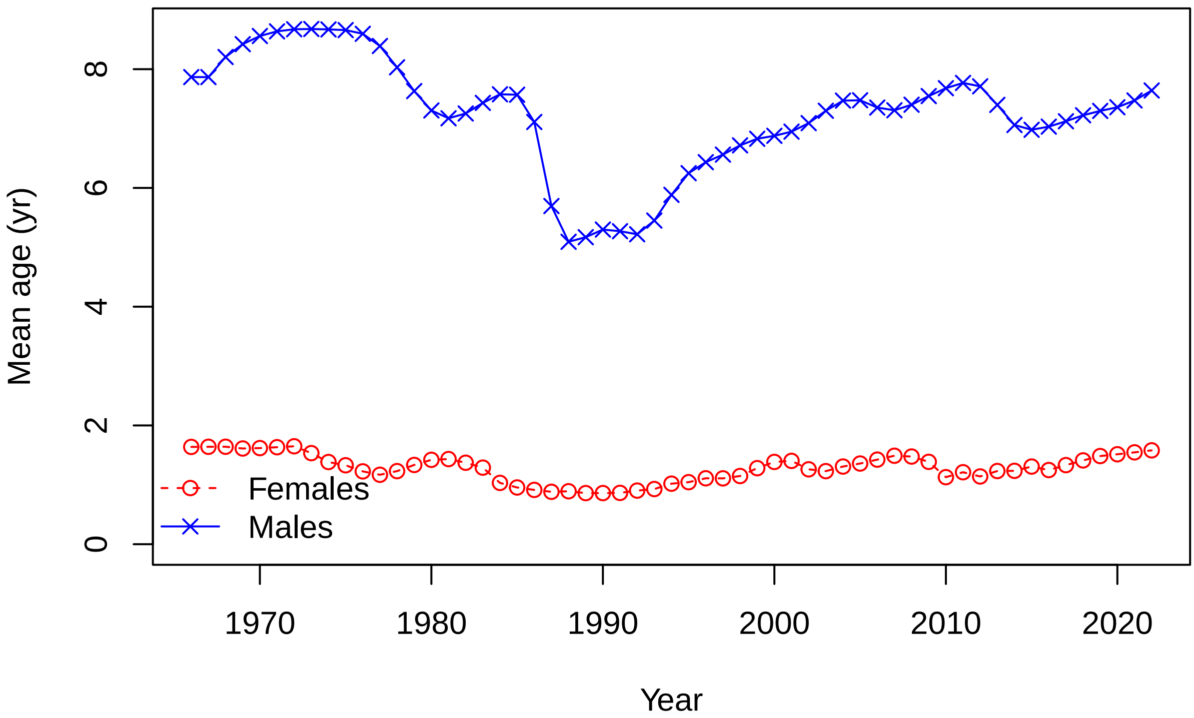
FISHERY
Standardized index
0.0
0.5
1.0
1.5
2015
2016
2017
2018
2019
2020
2021
2022
Year



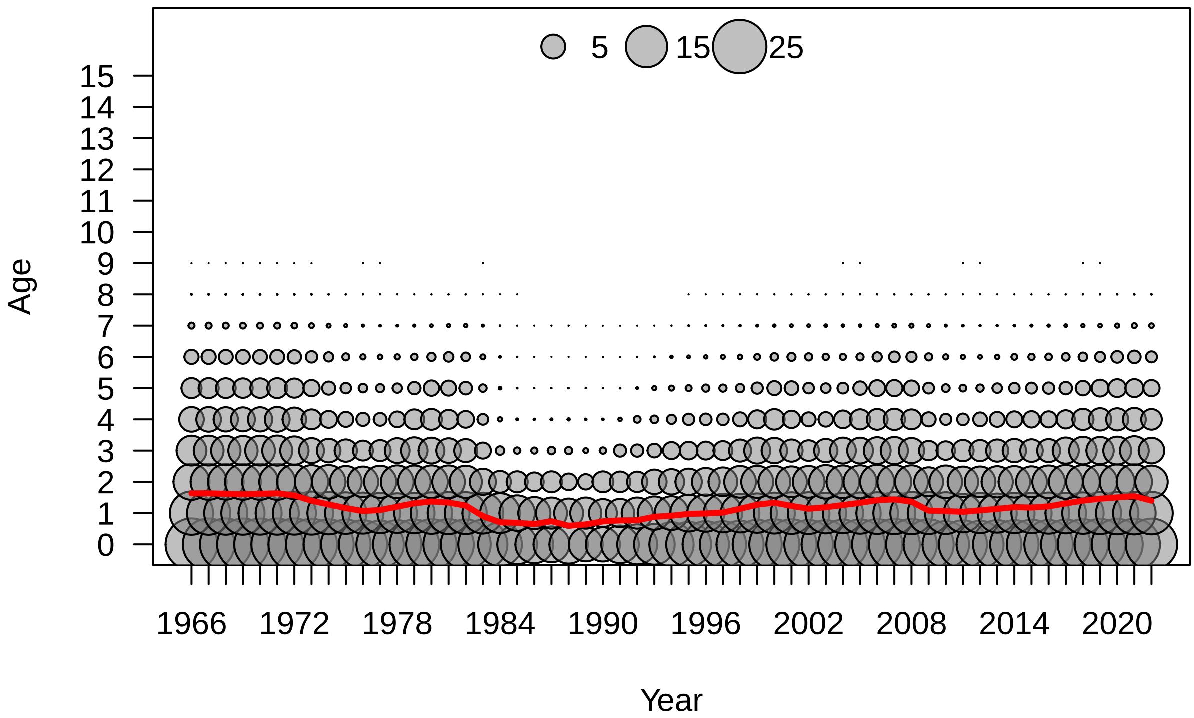
5
15
25
Age
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
1966
1972
1978
1984
1990
1996
2002
2008
2014
2020
Year



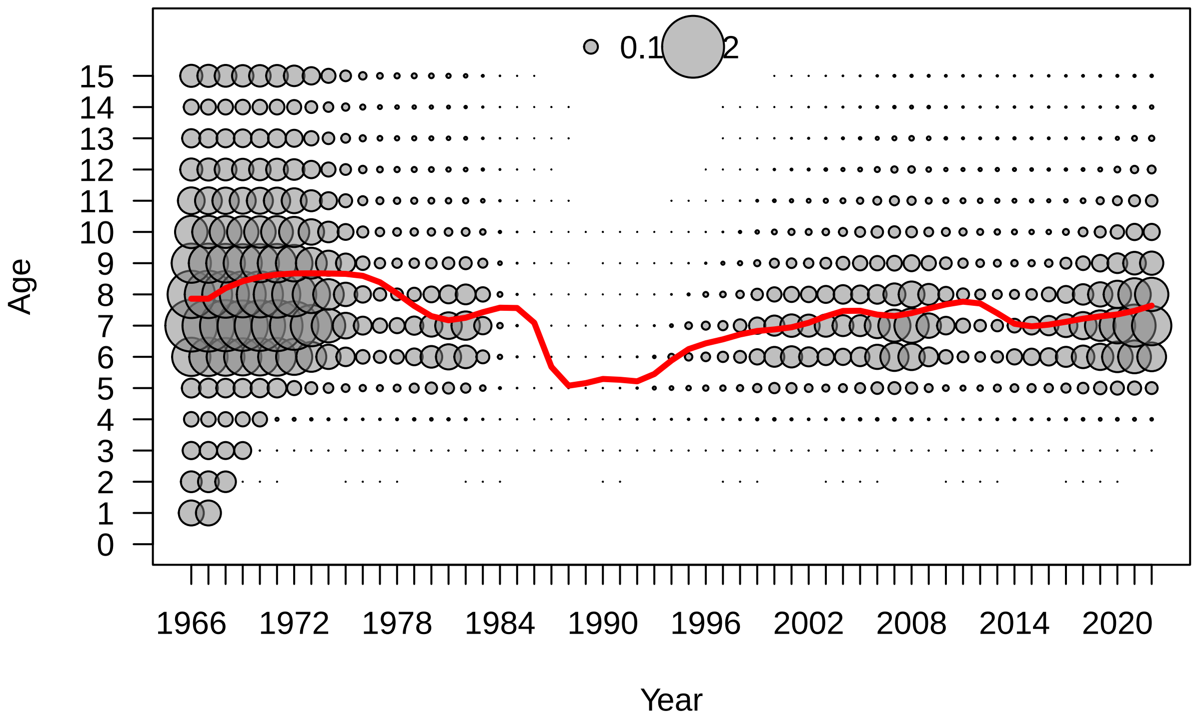
0.1
2
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
Age
1966
1972
1978
1984
1990
1996
2002
2008
2014
2020
Year



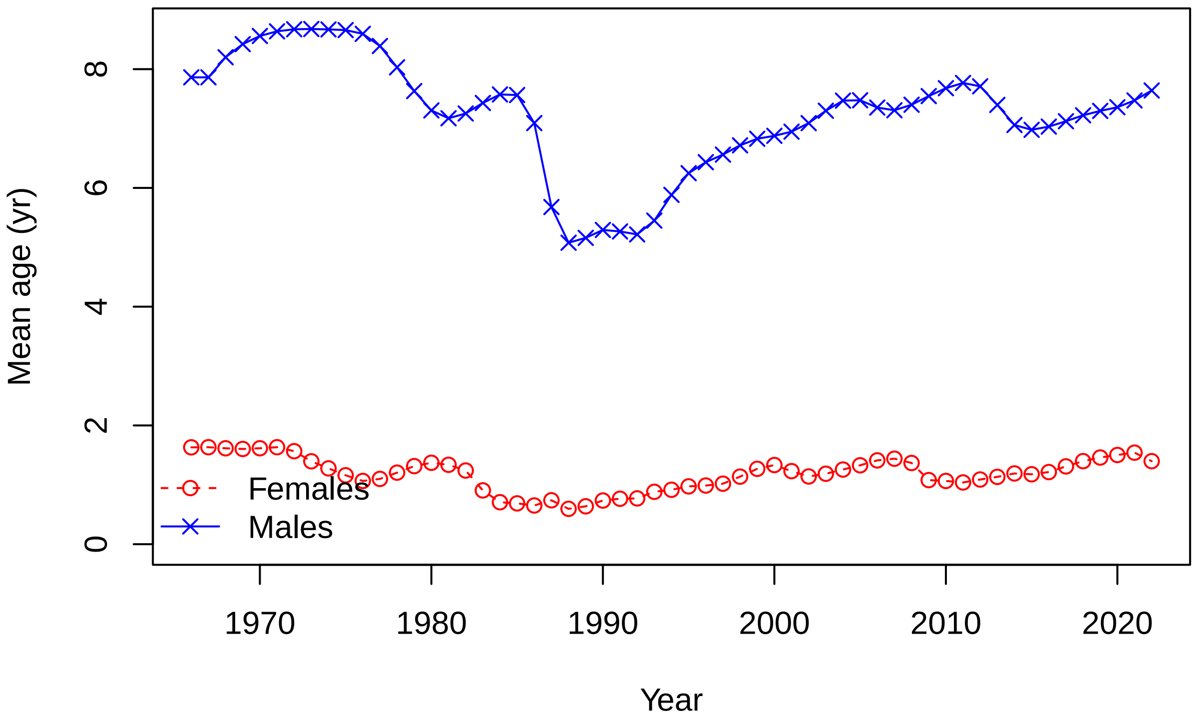
Mean age (yr)
0
2
4
6
8
Females
Males
1970
1980
1990
2000
2010
2020
Year

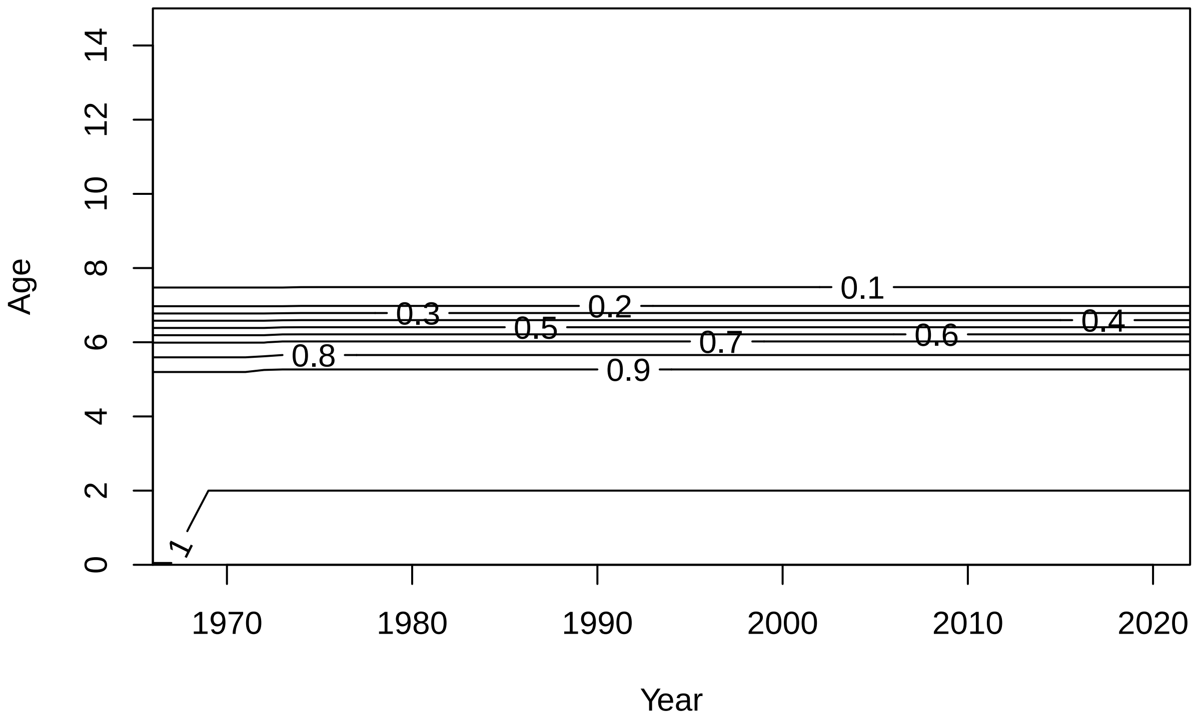


5
15
25
Age
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
1966
1972
1978
1984
1990
1996
2002
2008
2014
2020
Year

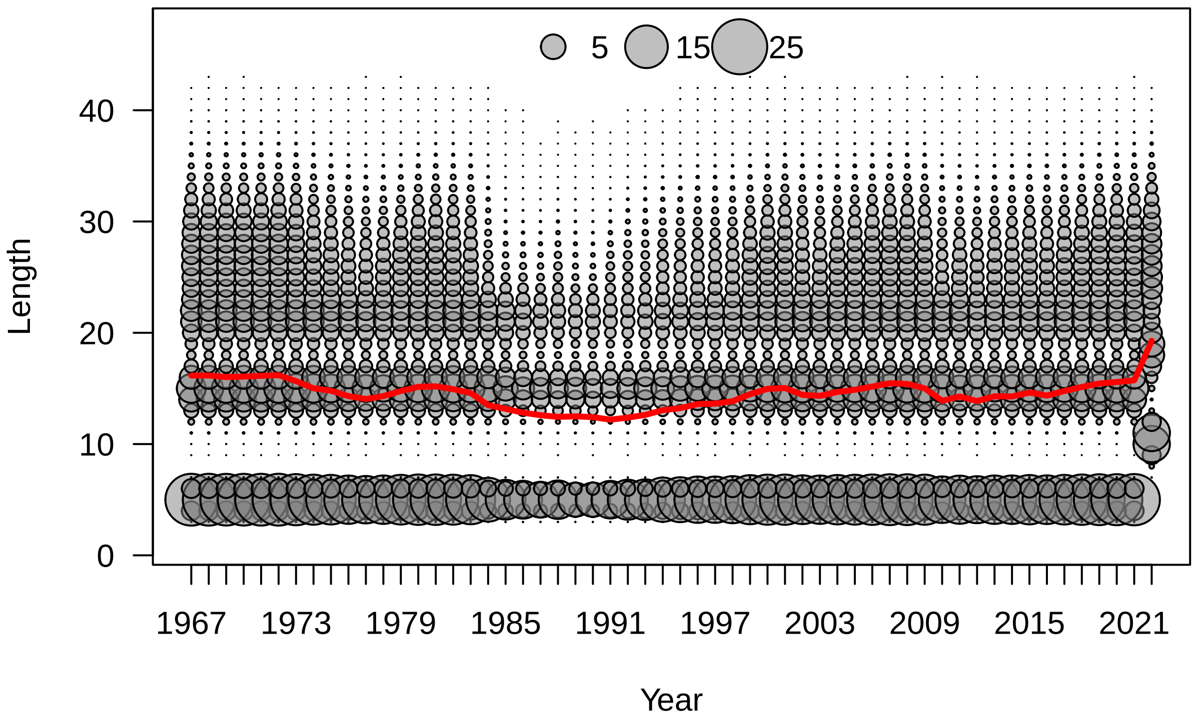


0.1
2
Age
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
1966
1972
1978
1984
1990
1996
2002
2008
2014
2020
Year

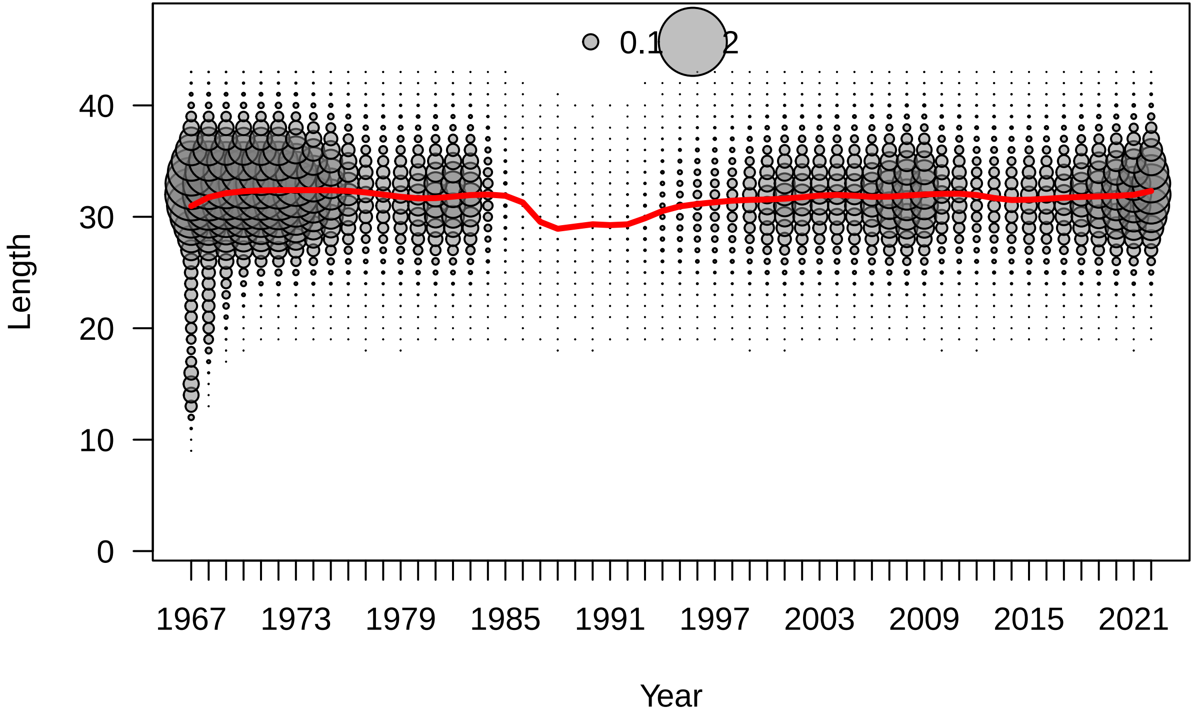

Mean age (yr)
0
2
4
6
8
Females
Males
1970
1980
1990
2000
2010
2020
Year



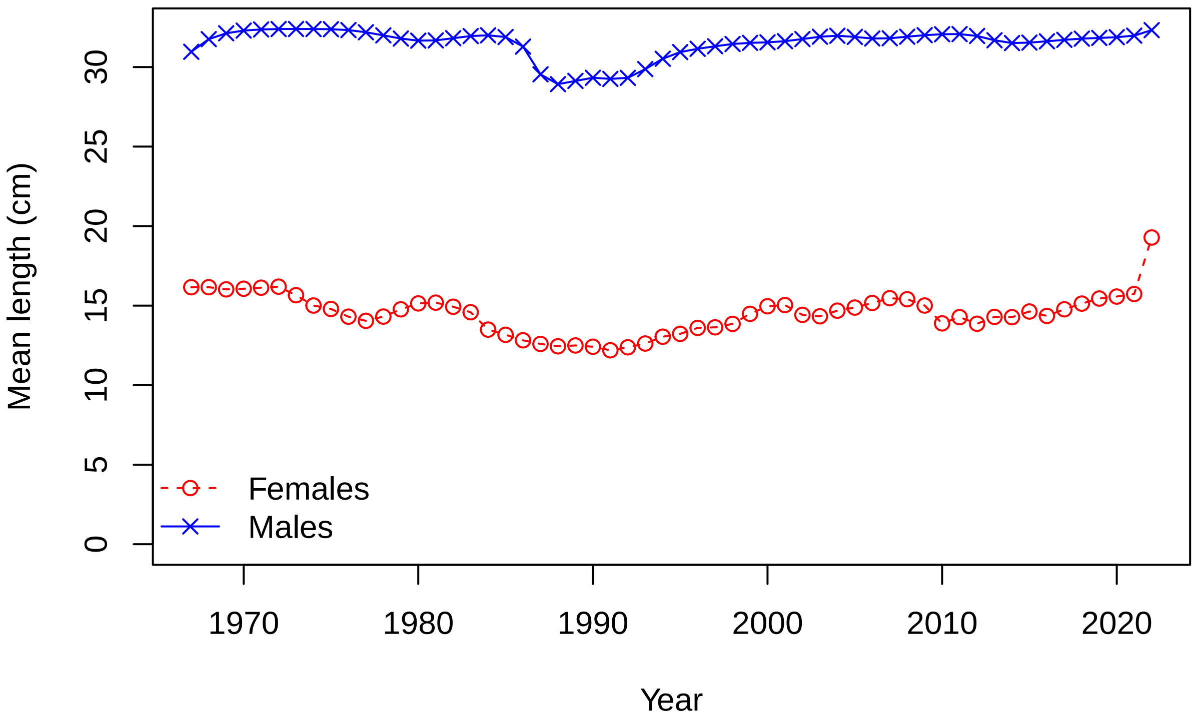
Age
0
2
4
6
8
10
12
14
Year
1970
1980
1990
2000
2010
2020
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1



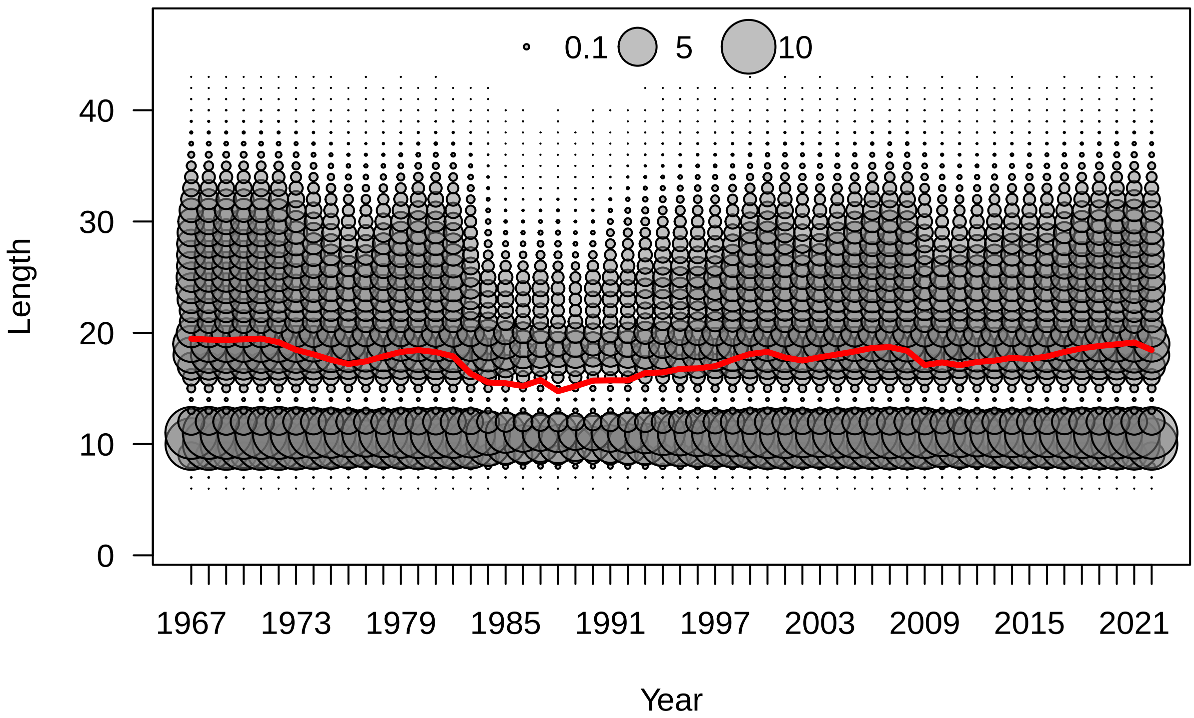
5
15
25
Length
40
30
20
10
0
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year



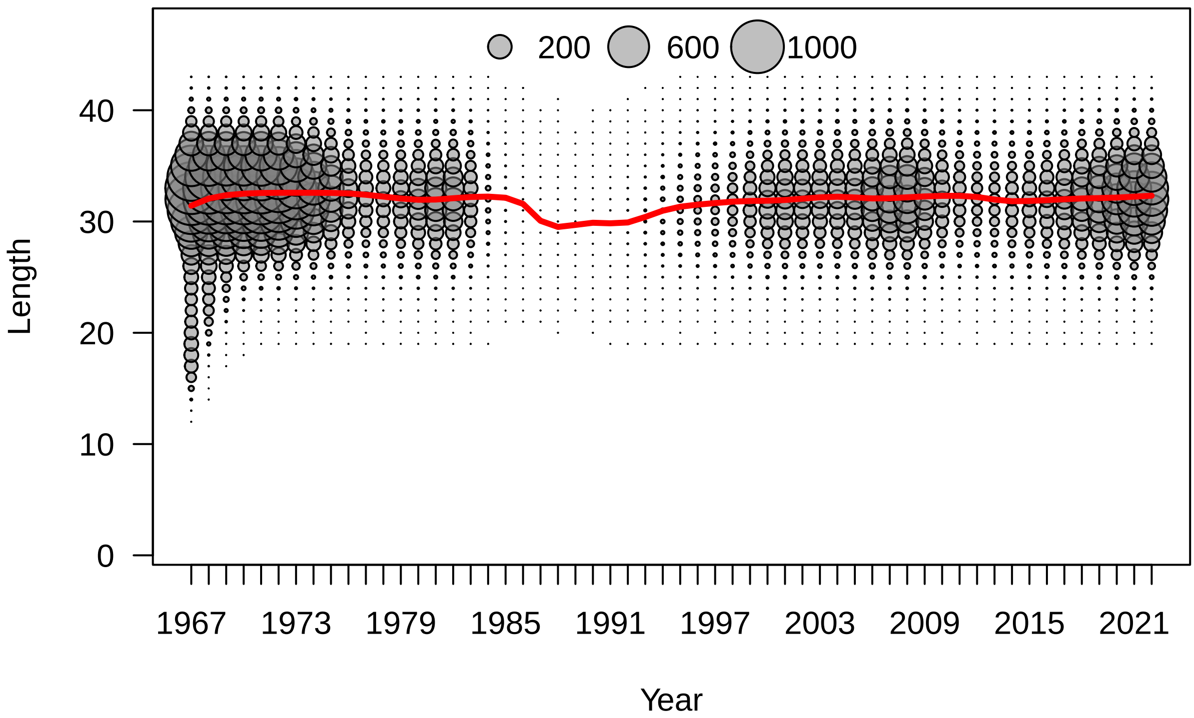
0.1
2
40
30
20
10
0
Length
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year



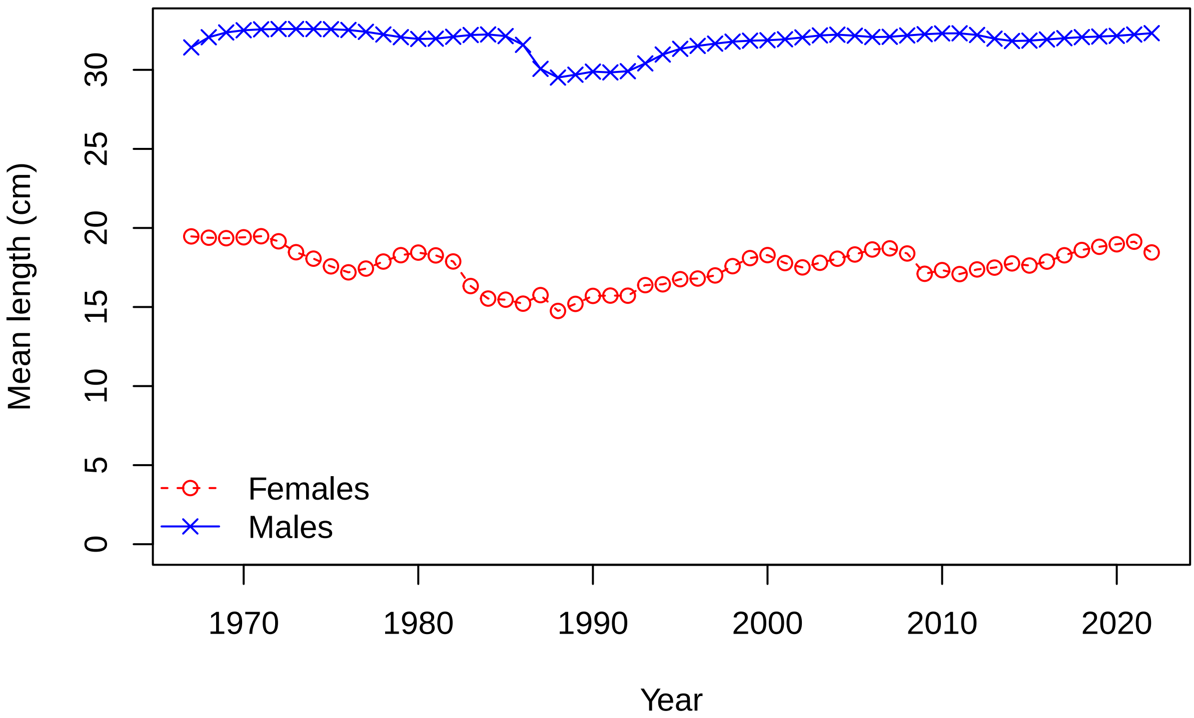
Mean length (cm)
0
5
10
15
20
25
30
1970
1980
1990
2000
2010
2020
Year
Females
Males



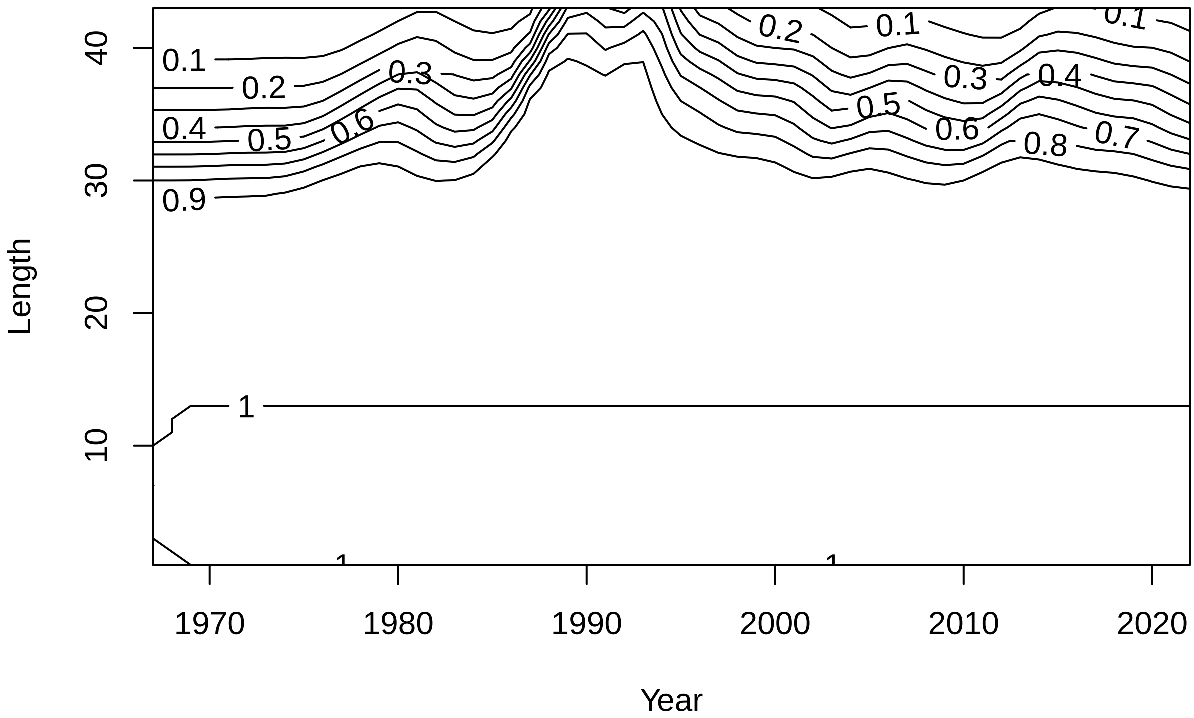
0.1
5
10
Length
40
30
20
10
0
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year



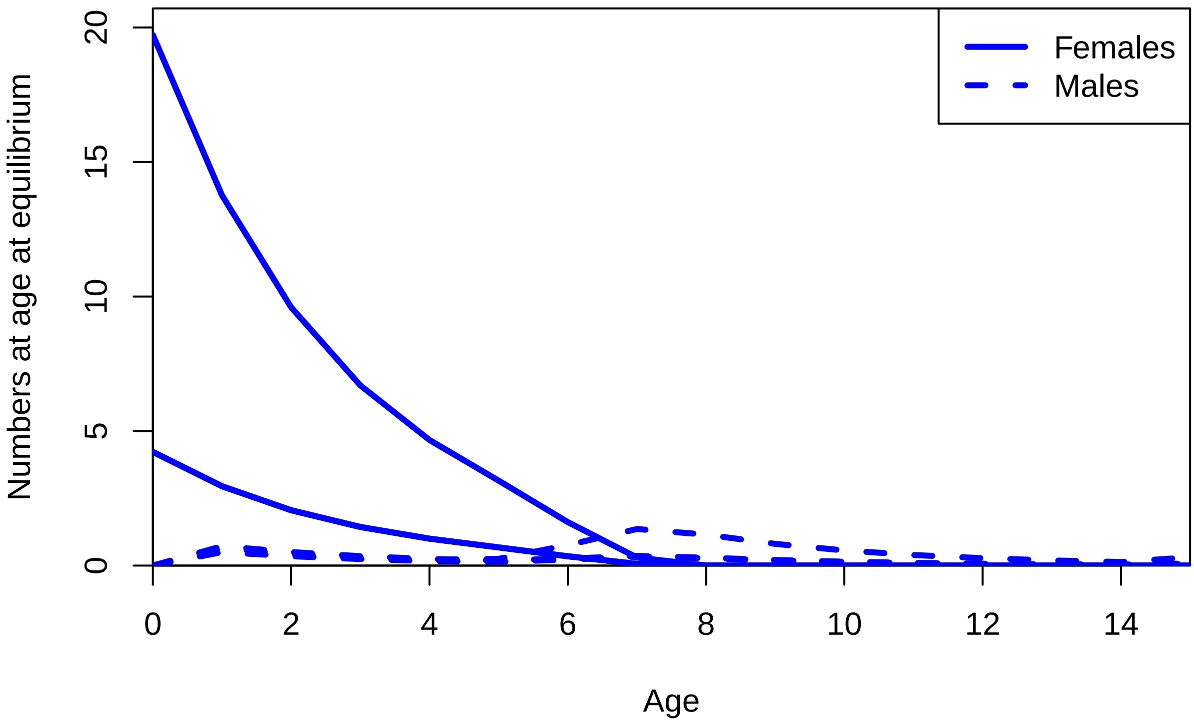
200
600
1000
40
30
20
10
0
Length
1967
1973
1979
1985
1991
1997
2003
2009
2015
2021
Year



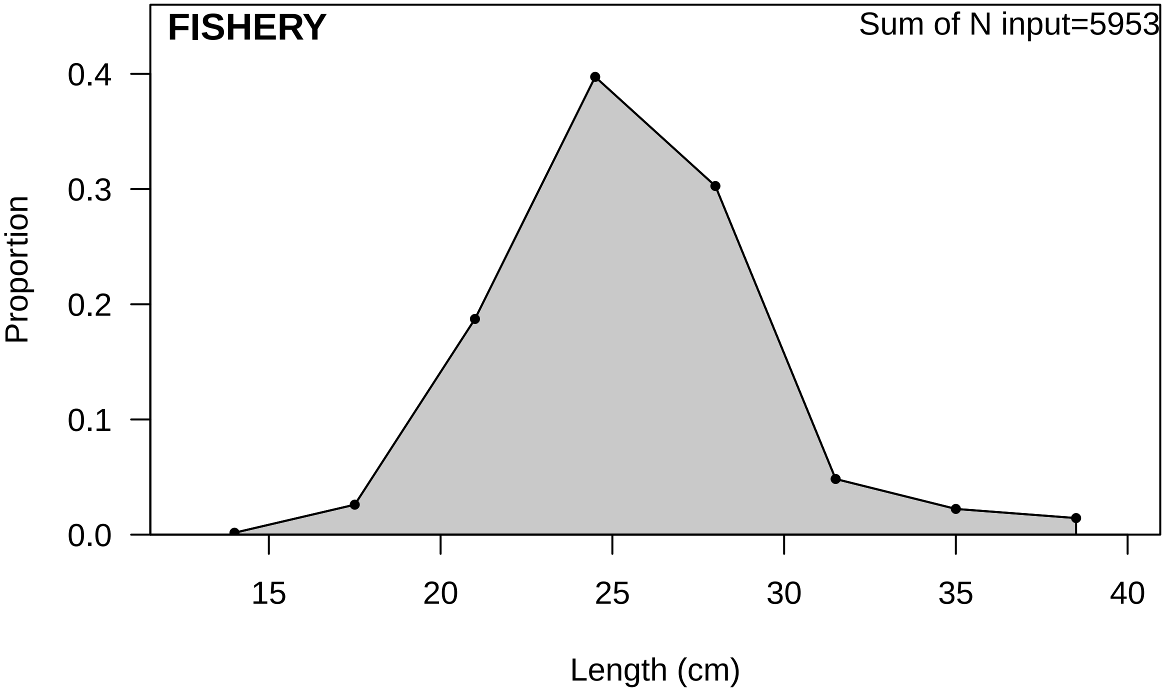
Mean length (cm)
0
5
10
15
20
25
30
1970
1980
1990
2000
2010
2020
Year
Females
Males

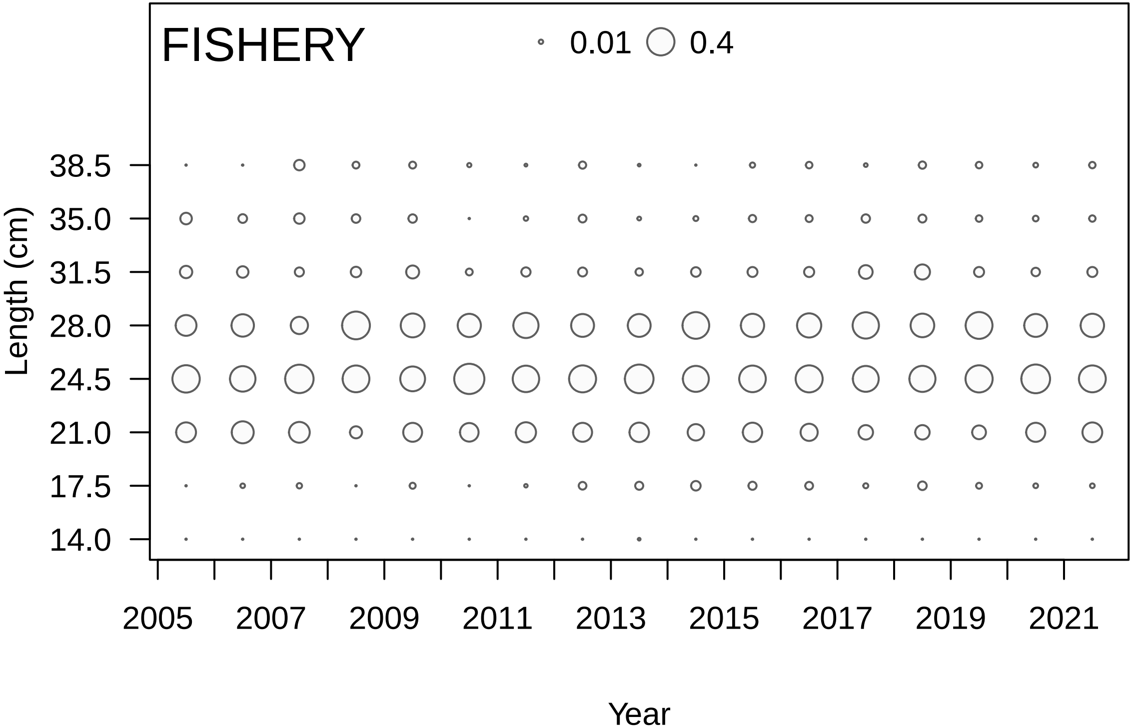


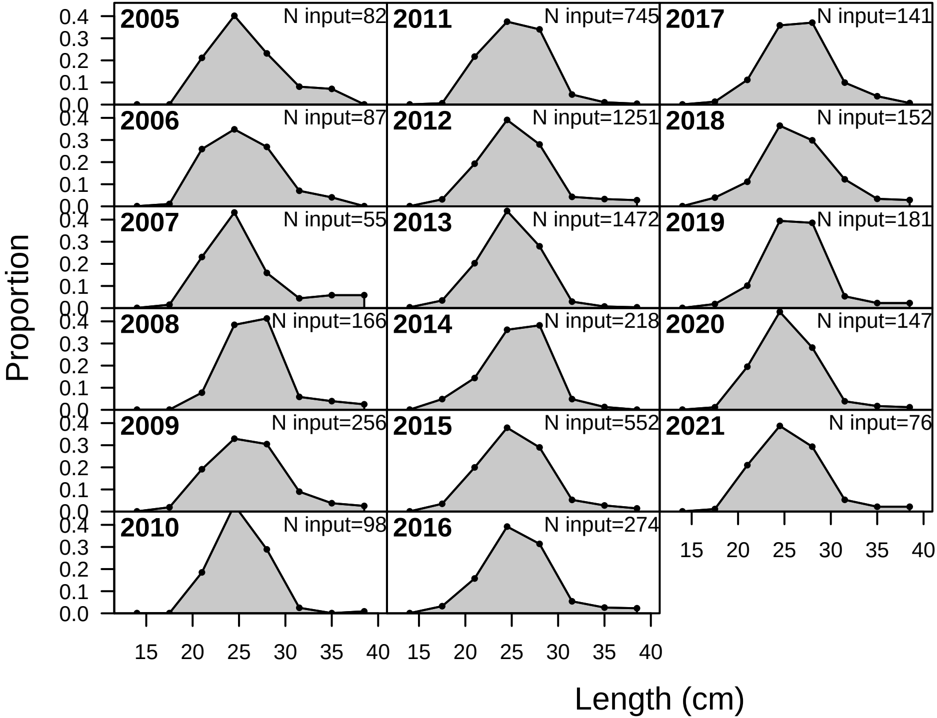
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Length
Year
10
20
30
40
1970
1980
1990
2000
2010
2020

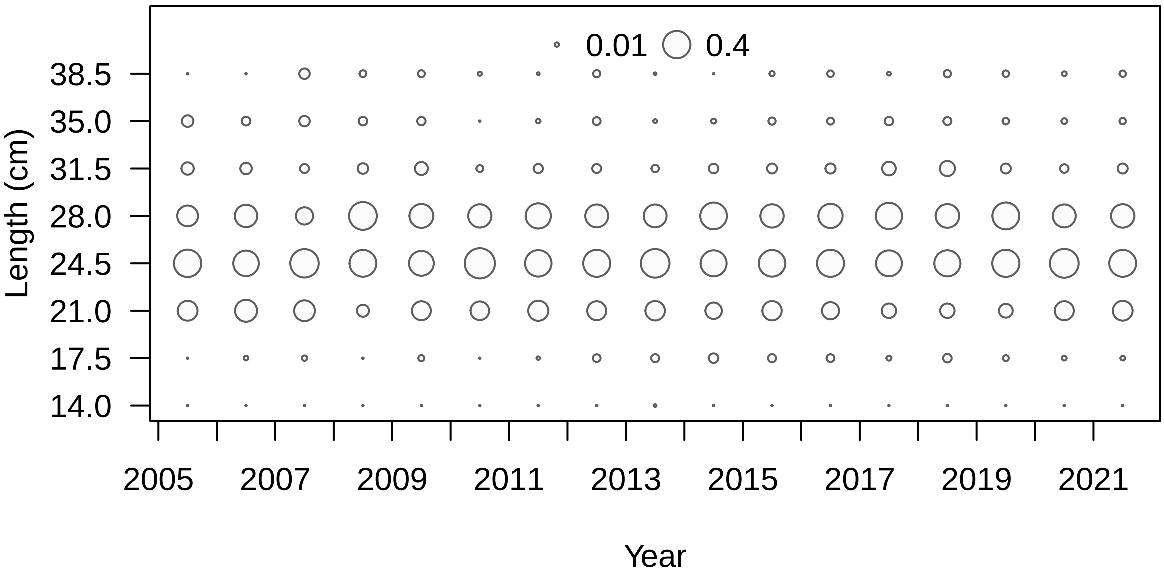


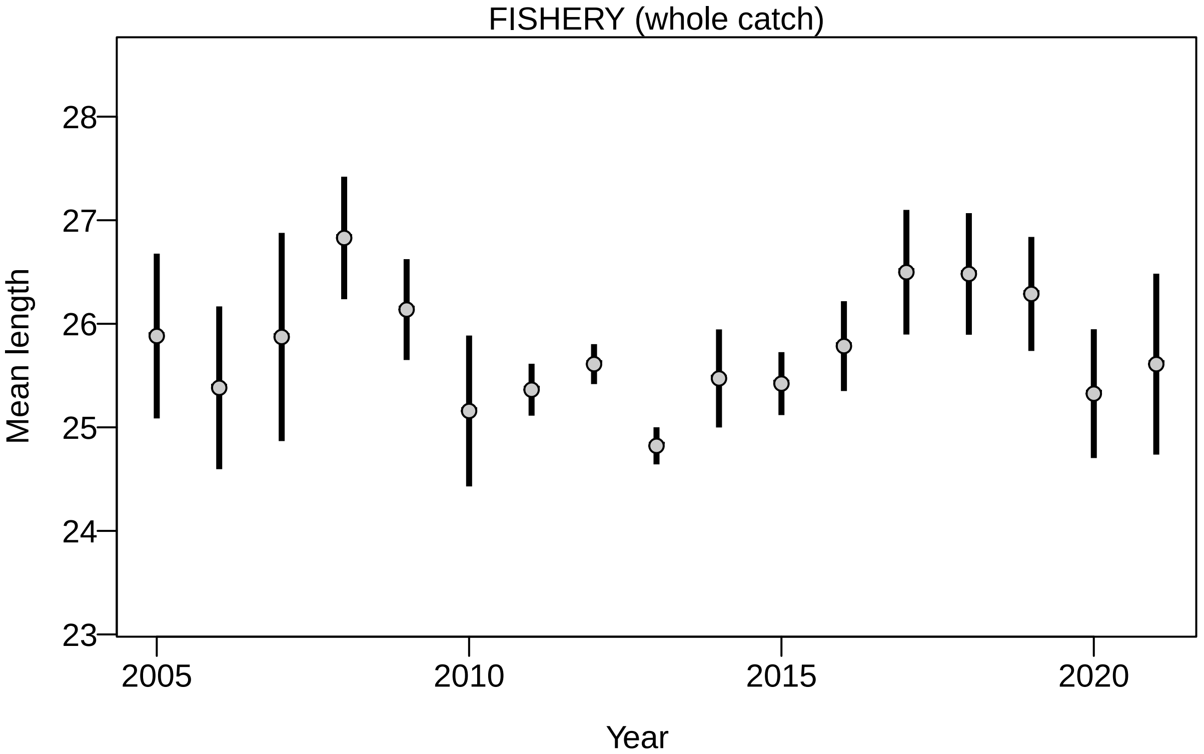
Numbers at age at equilibrium
0
5
10
15
20
0
2
4
6
8
10
12
14
Age
Females
Males

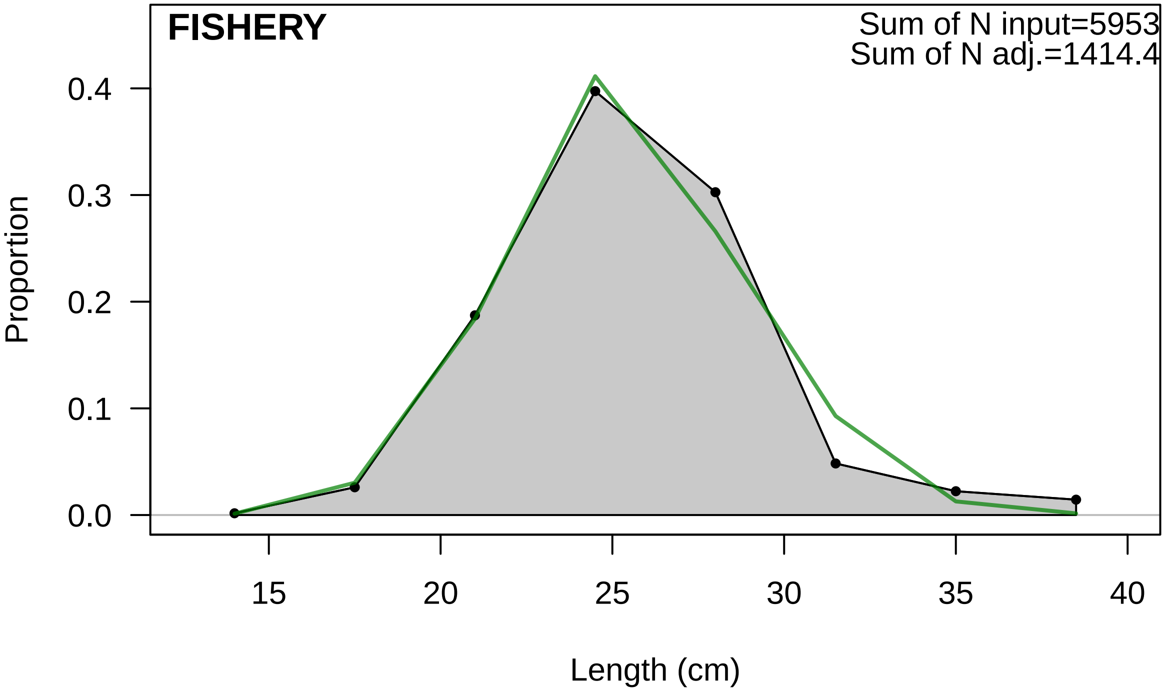

FISHERY
Sum of N input=5953
Proportion
0.0
0.1
0.2
0.3
0.4
15
20
25
30
35
40
Length (cm)

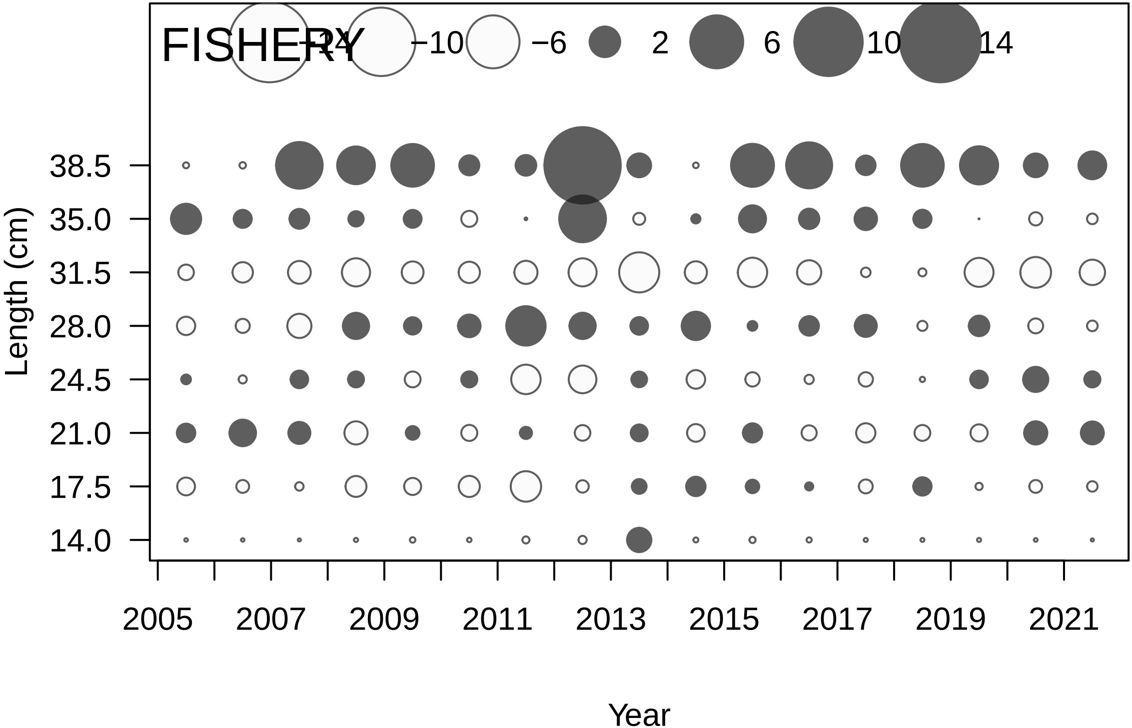

FISHERY
0.01
0.4
38.5
35.0
31.5
28.0
24.5
21.0
17.5
14.0
Length (cm)
2005
2007
2009
2011
2013
2015
2017
2019
2021
Year

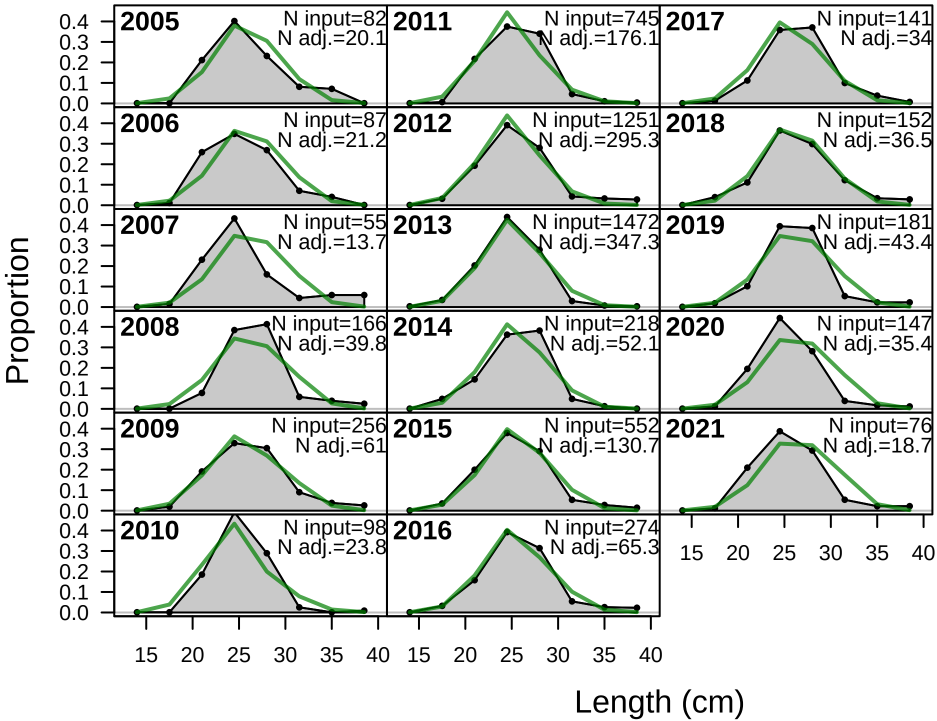

2005
N input=82
2011
N input=745
2017
N input=141
2006
N input=87
2012
N input=1251
2018
N input=152
2007
N input=55
2013
N input=1472
2019
N input=181
2008
N input=166
2014
N input=218
2020
N input=147
2009
N input=256
2015
N input=552
2021
N input=76
2010
N input=98
2016
N input=274
Proportion
0.0
0.1
0.2
0.3
0.4
15
20
25
30
35
40
Length (cm)


0.01
0.4
38.5
35.0
31.5
28.0
24.5
21.0
17.5
14.0
Length (cm)
2005
2007
2009
2011
2013
2015
2017
2019
2021
Year

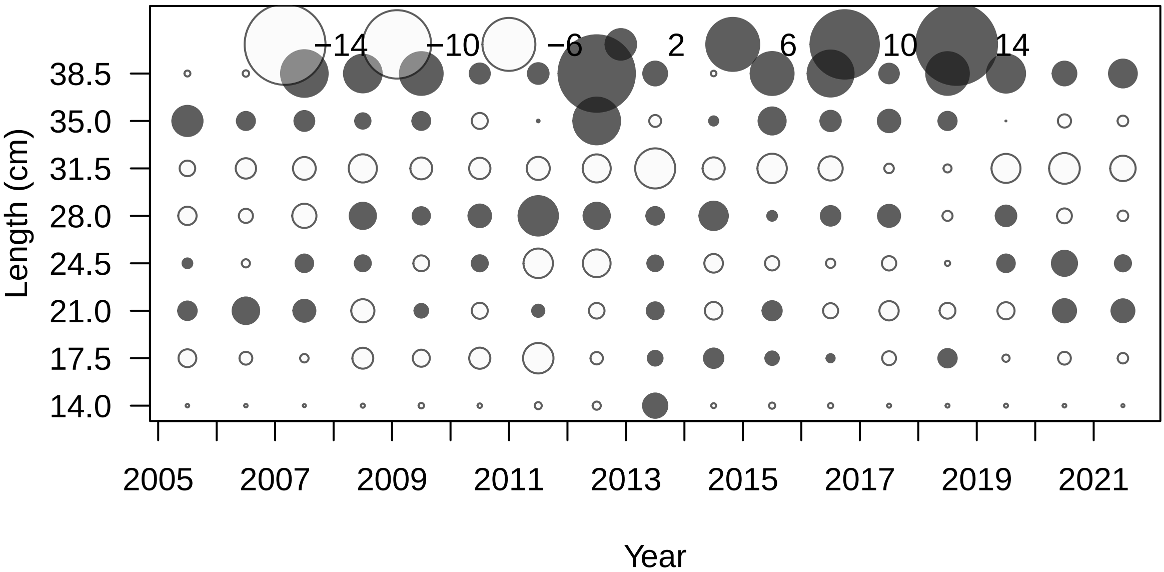

FISHERY (whole catch)
Mean length
28
27
26
25
24
23
2005
2010
2015
2020
Year

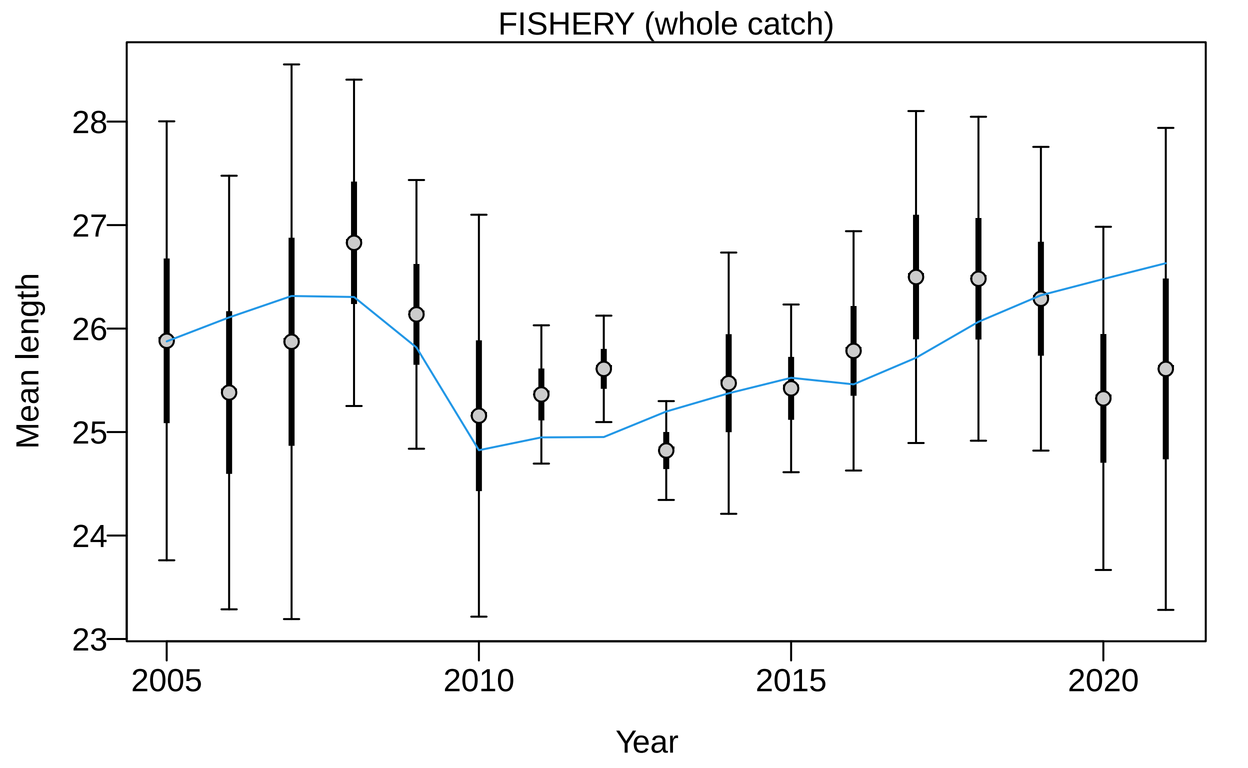

FISHERY
Sum of N input=5953
Sum of N adj.=1414.4
Proportion
0.0
0.1
0.2
0.3
0.4
15
20
25
30
35
40
Length (cm)

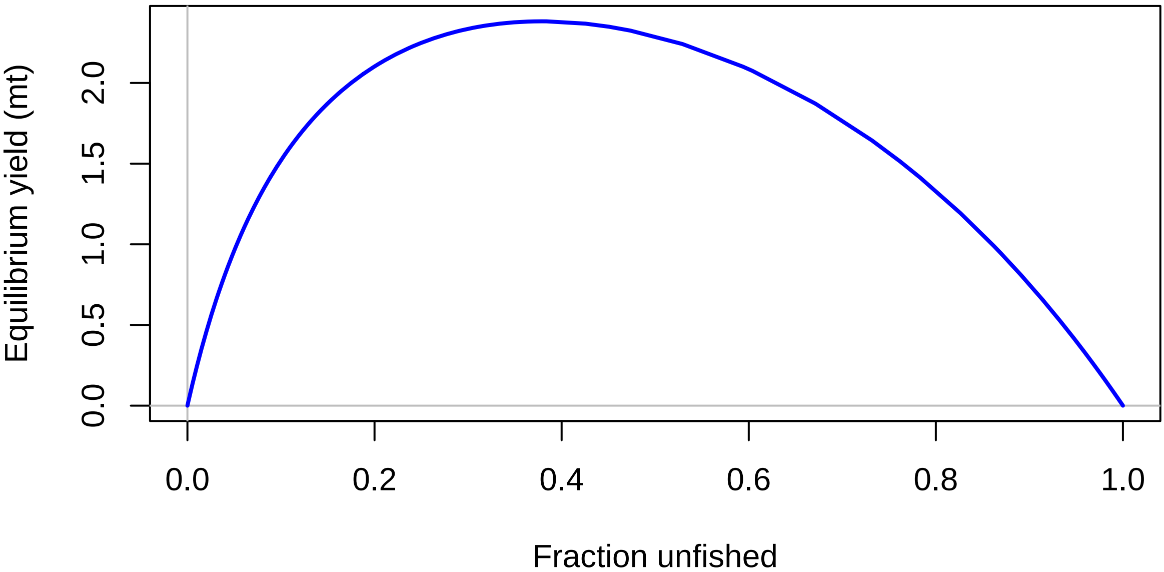

FISHERY
−14
−10
−6
2
6
10
14
Length (cm)
38.5
35.0
31.5
28.0
24.5
21.0
17.5
14.0
2005
2007
2009
2011
2013
2015
2017
2019
2021
Year



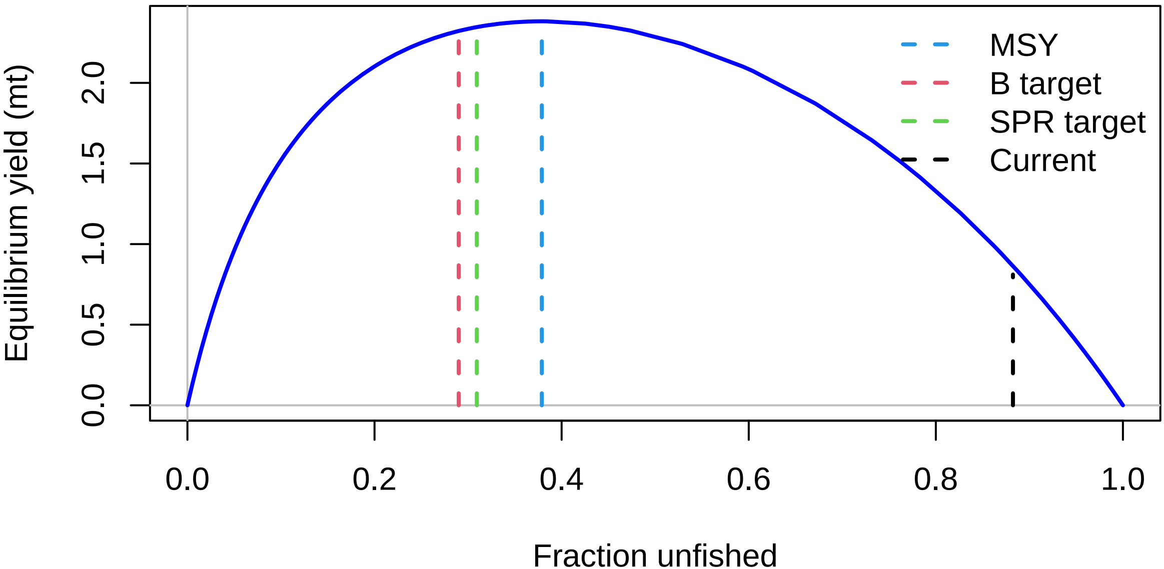
2005
N input=82
N adj.=20.1
2011
N input=745
N adj.=176.1
2017
N input=141
N adj.=34
2006
N input=87
N adj.=21.2
2012
N input=1251
N adj.=295.3
2018
N input=152
N adj.=36.5
2007
N input=55
N adj.=13.7
2013
N input=1472
N adj.=347.3
2019
N input=181
N adj.=43.4
2008
N input=166
N adj.=39.8
2014
N input=218
N adj.=52.1
2020
N input=147
N adj.=35.4
2009
N input=256
N adj.=61
2015
N input=552
N adj.=130.7
2021
N input=76
N adj.=18.7
2010
N input=98
N adj.=23.8
2016
N input=274
N adj.=65.3
Proportion
0.0
0.1
0.2
0.3
0.4
15
20
25
30
35
40
Length (cm)


−14
−10
−6
2
6
10
14
38.5
35.0
31.5
28.0
24.5
21.0
17.5
14.0
Length (cm)
2005
2007
2009
2011
2013
2015
2017
2019
2021
Year

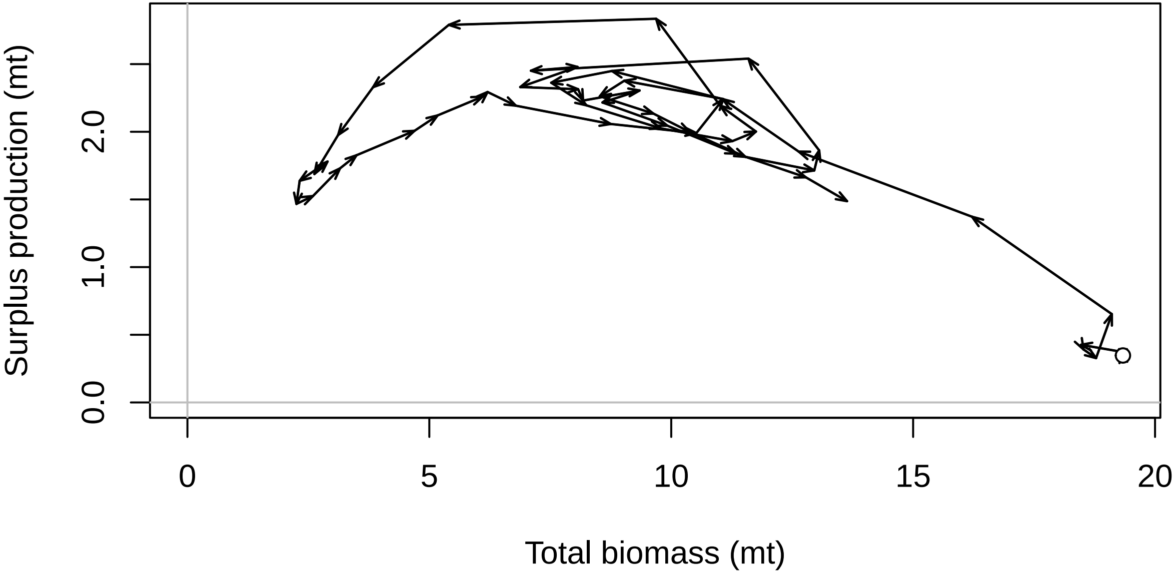

FISHERY (whole catch)
Mean length
28
27
26
25
24
23
2005
2010
2015
2020
Year



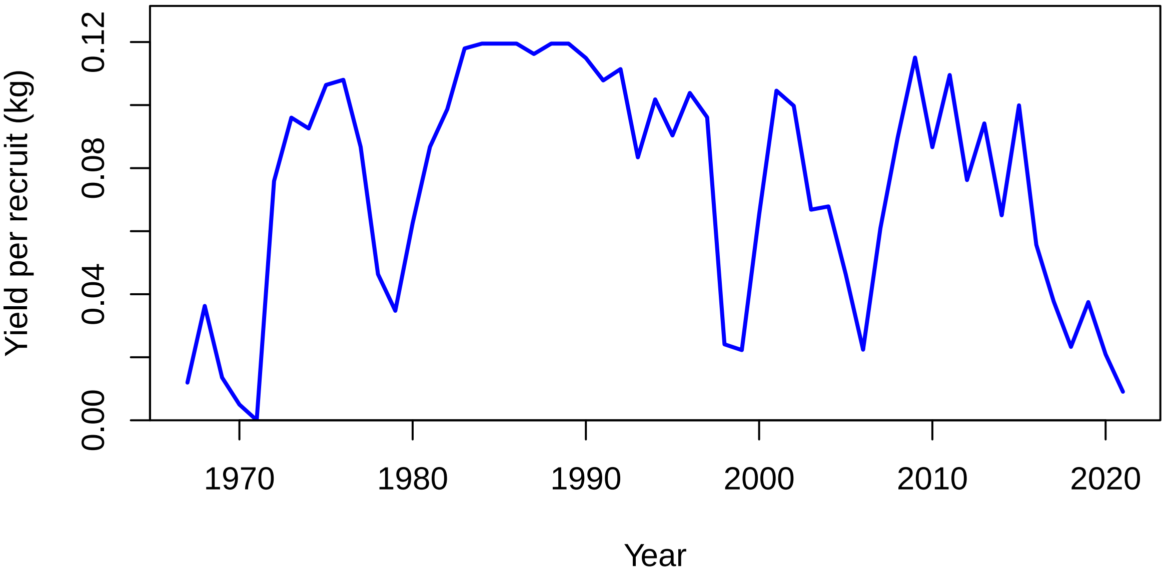
Equilibrium yield (mt)
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished



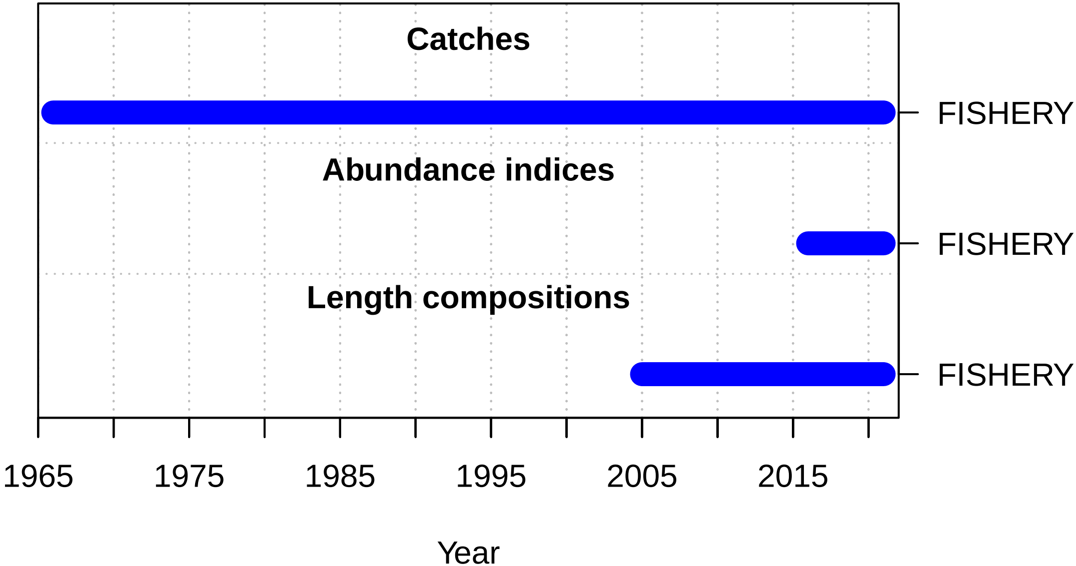
MSY
B target
SPR target
Current
Equilibrium yield (mt)
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Fraction unfished



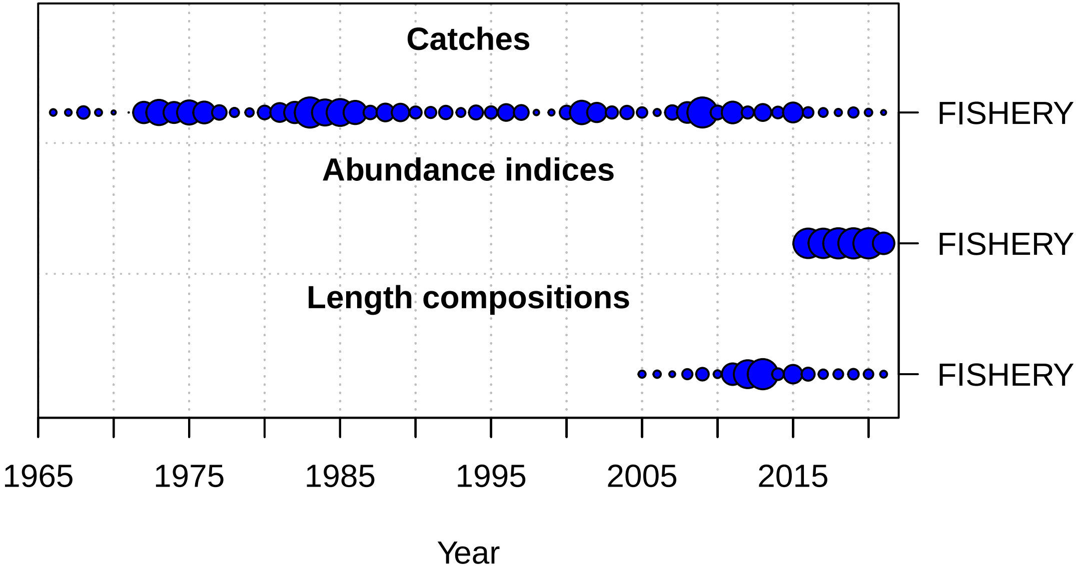
Surplus production (mt)
0.0
1.0
2.0
0
5
10
15
20
Total biomass (mt)

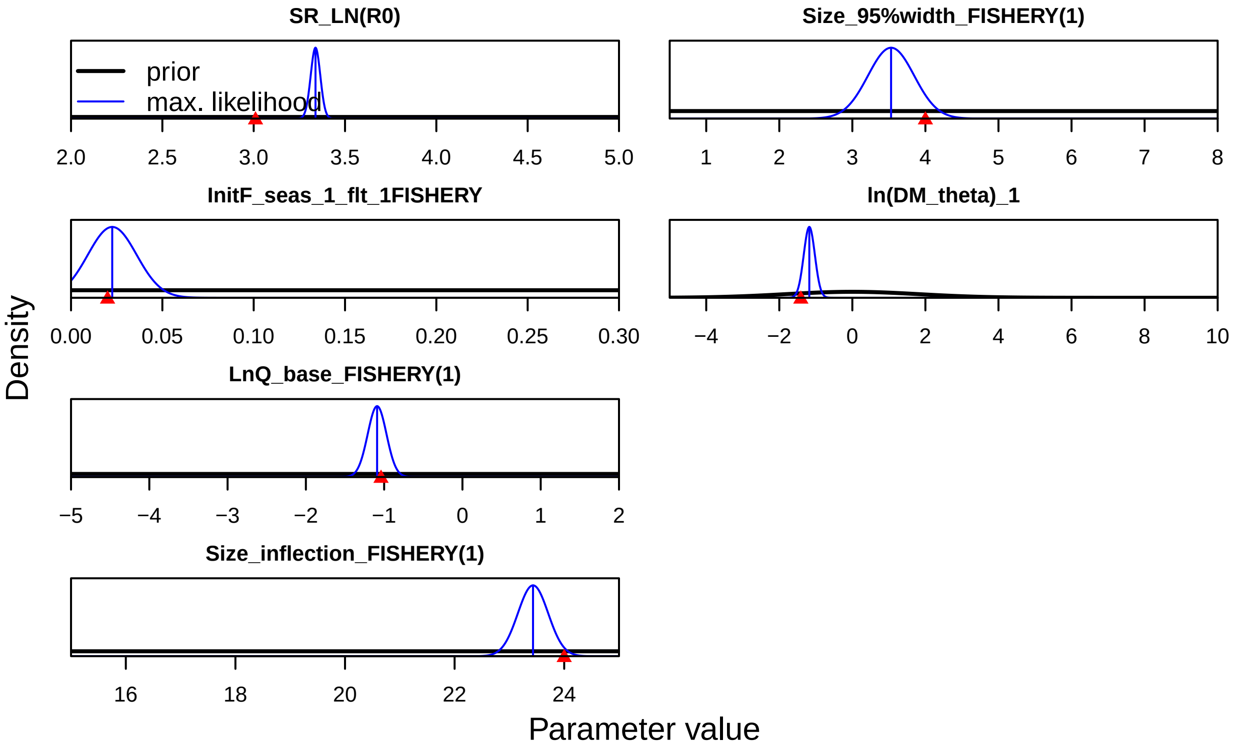


Yield per recruit (kg)
0.00
0.04
0.08
0.12
1970
1980
1990
2000
2010
2020
Year



Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1965
1975
1985
1995
2005
2015
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1965
1975
1985
1995
2005
2015
Year



SR_LN(R0)
prior
max. likelihood
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Size_95%width_FISHERY(1)
1
2
3
4
5
6
7
8
InitF_seas_1_flt_1FISHERY
0.00
0.05
0.10
0.15
0.20
0.25
0.30
ln(DM_theta)_1
−4
−2
0
2
4
6
8
10
LnQ_base_FISHERY(1)
−5
−4
−3
−2
−1
0
1
2
Size_inflection_FISHERY(1)
16
18
20
22
24
Density
Parameter value